

College as a Commitment Device: Parental Altruism and the Samaritan's Dilemma*

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Abstract

Why do parents invest so heavily in their children's college education? I argue that college acts as a commitment device against a Samaritan's dilemma: anticipating parental support, children under-save and over-consume, and college mitigates this by permanently raising the child's income, shrinking the set of states in which parents optimally transfer. Consistent with this mechanism, in matched parent-child data parents of college children consume more, save with a weaker precautionary motive, and transfer less often, though in larger amounts when they do. A dynastic model estimated by the simulated method of moments shows that the friction raises college enrollment above the level a parent who could commit would choose, because the anticipation of support subsidizes attendance. The welfare consequences of removing the friction—and of policies such as free college—depend on whether parents can commit, since the same transfers that distort the college decision also insure children against income risk.

JEL: D15, D64, I22, I23

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1 Introduction

Parents in the United States transfer substantial resources to their adult children. Inter-vivos transfers—cash gifts, tuition payments, housing assistance—flow overwhelmingly downward, from parents to children, and persist well into adulthood (McGarry, 1999, 2016). These transfers are infrequent—in the Health and Retirement Study, only 14% of parent-child pairs report a positive transfer in any given wave—but when they occur, they are large: conditional amounts average over \$8,000. Parental transfers are also a central input into children’s human capital formation: families remain the primary source of college financing in the United States, and the anticipation of parental financial support is a key determinant of whether children enroll (Belley and Lochner, 2007; Brown et al., 2012; Abbott et al., 2019). From a public policy perspective, this link matters: government financial aid crowds out private parental contributions, and this crowding out is sizable precisely because family transfers are responsive to the child’s enrollment decision (Abbott et al., 2019). Understanding why parents transfer, and how these transfers—both realized and anticipated—shape children’s choices, is therefore fundamental both for explaining college attendance and for designing effective education policy.

This paper studies these questions through the lens of the Samaritan’s dilemma (Lindbeck and Weibull, 1988). Altruistic parents who cannot credibly commit to withholding future support create an incentive problem: children who anticipate parental transfers under-save and over-consume, knowing that their parents will step in when resources are low. This strategic response induces parents to make larger and more frequent transfers than they would make under commitment. The dilemma therefore generates a strategic interaction between parental altruism and children’s choices: parents support their children *because* they care, but the anticipation of that support distorts the decisions that determine how much support will ultimately be needed. Because transfers are closely tied to the college decision, this strategic interaction has direct implications for human capital investment. Children who expect parental support face different incentives when choosing whether to attend college, and their college decision in turn reshapes the future pattern of transfers.

I argue that college education plays a distinctive role in this dynamic. From the parent’s perspective, a dollar spent on tuition is fundamentally different from a dollar transferred di-

rectly. Tuition generates a permanent shift in the child’s income level, moving the child away from the states of the world where parental support would be triggered. A direct transfer, by contrast, provides temporary relief and may itself exacerbate moral hazard by rewarding under-saving. College thus serves as a commitment device: by investing in the child’s human capital, parents can shrink the *transfer region*—the set of states where they would optimally provide support—thereby reducing the lifetime costs of the Samaritan’s dilemma. The net effect on college attendance, however, is theoretically ambiguous. Anticipated transfers during the college years subsidize enrollment, raising attendance, but anticipated lifetime transfers also provide implicit insurance to children who forgo college, reducing their incentive to invest in human capital. The structural model quantifies these opposing forces and shows that the sign of the moral hazard effect on college varies across the ability–wealth distribution.

I formalize this mechanism in a continuous-time, non-cooperative dynastic model, building on [Barczyk and Kredler \(2014a,b\)](#). Parents choose consumption and a non-negative transfer flow to their adult children; children choose consumption and, at the start of the dynasty, whether to attend college. The interaction between parent and child is non-cooperative: each agent optimizes taking the other’s policy as given, while the parent internalizes the child’s welfare through an altruism parameter. In equilibrium, the state space is endogenously partitioned into a transfer region, where parents actively support their children, and a no-transfer region, where the child is self-sufficient. The boundary between these regions—and hence the insurance value of parental support—is shaped by the child’s education, income, and accumulated wealth. College contracts the transfer region by permanently raising the child’s expected earnings, thereby reducing the insurance value of future parental transfers. The child’s college decision therefore depends not only on the human capital return, but also on how education reshapes the future financial relationship with the parent.

I document a set of stylized facts from the Panel Study of Income Dynamics (PSID) and the Health and Retirement Study (HRS) that are consistent with the model’s predictions about how parent-child financial interactions differ by child education. Throughout, I compare parents whose children attended college with parents whose children did not, conditioning on parent income to separate the role of child education from parental resources. *First*, parent consumption is 8% higher when children attend college—consistent with college

shrinking the transfer region and releasing resources that would otherwise be held as precautionary wealth or transferred to children. *Second*, parents whose children attend college appear to draw down wealth more slowly in retirement, consistent with the reduced lifetime transfer obligations implied by a smaller transfer region—though this difference is imprecisely estimated and I treat it as suggestive. *Third*, following the framework of [Boar \(2021\)](#), I examine whether college attenuates dynastic precautionary saving. The point estimate implies that the negative effect of child income uncertainty on parent consumption is 62% smaller for college-educated children. Although this estimate is not statistically significant, its magnitude is economically meaningful and suggests that college reduce parents’ exposure to their children’s income risk. *Fourth*, within families, college reduces the probability of any transfer by 21%, but conditional on transferring, amounts are weakly larger for college children—consistent with the model’s prediction that only worse shocks push college children into the transfer region. College transforms the financial relationship between parents and adult children by reducing the frequency and precautionary burden of parental support.

I develop a dynastic model of parent-child financial interactions in which parental altruism, the anticipation of support, and human capital investment are jointly determined. The model distinguishes college’s role as a commitment device—reducing the moral hazard costs of the Samaritan’s dilemma—from its standard human capital return, and generates predictions about how education reshapes transfers, consumption, and wealth accumulation within the family. I estimate the model by the simulated method of moments, targeting college enrollment rates by ability and wealth, the college premium, transfer patterns, and intergenerational wealth dynamics. I use the estimated model to quantify the role of the Samaritan’s dilemma in shaping both college decisions and the lifetime pattern of parent-child financial interactions. Relative to an efficient benchmark in which the parent can commit to education-contingent transfers, the lack of commitment leads parents to over-enroll their children in college: the anticipation of ongoing support subsidizes attendance and raises enrollment above the level a committed parent would choose, with the over-enrollment concentrated among medium- and high-ability children. Removing the friction is not unambiguously beneficial, because the transfers that distort the college decision also insure children against income risk; dynastic welfare rises only when commitment preserves that insurance rather than severing

the transfer relationship altogether. I show that the welfare effects of a free-college policy likewise depend critically on whether parents can commit. These findings carry implications for the design of financial aid and transfer policies.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model. Section 4 documents the stylized facts. Section 5 describes the estimation strategy. Section 6 discusses the model fit and the role of parental altruism. Section 7 introduces the full commitment benchmark, formalizes the two opposing forces of moral hazard on college, and presents the quantitative decomposition. Section 9 concludes.

2 Related Literature

This paper draws on several literatures: intergenerational transfers and family insurance, the role of parental resources in major life-cycle investments, saving and wealth decumulation in retirement, and the determinants of college attainment.

A large body of work studies how families insure members against income risk. Since [Becker and Tomes \(1979, 1986\)](#), two broad motives for intergenerational transfers have been proposed: altruism, supported by evidence that transfers flow disproportionately to less well-off children ([McGarry, 1999, 2016](#)), and strategic exchange, as in [Bernheim et al. \(1985\)](#). The seminal tests of [Altonji et al. \(1992\)](#) and [Hayashi et al. \(1996\)](#) reject perfect insurance within the extended family, but [Attanasio et al. \(2018\)](#) finds substantial insurance potential between parents and adult children, and [Kaplan \(2012\)](#) shows that the option to move back home insures young workers against labor market risk. Most directly related to this paper, [Boar \(2021\)](#) identifies dynastic precautionary saving as a distinct insurance channel: parents accumulate wealth to buffer their children against permanent income uncertainty. I build on this framework and extend it by showing that college education attenuates this dynastic precautionary motive—a prediction I test empirically and embed in the structural model.

Intergenerational transfers, however, are not limited to insurance against income shocks. Parents also provide resources for large, lumpy investments such as college attendance and home purchases, and they transmit wealth through inter-vivos gifts and bequests. This broader view is important because parental transfers may affect children’s outcomes not only by smoothing consumption, but also by relaxing borrowing constraints, changing portfolio

choices, and shaping exposure to risk. [Nishiyama \(2002\)](#) studies altruistic and accidental bequests in an overlapping-generations model and shows how bequest motives affect wealth accumulation. [Barczyk et al. \(2023\)](#) emphasize the interaction between housing, family insurance, and the timing of transfers, showing that illiquid housing can help explain why many transfers are delayed until death. Closest to this broader mechanism, [Brandsaas \(2025\)](#) develops a life-cycle model with housing and parent-child transfers and finds that parental transfers account for a sizable share of young-adult homeownership by relaxing down-payment constraints, reducing housing risk, and interacting with commitment frictions. This paper builds on the insight that family transfers affect children’s long-run investment, but studies this mechanism in the context of human capital accumulation.

Because parents here hold wealth in part to stand ready to support adult children, the paper also connects to the literature on saving and wealth decumulation in retirement, where older households draw down wealth far more slowly than a life-cycle model without family motives predicts, and the income-rich decumulate most slowly of all. [De Nardi et al. \(2010\)](#) trace much of this slow drawdown to longevity and medical-expense risk, while [Love et al. \(2009\)](#) show that annualized wealth is roughly flat or rising over retirement and that intended bequests are needed to match the profiles of higher-income households in the Health and Retirement Study. Family links do much of the remaining work: [De Nardi \(2004\)](#) shows that voluntary bequests are required to reproduce both the slow late-life drawdown and the concentration of wealth at the top, [De Nardi and Yang \(2014\)](#) that they generate the dispersion of wealth observed at retirement, and [Lockwood \(2018\)](#) that a bequest motive can rationalize why retirees decline to insure late-life risks. Closest to the motive I emphasize, [Ameriks et al. \(2016\)](#) separate bequest from precautionary saving and find that parents hold wealth to assist children when the need is greatest. This literature models the family motive largely as a terminal bequest or treats old-age risk as exogenous; I instead micro-found a forward-looking inter-vivos motive—parents retain wealth because of the option value of an endogenous transfer region in which supporting a child is optimal—and show that this region, and hence the predicted pace of decumulation, depends on the child’s college. Consistent with this account, the parents in my sample reproduce the slow, income-graded drawdown these studies document, with a (statistically imprecise) college differential layered on top

(Section 4.3).

A separate literature asks whether financial constraints bind for college attendance. [Cameron and Heckman \(1998\)](#) and [Keane and Wolpin \(2001\)](#) found that, conditional on ability, family income played a small role, but [Belley and Lochner \(2007\)](#) document a growing importance of family resources in the 2000s, and [Lochner and Monge-Naranjo \(2011\)](#) provide evidence of binding constraints for some students. [Brown et al. \(2012\)](#) emphasize that the expected family contribution is neither legally guaranteed nor universally provided, implying that students with similar observed family resources may face very different access to parental support. [Hotz et al. \(2018\)](#) provide direct evidence that parental income and housing wealth affect college attendance, parental financing, and graduation, and that parental financing may affect the subsequent indebtedness of both parents and children. On the quantitative side, [Restuccia and Urrutia \(2004\)](#) study how parental investments drive intergenerational earnings persistence, [Caucutt and Lochner \(2020\)](#) analyze the interaction between borrowing constraints and the timing of human capital investments, and [Heckman and Mosso \(2014\)](#) provide a comprehensive treatment of how family environments shape skill development. Closest to my setting, [Abbott et al. \(2019\)](#) develop a general equilibrium model with parental transfers and show that crowding out of private contributions by government financial aid is quantitatively important. My paper adds to this work by providing a mechanism—moral hazard reduction through the Samaritan’s dilemma—through which parental wealth affects college enrollment beyond direct cost subsidies and paternalism.

The model builds on the quantitative literature on family dynamics without commitment, including [Laitner \(1988\)](#) and [Lindbeck and Weibull \(1988\)](#). Within this tradition, [Nishiyama \(2002\)](#) studies how bequests and inter-vivos transfers jointly shape the wealth distribution, and [Lee and Seshadri \(2019\)](#) develop a dynastic model with parental investments in children’s human capital to study the intergenerational transmission of economic status. I adopt the continuous-time framework of [Barczyk and Kredler \(2014a,b\)](#), who show that modeling simultaneous consumption and transfer decisions in continuous time yields a well-defined equilibrium with an endogenous transfer region—the set of states where the parent optimally provides support. [Barczyk and Kredler \(2018\)](#) and [Barczyk et al. \(2023\)](#) apply this framework to long-term care and housing decisions; I extend it by embedding endogenous

college decisions, which allows me to study how the transfer region—and the moral hazard it generates—interacts with education choice.

The Samaritan’s dilemma that drives this moral hazard was formalized and analyzed in dynamic settings by [Lindbeck and Weibull \(1988\)](#) and [Bruce and Waldman \(1990\)](#). In my model, the distortion takes the form of strategic under-saving by the child, which raises the parent’s lifetime transfer costs. College mitigates the dilemma by permanently shifting the child’s income, reducing the frequency with which the child enters the transfer region. The net effect on college attendance, however, is ambiguous, because anticipated transfers both subsidize enrollment and cushion the non-college path.

3 Model

This section presents a continuous-time, non-cooperative dynastic model without commitment that captures the interactions between altruistic parents and their adult children. The model extends the framework of [Barczyk and Kredler \(2014a,b\)](#) by incorporating endogenous college decisions, allowing me to study how family transfers affect college financial support, graduation, and parent retirement consumption. I first describe the demographic structure and income processes, then present the coupled parent-child optimization problem, the parent’s transfer decision, and the college decision.

3.1 Demographics and Timing

Time is continuous. Each dynasty consists of one parent and one child who overlap for a finite period, as illustrated in [Figure 1](#). The child enters the model at age 18, at which point the parent is 42. Parent and child are coupled for the parent’s entire remaining life: the overlap runs from child age 18 / parent age 42 to the parent’s death at age 72, when the child is 48. The parent works until age 66 ($t = T^{\text{ret}}$), earning labor income $y_p(t)$, and then receives social security income $\text{SS}(e_p)$ until death at age 72 ($t = T$, where $T = T^{\text{ret}} + 6$). The retired parent remains coupled with the child, so transfers $\tau(t) \geq 0$ are available throughout the overlap $[0, T]$, including during the parent’s retirement; the coupled problem is solved over the parent’s full remaining life. Death is deterministic at age 72 (the model has no mortality risk). At the parent’s death, a fraction α_{beq} of remaining parent wealth transfers to the child

as a bequest; the remaining $(1 - \alpha_{\text{beq}})$ share is not transmitted.¹ The child, now 48, then continues alone—working until its own retirement at 66 and consuming social security until death at 72.

At the beginning of the dynasty, the child makes a one-time, irreversible college decision. Children who attend college spend four years (ages 18–22) earning a fraction ℓ_C of full-time income while paying annual tuition ϕ . After college, the child enters the labor force with a permanently higher income process. Children who do not attend college enter the labor market immediately at age 18.

Both parent and child can save in a risk-free bond paying interest rate r . The only debt instrument is the student loan: during and after college the child pays $r + \iota_s$ on any negative balance ($a_c < 0$), and this college debt must be amortized to zero by the time the child becomes a parent (age 42). The child cannot otherwise borrow as an adult, so outside the student-loan window $a_c \geq 0$. The parent faces a non-negativity constraint $a_p \geq 0$ and always earns r . Markets are incomplete: no state-contingent insurance is available against the child’s income risk. The parent and child are connected through the transfer flow during the overlap period and through a bequest at the parent’s death: a fraction α_{beq} of remaining parent wealth transfers to the child. Inter-vivos transfers and bequests are the two channels of intergenerational resource transmission.

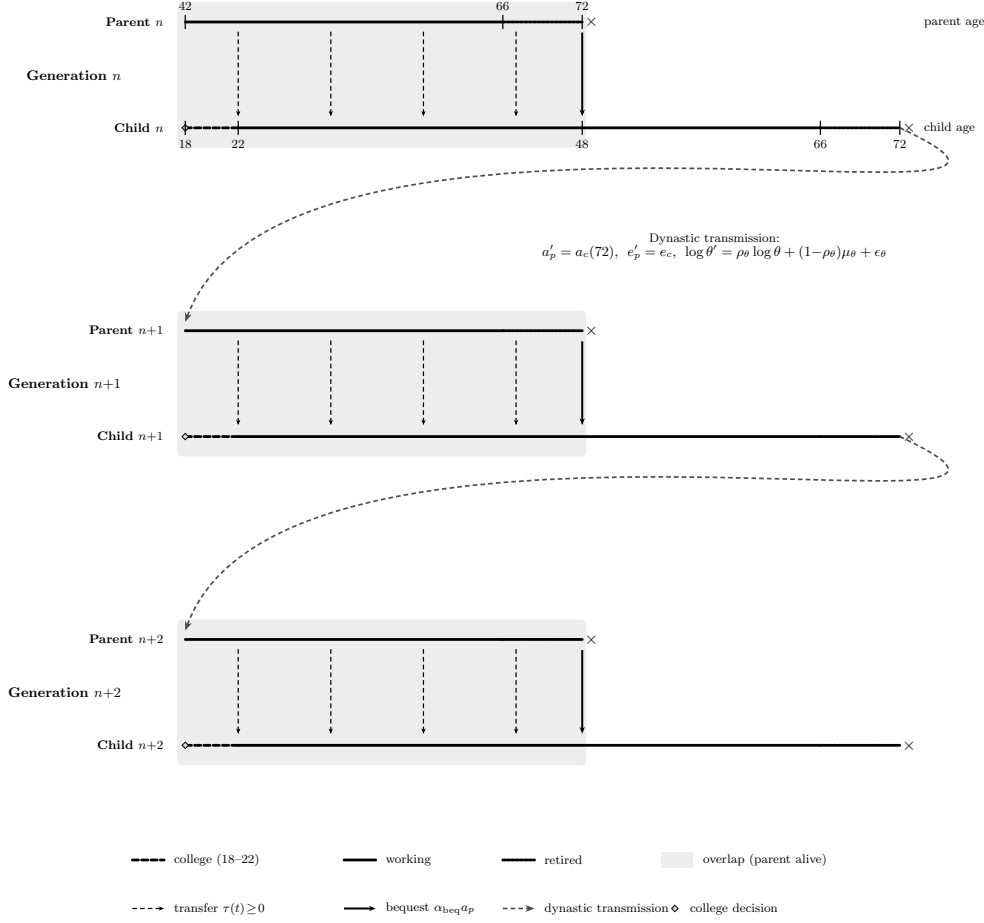
Three features of this demographic structure deserve comment. First, altruism is one-sided: the parent cares about the child’s welfare, but the child does not internalize the parent’s utility. Second, transfers flow only downward—from parent to child—and the overlap is finite, ending at the parent’s death (age 72). The dynasty is closed as a stationary recursion: the wealth a child accumulates over its life becomes the initial wealth of the next generation’s parent, while the share of the deceased parent’s wealth not bequeathed to the child is not transmitted. Third, the child does not support the parent in old age. A natural alternative motive for financing college is that parents invest in their children’s human capital expecting financial support in return—an exchange motive in the tradition of [Bernheim et al. \(1985\)](#). I abstract from upward transfers because they are empirically small relative to downward

¹The bequest fraction α_{beq} is estimated by SMM. The warm-glow motive ϕ_b gives the parent direct utility from terminal wealth, while α_{beq} governs the actual resource transfer. De Nardi (2004) estimates bequest motives implying $\alpha_{\text{beq}} \approx 0.6$ –0.8; Boar (2021) implies ~ 0.75 in her altruism calibration.

flows, as the next paragraph documents: parent-to-child transfers are the first-order feature of parent-child finances and the natural object for a model of college support. This restriction does not separate altruism from the exchange motive—the two are not separately identified in my setting—but it keeps the analysis focused on the Samaritan’s dilemma, the friction this paper quantifies.

Upward transfers are empirically minor. In the United States, intergenerational transfers are overwhelmingly downward: [McGarry \(1999\)](#) documents that inter-vivos transfers from parents to children are an order of magnitude larger than transfers in the reverse direction, and [McGarry \(2016\)](#) confirms this asymmetry persists over the life cycle. Old-age consumption in the US is largely financed through Social Security, employer pensions, and private savings rather than through children’s contributions—a pattern the model captures through private retirement savings and the social security income $SS(e_p)$ during retirement. The exchange motive is an interesting and complementary channel, but incorporating it would require modeling two-sided altruism or strategic bargaining over old-age support, substantially complicating the equilibrium characterization without directly informing the mechanism this paper studies.

Figure 1. Dynasty Timeline



Notes: Three consecutive generations of the dynasty. Within each generation, the parent (upper line) and child (lower line) are coupled for the parent's entire remaining life (parent ages 42–72, child ages 18–48). The shaded region is this overlap, during which transfers $\tau(t)$ flow from parent to child (vertical dashed arrows); the retired parent (ages 66–72) remains coupled and can still transfer. The dashed segment on each child line denotes the college phase (ages 18–22), preceded by the college-decision marker ($\diamond$). At the parent's death (child age 48), a fraction α_{beq} of remaining wealth transfers to the child as a bequest (solid vertical arrow); the rest is not transmitted. The child then lives alone (ages 48–72). The large dashed arrows show the dynastic recursion that closes the model: the child's terminal state at age 72 seeds the next generation's parent—wealth ($a'_p = a_c(72)$), education ($e'_p = e_c$), and ability transmitted via the AR(1) process $\log \theta' = \rho_\theta \log \theta + (1 - \rho_\theta)\mu_\theta + \epsilon_\theta$. Ages are labeled once, on generation n .

3.2 State Variables and Income Processes

The state of a dynasty at any instant t during the overlap phase is $(a_p(t), a_c(t), z(t); \theta, z_p, e_p, e_c)$, where a_p and a_c are parent and child wealth, z is the child's stochastic productivity shock, θ is the child's cognitive ability, z_p is the parent's frozen persistent income type (the productivity state the parent inherited as the previous generation's child), and $e_p, e_c \in \{HS, C\}$ denote the education levels of parent and child. The variables θ, z_p, e_p , and e_c are fixed within a dynasty and index separate value-function problems; (a_p, a_c, z) are the dynamic states.

Child income. The child's labor income at time t is:

$$y_c(t) = w_{e_c} \cdot \left(\frac{\theta}{\bar{\theta}} \right)^{\lambda_{e_c}} \cdot \exp(\gamma(t, e_c) + z(t))$$

where $h(\theta, e_c) = (\theta/\bar{\theta})^{\lambda_{e_c}}$ maps ability into human capital with an education-specific ability gradient λ_{e_c} , following [Abbott et al. \(2019\)](#), and $\bar{\theta} = \exp(\mu_\theta)$ is the population mean ability.² The gradient $\lambda_C > \lambda_{HS}$ implies that cognitive ability and college education are complements: higher-ability individuals benefit more from college. The education-specific wage w_{e_c} captures the relative price of education-specific labor: $w_{HS} = w$ and $w_C = w \cdot \omega_C$. In the model, the college premium is generated by three components jointly: the higher ability gradient ($\lambda_C > \lambda_{HS}$), the steeper age-income profile ($\gamma_{1,C} > \gamma_{1,HS}$), and the relative wage ω_C . The parameter ω_C is estimated via SMM to match the empirical college premium. The deterministic age-income profile $\gamma(t, e_c) = \gamma_{1,e} t - \gamma_{2,e} t^2$ is taken from [Abbott et al. \(2019\)](#), and $z(t)$ follows an Ornstein-Uhlenbeck process:

$$dz = -\kappa_{e_c} z dt + \sigma_{e_c} dW$$

with education-specific mean-reversion rate κ_{e_c} and diffusion σ_{e_c} . Both the mean-reversion rates and the diffusion coefficients are set externally from the male estimates of [Abbott et al. \(2019\)](#): κ_{e_c} is mapped from the discrete-time persistence and σ_{e_c} is chosen so that the stationary variance $\sigma_{e_c}^2/(2\kappa_{e_c})$ reproduces their persistent-shock variance (Appendix D). The channel through which college reshapes the child's income distribution, and hence the transfer

²The normalization by $\bar{\theta}$ ensures that the human capital component equals one at mean ability for both education groups, so that the average college premium comes entirely from the life-cycle profile and wage price ω_C , not from the level effect of $\lambda_C \neq \lambda_{HS}$ evaluated at a non-zero mean $\log \theta$.

region, is the income *level*: a higher ability gradient ($\lambda_C = 0.797$ vs. $\lambda_{HS} = 0.517$), a steeper age-income profile ($\gamma_{1,C} > \gamma_{1,HS}$), and the relative wage ω_C jointly raise expected income and keep college-educated children far from the states where parental transfers would be triggered. The persistent-shock process is, if anything, slightly more dispersed for college graduates in the stationary distribution—the [Abbott et al. \(2019\)](#) estimates imply a stationary standard deviation of 0.51 for college versus 0.43 for high school—so the model does not rely on a reduction in income risk to generate the transfer-region contraction.³ During college ($t \in [0, 4]$ for $e_c = C$), child income is reduced to a fraction ℓ_C of full-time earnings, reflecting part-time work while enrolled.

Parent income. Parent income depends on education, age, and a frozen persistent income type z_p . Before retirement, $y_p(t) = w_{e_p} \cdot (\theta/\bar{\theta})^{\lambda_{e_p}} \cdot \exp(\gamma(t, e_p) + z_p \nu(t))$, and $y_p(t) = SS(e_p)$ after retirement at age 66. The parent draws no *new* idiosyncratic income shocks; instead it carries the fixed type z_p —the persistent productivity state it held as the previous generation’s child at the dynastic hand-off (age 42)—whose contribution fans out deterministically over the parent’s working life, $\nu(t) = \sqrt{1 + g_{zp} \max(0, \text{age}_p(t) - 42)}$ with $\nu = 1$ at age 42. This generates realistic cross-sectional dispersion in parent income and wealth while keeping the child’s stochastic income z as the only source of *within-dynasty* uncertainty that drives transfers. The fanning rate g_{zp} is fixed externally.

Ability transmission. Cognitive ability is transmitted across generations through a log-normal AR(1) process: $\log \theta' = \rho_\theta \log \theta + \mu_\theta(1 - \rho_\theta) + \epsilon_\theta$, with $\epsilon_\theta \sim N(0, \sigma_\theta^2)$ and μ_θ the unconditional mean of log ability. The drift term $\mu_\theta(1 - \rho_\theta)$ ensures that the stationary distribution of ability has mean μ_θ , generating realistic intergenerational persistence while allowing for regression to the mean.

Intergenerational earnings transmission. In addition to ability and bequests, the model includes a third intergenerational link operating through initial earning capacity. A child whose parent attended college enters the labor market with a permanently higher initial

³Lower lifetime earnings *growth* risk for college graduates (e.g. [Meghir and Pistaferri, 2004](#)) is a complementary channel, but it concerns the variance of earnings growth rather than the level dispersion that enters here, and it is not captured by the [Abbott et al. \(2019\)](#)-calibrated income process.

productivity draw: $z_0 \sim N(\zeta_{pe} \mathbf{1}\{e_p = C\}, \sigma_{z_0, e_c}^2)$, where $\zeta_{pe} \geq 0$ is an estimated parent-education earnings premium and σ_{z_0, e_c} is an education-specific entry dispersion taken from [Abbott et al. \(2019\)](#). Because the early-career income advantage accumulates permanently into the wealth stock even as the mean-reverting shock z decays, this channel routes part of the parent-child wealth correlation through inherited earning capacity rather than liquid transfers, leaving the child’s age-18 college-financing constraint untouched.

Asset-return risk. Following [Barczyk and Kredler \(2014a\)](#), each agent’s wealth carries a small multiplicative shock,

$$da_p = \dot{a}_p dt + \sigma_a a_p dB_p, \quad da_c = \dot{a}_c dt + \sigma_a a_c dB_c,$$

where B_p and B_c are independent standard Brownian motions, the drifts $\dot{a}_p, \dot{a}_c$ are given in (6)–(7), and the volatility vanishes at the borrowing constraint ($\sigma_a a = 0$ at $a = 0$). The shock admits an interpretation as modest idiosyncratic asset-return risk, but its essential role is to regularize the equilibrium. [Barczyk and Kredler \(2014a\)](#) show that a diffusion in the *wealth* state renders the value functions C^2 almost everywhere and the transfer boundary smooth, restoring existence of a pure-strategy Markov equilibrium and eliminating the non-smoothness—the “blast from the past” of [Barczyk and Kredler \(2021\)](#)—that arises in deterministic and discrete-time formulations. The income shock z alone does not deliver this: it does not smooth the transfer boundary, which lives in the wealth state. I set σ_a to the minimal magnitude (about three percent per year) that suffices for the regularization, so the wealth distribution is shaped by saving and bequests rather than by return risk.

3.3 The Coupled Parent-Child Problem

During the overlap period, parent and child simultaneously choose consumption and savings, while the parent additionally chooses a non-negative transfer flow $\tau(t) \geq 0$. The interaction is non-cooperative: each agent optimizes taking the other’s policy as given, but the parent internalizes the child’s welfare through the altruism parameter $\eta > 0$.

The continuous-time formulation follows [Barczyk and Kredler \(2014a\)](#), who show that simultaneity of moves in continuous time yields a well-defined equilibrium—an advantage

over discrete-time formulations where the timing protocol (who moves first within a period) can affect the equilibrium outcome.

3.3.1 Parent's Problem

The parent chooses consumption c_p and transfer flow $\tau \geq 0$ to maximize lifetime utility, weighting the child's instantaneous utility by η . The parent's value function $V^p(a_p, a_c, z; t)$ satisfies:

$$\begin{aligned} \rho V^p = \max_{c_p, \tau \geq 0} & \left\{ u(c_p) + \eta u(c_c^*) + V_{a_p}^p [r a_p + y_p - c_p - \tau] + V_{a_c}^p [\tilde{r}_c a_c + y_c - c_c^* + \tau] \right. \\ & \left. + V_z^p(-\kappa z) + \frac{1}{2} \sigma^2 V_{zz}^p + \frac{1}{2} \sigma_a^2 a_p^2 V_{a_p a_p}^p + \frac{1}{2} \sigma_a^2 a_c^2 V_{a_c a_c}^p + V_t^p \right\} \end{aligned} \quad (1)$$

where ρ is the continuous-time discount rate, $u(c) = c^{1-\gamma}/(1-\gamma)$ is CRRA utility with $\gamma = 1.5$, and c_c^* is the child's equilibrium consumption policy, taken as given by the parent. Consumption is bounded below by a floor $\underline{c} = \$3,900$ (in 2016 dollars), representing the effective consumption guaranteed by means-tested government transfers (SSI, SNAP, Medicaid), following [De Nardi \(2004\)](#). The terms involving V_z^p and V_{zz}^p capture the effect of income uncertainty on the parent's value function.

3.3.2 Child's Problem

The child chooses consumption c_c to maximize own lifetime utility:

$$\begin{aligned} \rho V^c = \max_{c_c} & \left\{ u(c_c) + V_{a_c}^c [\tilde{r}_c a_c + y_c - c_c + \tau^*] + V_{a_p}^c [r a_p + y_p - c_p^* - \tau^*] \right. \\ & \left. + V_z^c(-\kappa z) + \frac{1}{2} \sigma^2 V_{zz}^c + \frac{1}{2} \sigma_a^2 a_p^2 V_{a_p a_p}^c + \frac{1}{2} \sigma_a^2 a_c^2 V_{a_c a_c}^c + V_t^c \right\} \end{aligned} \quad (2)$$

where τ^* and c_p^* are the parent's equilibrium policies, taken as given. The child tracks parent wealth a_p because it affects expected future transfers: wealthier parents are more likely to provide support.

3.3.3 Transfer Decision

The parent's optimal transfer is characterized by a complementarity condition:

$$\tau^* \geq 0, \quad V_{a_p}^p \geq V_{a_c}^p, \quad \tau^* (V_{a_p}^p - V_{a_c}^p) = 0 \quad (3)$$

This partitions the state space into two regions. In the *no-transfer region* ($\mathcal{N}$), the parent's marginal value of own wealth exceeds the marginal value of child wealth ($V_{a_p}^p > V_{a_c}^p$), and $\tau^* = 0$. In the *transfer region* ($\mathcal{T}$), these marginal values are equalized ($V_{a_p}^p = V_{a_c}^p$) and $\tau^* > 0$. The boundary between $\mathcal{N}$ and $\mathcal{T}$ is a free boundary, endogenously determined as part of the equilibrium. The parent transfers when the child is relatively poor (low a_c), faces negative productivity shocks (low z), or when the parent is relatively wealthy (high a_p).

This complementary-slackness condition is the continuous-time analog of the transfer decision in discrete-time dynastic models, but with two key advantages. First, transfers are a continuous flow rather than lump-sum payments, eliminating the timing discreteness that affects transfer amounts in discrete models. Second, the boundary of the transfer region provides a clean characterization of the states where moral hazard is operative.

3.3.4 Consumption Policies and Wealth Dynamics

The first-order conditions for consumption are:

$$u'(c_p^*) = V_{a_p}^p, \quad (4)$$

$$u'(c_c^*) = V_{a_c}^c. \quad (5)$$

Under CRRA utility, these invert to $c_p^* = (V_{a_p}^p)^{-1/\gamma}$ and $c_c^* = (V_{a_c}^c)^{-1/\gamma}$. The drifts of the two wealth processes (the deterministic part of the laws of motion $da_p = \dot{a}_p dt + \sigma_a a_p dB_p$ and $da_c = \dot{a}_c dt + \sigma_a a_c dB_c$) are:

$$\dot{a}_p = r a_p + y_p(t) - c_p - \tau, \quad (6)$$

$$\dot{a}_c = \tilde{r}_c a_c + y_c(t, z) - c_c + \tau, \quad (7)$$

where $a_p \geq 0$ and the child's effective interest rate is

$$\tilde{r}_c = r + \iota_s \mathbf{1}\{a_c < 0\},$$

so the child earns r on savings and pays the student-loan rate $r + \iota_s$ on any negative balance. The student loan is the only debt instrument: during college the constraint is relaxed to $a_c \geq -\bar{b}$ (see Section 3.7), and the accumulated balance is repaid so that $a_c \geq 0$ by the time the child becomes a parent (age 42); the child cannot otherwise borrow as an adult.

3.4 Child-Alone Problem and the Dynastic Link

When the parent dies (parent age 72, child age 48), the child solves a consumption-savings problem with their own accumulated wealth—augmented by the bequest—for the remainder of their life:

$$\rho V^{c,\text{alone}} = \max_{c_c} \left\{ u(c_c) + V_{a_c}^{c,\text{alone}} [\tilde{r}_c a_c + y_c(t, z) - c_c] + V_z^{c,\text{alone}} (-\kappa z) + \frac{1}{2} \sigma^2 V_{zz}^{c,\text{alone}} + \frac{1}{2} \sigma_a^2 a_c^2 V_{a_c a_c}^{c,\text{alone}} + V_t^{c,\text{alone}} \right\} \quad (8)$$

The child lives alone until its own death at age 72, working until 66 and then drawing social security. The dynasty is then closed as a stationary recursion: a new dynasty begins with a parent whose initial wealth equals the child's terminal wealth a_c at age 72, whose education is $e'_p = e_c$, and whose child has ability θ' drawn from the intergenerational transmission process (Section 3). The solution of (8) at $t = 0$ —the moment of the parent's death—provides the terminal condition for the coupled system.

3.5 Terminal Conditions

At the parent's death ($t = T$, parent age 72, child age 48), the coupled system terminates and the child transitions to the child-alone problem. The boundary conditions at $t = T$ are:

$$V^p(a_p, a_c, z; T) = \phi_b \frac{(a_p + \underline{a}_{\text{beq}})^{1-\gamma}}{1-\gamma} + \eta \cdot V^{c,\text{alone}}(a_c + \alpha_{\text{beq}} a_p, z; 0), \quad (9)$$

$$V^c(a_p, a_c, z; T) = V^{c,\text{alone}}(a_c + \alpha_{\text{beq}} a_p, z; 0). \quad (10)$$

The child's value function transitions to the child-alone problem (8), evaluated at $t = 0$ of the child-alone clock—the moment of the parent's death. Equation (10) states this continuity: the child carries forward its own wealth a_c plus the bequest $\alpha_{\text{beq}} a_p$.

The parent's terminal value (9) has two components. The first, $\phi_b (a_p + \underline{a}_{\text{beq}})^{1-\gamma}/(1-\gamma)$ with $\phi_b \geq 0$, is a warm-glow term following De Nardi (2004) that captures the parent's direct utility from holding wealth at the end of the overlap period. The shifter $\underline{a}_{\text{beq}} \geq 0$ makes bequests a luxury good in the sense of De Nardi (2004): with $\underline{a}_{\text{beq}} > 0$ the marginal propensity to bequeath rises with wealth, so poor parents leave little while the rich bequeath a rising share—the mechanism that generates the fat right tail of the wealth distribution. Setting $\underline{a}_{\text{beq}} = 0$ recovers the pure warm-glow CRRA form. The warm-glow term captures the parent's direct utility from the wealth held at death. The second component, $\eta \cdot V^{c,\text{alone}}(a_c + \alpha_{\text{beq}} a_p, z; 0)$, reflects the parent's altruistic concern for the child's continuation utility after the parent's death. The child's continuation value is evaluated at effective wealth $a_c + \alpha_{\text{beq}} a_p$, where $\alpha_{\text{beq}} \in [0, 1]$ is the fraction of parent terminal wealth that transfers to the child as a bequest at death. This bequest channel creates two motives for parental wealth accumulation: the warm-glow motive ϕ_b and the altruistic motive through α_{beq} , which disciplines the intergenerational wealth correlation. The parent also has a forward-looking motive for inter-vivos transfers during the overlap: by transferring resources while alive, the parent raises a_c at $t = T$ and thereby improves the child's post-overlap continuation value.

3.6 College Decision

At the beginning of the dynasty ($t = 0$), the child decides whether to attend college. Let V_C^c , V_{HS}^c denote the child's equilibrium continuation values and V_C^p , V_{HS}^p the parent's continuation values under college and high school, evaluated at the initial state ($a_p, a_c = 0, z = 0$). The child weighs the paternalism-adjusted value gain from attending,

$$G(a_p, \theta) \equiv (V_C^c - V_{HS}^c) + \psi (V_C^p - V_{HS}^p),$$

against the psychic cost of college, which follows the [Abbott et al. \(2019\)](#) linear-in-log specification with an idiosyncratic shock:

$$\kappa(\theta, \varepsilon_\kappa) = (\kappa_0 + \kappa_\theta \log \theta + \sigma_\kappa \varepsilon_\kappa) \cdot \bar{V}, \quad \varepsilon_\kappa \sim N(0, 1),$$

where $\psi \geq 0$ is a paternalism weight that captures the child's internalization of parental welfare when choosing education, following [Abbott et al. \(2019\)](#);⁴ ε_κ is drawn once per individual at $t = 0$; and $\bar{V} = \text{mean}_{a_p, \theta} |V_C^c - V_{HS}^c|$ is a normalization constant that makes the dimensionless parameters κ_0 , κ_θ , and σ_κ comparable in scale to the college value gap. The gradient $\kappa_\theta < 0$ implies that higher-ability individuals face lower psychic costs (following [Cunha et al. 2005](#) and [Heckman et al. 2006](#)). The child attends whenever the value gain exceeds the realized cost, $G(a_p, \theta) \geq \kappa(\theta, \varepsilon_\kappa)$. Integrating over the normally distributed cost shock yields a probit college-attendance probability:

$$\Pr(e_c = C \mid a_p, \theta) = \begin{cases} \Phi\left(\frac{G(a_p, \theta)/\bar{V} - \kappa_0 - \kappa_\theta \log \theta}{\sigma_\kappa}\right) & \text{if } G(a_p, \theta) > 0, \\ 0 & \text{otherwise,} \end{cases} \quad (11)$$

where Φ is the standard normal CDF and σ_κ governs the dispersion of college choices: a larger σ_κ flattens the mapping from the value gap to attendance and, through favorable cost draws, raises enrollment among low-ability types.

The college decision is endogenous in two senses. First, parental altruism affects the child's continuation value: parents who anticipate supporting their child during and after college raise the value of attending. Second, the continuation values under $e_c = C$ and $e_c = HS$ incorporate the entire future trajectory of transfers, which depends on the child's income path and hence on education. This creates a channel through which parent wealth affects college enrollment beyond the direct cost channel. Importantly, anticipated transfers affect both V_C and V_{HS} : transfers subsidize college enrollment (raising V_C) but also cushion non-college children against income risk (raising V_{HS}). The net effect of the Samaritan's

⁴The paternalism weight ψ enters only the college decision, not the child's consumption-savings problem during the overlap period. It captures the idea that children partially internalize the cost their education choice imposes on parents—for example, through family pressure or social norms—without altering the non-cooperative structure of the within-period transfer game.

dilemma on the college decision is therefore theoretically ambiguous; Section 7.1 formalizes this result.

3.7 College Financial System

I model the college financial system following Abbott et al. (2019). The annual cost of attending college is $\phi = \$6,700$, reflecting average net tuition (after institutional discounts) from the NLSY97. During college, students work a fraction $\ell_C = 0.56$ of full-time hours.

Financial aid is modeled as a continuous expected-family-contribution (EFC) schedule keyed on parental *income* at college entry, following Abbott et al. (2019). Let $y_p^0(\theta, e_p)$ denote the parent's income when the child enters college (parent age 42). The annual grant is:

$$g(y_p^0) = \bar{g} \cdot \max\left(0, 1 - \frac{y_p^0}{\bar{y}_{\text{efc}}}\right)$$

where $\bar{g}$ is the maximum grant (for zero-income families) and $\bar{y}_{\text{efc}}$ is the parental-income level at which aid phases out completely. Both $\bar{g}$ and $\bar{y}_{\text{efc}}$ are set externally to the Abbott et al. (2019) values rather than estimated, and reproduce their net-tuition schedule. This continuous specification captures the essential feature of the federal financial aid system: grants decline smoothly with family income, generating an income-dependent effective price of college.

During college, students can borrow up to a cumulative limit $\bar{b}(a_p) = \bar{b}_0 + b_{\text{slope}} \cdot a_p$ at the student loan rate $r + \iota_s$, following Abbott et al. (2019). The base limit $\bar{b}_0 = \$23,000$ reflects the average federal student loan cap; the collateral channel $b_{\text{slope}} \geq 0$ allows children of wealthier parents to borrow more (e.g., through parental co-signing), and is estimated by SMM. This is implemented by allowing $a_c \geq -\bar{b}(a_p)$ during $t \in [0, 4]$; the accumulated balance then amortizes so that $a_c \geq 0$ by the time the child becomes a parent (age 42). The child's wealth dynamics during college are:

$$\dot{a}_c = (r + \iota_s) a_c + \ell_C \cdot y_c(t, z) - c_c - \phi_{\text{net}}(y_p^0) + \tau, \quad a_c \geq -\bar{b}(a_p)$$

where $\phi_{\text{net}}(y_p^0) = \phi - g(y_p^0)$ is net tuition. This system generates heterogeneity in the effective cost of college across the parental income distribution. Parental transfers supplement

institutional aid, and as [Abbott et al. \(2019\)](#) document, government grants can crowd out private contributions.

3.8 Equilibrium

A Markov-Perfect Equilibrium (MPE) consists of value functions V^p , V^c , consumption policies c_p^* , c_c^* , a transfer policy τ^* , and a college enrollment rule e_c^* , for each (θ, e_p, e_c) pair, such that: (i) given c_c^* and τ^* , the parent’s HJB (1) and the complementary-slackness condition (3) are satisfied; (ii) given c_p^* and τ^* , the child’s HJB (2) is satisfied; (iii) after the parent’s death ends the overlap ($t > T$), $V^{c,\text{alone}}$ satisfies (8); (iv) the college rule follows from (11); and (v) boundary conditions (9)–(10) and borrowing constraints hold at all times.

The equilibrium is solved backward: first the child-alone problem, then the coupled system during the overlap period, and finally the college choice at $t = 0$. Within the coupled system, the transfer is obtained in closed form at each state—the one-sided bounded gift of Section 3.9—so the transfer region is determined pointwise rather than by iterating on the complementary-slackness condition, and the value functions are updated by implicit time-stepping. Appendix I derives the equilibrium properties, including the Samaritan’s dilemma distortion in continuous time, and Appendix J describes the numerical solution algorithm.

3.9 Model Predictions: How College Affects the Transfer Region

The model generates predictions about how college alters the financial relationship between parents and adult children. The central mechanism operates through the transfer region $\mathcal{T}$ —the set of states where the parent optimally transfers resources to the child. College education shifts the boundary of this region, and the consequences propagate to consumption, wealth, and precautionary behavior.

Why college shrinks the transfer region. Recall that the parent transfers when the marginal value of own wealth equals the marginal value of child wealth ($V_{a_p}^p = V_{a_c}^p$). Because the parent weighs the child’s utility by η in the objective (1), $V_{a_c}^p$ is increasing in η : more altruistic parents have a larger transfer region. This condition is more likely to bind when the child is relatively poor or faces negative income shocks. College raises the child’s expected

income through two channels: a higher ability gradient ($\lambda_C > \lambda_{HS}$) and a steeper life-cycle earnings profile. The resulting increase in child wealth pushes states away from the transfer boundary, so college-educated children spend less time in $\mathcal{T}$. Higher expected income therefore shrinks $\mathcal{T}$ for college-educated children relative to high-school-educated children.

Figure 2 illustrates this mechanism in the (a_c, a_p) state space. The boundary $\partial\mathcal{T}$ separates states where transfers are positive (to the left) from states where the parent saves independently (to the right). College shifts this boundary to the left: at any given level of parent wealth, the child needs to be poorer before transfers become optimal. The region $\mathcal{T}_{HS} \setminus \mathcal{T}_C$ —states where a high-school child would receive transfers but a college child would not—measures how far college retracts the transfer region. Together with the overconsumption zone bordering it—where the child under-saves in anticipation of entering $\mathcal{T}$ (Appendix I)—this contraction captures the scope for moral hazard that college eliminates.

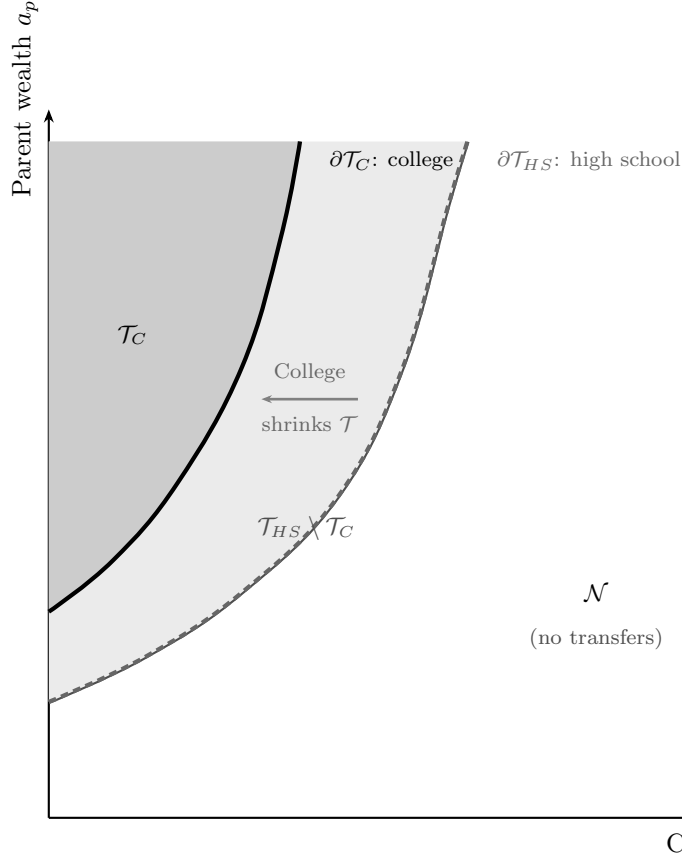


Figure 2. Transfer Region Boundary: College vs. High School

Notes: Schematic illustration of the transfer region $\mathcal{T}$ in the (a_c, a_p) state space. In the transfer region (shaded), the parent optimally transfers to the child ($\tau > 0$). The solid boundary corresponds to a college-educated child; the dashed boundary corresponds to a high-school-educated child. College shrinks $\mathcal{T}$ because higher expected income keeps child wealth further from the transfer boundary.

Four testable predictions. The contraction of the transfer region generates specific predictions about observable outcomes, all conditional on parent income:

1. **Consumption release.** *Parent consumption is higher when the child attends college.*
A smaller transfer region means that parents spend less of their lifetime making transfers. The resources that would have flowed to the child under the high-school path are instead available for own consumption.
2. **Wealth decumulation.** *Parent wealth paths diverge by child education.* In the short run, parents of college children pay tuition costs, reducing their wealth. In the long

run, however, the smaller transfer region means fewer ongoing transfers, and parent wealth recovers and eventually exceeds the high-school path.

3. Precautionary attenuation. *College attenuates the precautionary saving motive.*

Parents hold precautionary wealth against the contingency of supporting their children after negative income shocks. Because college raises the child’s income level, parents of college children face a smaller precautionary burden. This prediction links directly to the option value of the transfer region: even if transfers are rarely observed, the *possibility* of having to transfer affects parent saving behavior.

4. Transfer margins. *College reduces the probability of entering the transfer region, but transfers conditional on entry may be larger.*

Because college shifts the boundary of $\mathcal{T}$ to the left—requiring a larger negative shock to trigger transfers—the probability of entering $\mathcal{T}$ falls. However, conditional on entry, the states reached are further from the boundary (the child must be in a worse position for the transfer condition to bind), so the optimal transfer τ^* may be weakly larger. The model therefore predicts a negative extensive-margin effect and a zero or slightly positive intensive-margin effect. Combined with the precautionary-attenuation prediction, this implies that the consumption and wealth gains from college far exceed the reduction in realized transfer flows—the gap reflects the option-value channel operating through precautionary behavior rather than through observed transfers.

I now turn to the data to test these predictions.

4 Empirical Evidence

This section documents a set of stylized facts from matched parent-child data (PSID, HRS, and NLSY97) that are consistent with the model’s predictions. Each stylized fact compares outcomes for parents whose children attended college with those whose children did not, conditioning on parent income to isolate the role of the child’s education from the parent’s own resources. I begin with the relationship between child college status and parent consumption, then examine wealth accumulation, the dynastic precautionary saving channel, and finally the structure of inter-vivos transfers.

4.1 Data

The empirical analysis draws on three data sources. The Panel Study of Income Dynamics (PSID) provides matched parent-child panels with consumption, income, wealth, and inter-vivos transfer data from 1999 onward. I link parents to adult children using the Family Identification Mapping System (FIMS) and restrict the sample to parents aged 50+ and children aged 26+, yielding 8,944 observations across 2,338 parent-child pairs. The Health and Retirement Study (HRS) offers a larger sample of older parent-child pairs (19,179 pairs, 98,861 observations) with detailed transfer information, which I use to study the extensive and intensive margins of inter-vivos transfers. The National Longitudinal Survey of Youth 1997 (NLSY97) provides cognitive ability scores (ASVAB) and parental wealth for 5,400 individuals, which I use to construct college graduation rates by ability and wealth quartile for the SMM estimation (Section 5). Throughout, the key explanatory variable is child college attainment—an indicator for holding a bachelor’s degree or higher—and I condition on within-cohort parent income quartiles to isolate the role of child education from parental resources. All monetary variables are in 2016 dollars, winsorized at the 1st and 99th percentiles. Appendix A documents the sample construction pipeline, and Appendix A.5 provides detailed variable definitions.

4.1.1 Summary Statistics

Tables 1–2 report summary statistics for the PSID sample, stratified by average household income group. Table 1 presents parent demographics and key financial variables: parents in higher income groups are older on average, more likely to hold a college degree, more likely to own their home, and hold substantially more wealth. Total wealth ranges from \$103,000 for the lowest income group to \$966,000 for the highest, with home equity accounting for roughly half of wealth for low-income families but a smaller share for wealthy families. Total consumption expenditures increase monotonically with income, from \$36,000 to \$73,000. A detailed breakdown of the wealth portfolio is provided in Appendix Table E.1, and a consumption decomposition by expenditure category in Appendix Table F.1. Table 2 presents child-level statistics, including the college attendance rate among 18–24 year olds by parental income group, which rises steeply with parental resources.

Table 1. Parent Demographics and Financial Characteristics by Income Group (PSID)

	By Average HH Income Group				
	\$30–60	\$60–100	\$100–160	\$160+	All
<i>Panel A: Demographics</i>					
Age of head	45.6 (16.2)	45.2 (14.4)	46.8 (13.4)	49.1 (13.2)	46.3 (14.6)
Children in household	1.3 (1.4)	1.2 (1.3)	1.1 (1.2)	1.0 (1.1)	1.1 (1.3)
Total children	2.1 (1.3)	2.0 (1.0)	2.1 (1.0)	2.2 (1.0)	2.1 (1.1)
Head has college degree	0.07 (0.26)	0.19 (0.39)	0.42 (0.49)	0.69 (0.46)	0.29 (0.45)
White	0.44 (0.50)	0.63 (0.48)	0.77 (0.42)	0.85 (0.36)	0.64 (0.48)
Black	0.45 (0.50)	0.29 (0.45)	0.18 (0.38)	0.09 (0.28)	0.28 (0.45)
Homeowner	0.50 (0.50)	0.71 (0.45)	0.84 (0.36)	0.85 (0.36)	0.70 (0.46)
<i>Panel B: Income, Wealth, and Consumption</i>					
Household income	44,223 (21,535)	76,829 (37,964)	120,169 (51,447)	242,277 (253,591)	102,569 (121,904)
Head labor income	23,523 (19,369)	40,976 (28,162)	64,728 (42,902)	123,504 (123,970)	53,930 (64,202)
Total wealth	102,555 (230,260)	200,522 (377,059)	367,076 (536,534)	965,941 (1,052,497)	326,288 (612,283)
Total consumption	36,426 (18,374)	46,131 (20,389)	57,243 (24,489)	73,130 (33,277)	50,076 (26,252)
Observations	15 096	17 356	13 011	7863	53 326
Unique parent households	2718	2770	1833	1041	8362

Notes: Means with standard deviations in parentheses. Income groups defined by parent average real household income (2016 dollars) observed between ages 25 and 60. Panel A reports demographics; Panel B reports income, wealth, and consumption. “Children in household” is the number of children under age 18 in the parent’s PSID family unit; “Total children” is the total number of children linked to the parent household via the Family Identification Mapping System (FIMS). All monetary variables in 2016 dollars, winsorized at the 1st and 99th percentiles. A detailed wealth portfolio breakdown is in Appendix Table E.1.

Table 2. Children Descriptive Statistics by Parent Income Group (PSID)

	By Parent’s Average HH Income Group				
	\$30–60	\$60–100	\$100–160	\$160+	All
<i>Panel A: Child Demographics</i>					
Age	17.9 (19.0)	15.9 (15.7)	16.7 (13.6)	17.4 (13.4)	16.9 (16.0)
Number of siblings in sample	1.9 (1.7)	1.6 (1.2)	1.5 (1.2)	1.6 (1.1)	1.6 (1.4)
College-age (18–24)	0.14 (0.34)	0.14 (0.35)	0.16 (0.36)	0.17 (0.38)	0.15 (0.36)
<i>Panel B: Education</i>					
Years of education (age 18+)	13.7 (11.8)	13.7 (9.9)	13.8 (7.8)	14.4 (8.1)	13.9 (9.7)
Max years of education (age 25+)	16.1 (14.8)	15.7 (11.5)	15.9 (9.9)	16.5 (8.9)	16.0 (11.9)
Currently in college	0.05 (0.21)	0.06 (0.23)	0.08 (0.27)	0.10 (0.29)	0.07 (0.25)
Ever attains post-secondary ed.	0.33 (0.47)	0.37 (0.48)	0.47 (0.50)	0.56 (0.50)	0.41 (0.49)
<i>Panel C: College Rate Among 18–24 Year Olds</i>					
College attendance rate	0.32 (0.47)	0.40 (0.49)	0.47 (0.50)	0.54 (0.50)	0.42 (0.49)
College-age observations	4308	4984	4270	2907	16 469
Total child–year observations	31 731	35 484	27 026	16 962	111 203
Unique children	5680	5560	3784	2186	17 210

Notes: Means with standard deviations in parentheses. Each observation is a child-year. Children linked to parents via FIMS. Income groups refer to parent average real household income (2016 dollars). “Years of education” reported conditional on age ≥ 18 ; “Max years of education” conditional on age ≥ 25 . “College attendance rate” computed conditional on the child being aged 18–24. “Ever attains post-secondary ed.” indicates maximum observed years of education exceeds 12. Years of education directly observed only from 2013 onward; for earlier waves, college attendance inferred from an age-education proxy.

4.2 Parent Consumption and Child College Status

The model predicts that, conditional on their own income and wealth, parents whose children attend college should consume more (the consumption-release prediction of Section 3.9). College shifts children out of the transfer region $\mathcal{T}$ by permanently raising their income, freeing parents from contingent obligations and raising their consumption even when no transfers are actually flowing.

I test this prediction using the following regression on the PSID sample at the parent-year level:

$$C_{p,t} = \beta_0 + \beta_1 \text{ShareCollege}_{p,t} + \beta_2 N_{p,t} + \delta_{Q_p^{inc}} + \beta_X \mathbf{X}_{\mathbf{p},t} + \varepsilon_t + \epsilon_{p,t}$$

where $C_{p,t}$ is household consumption (in 2016 dollars) of parent p at time t , $\text{ShareCollege}_{p,t}$ is the fraction of parent p 's children with a bachelor's degree or higher, $N_{p,t}$ is the total number of children, $\delta_{Q_p^{inc}}$ are parent income quartile fixed effects (within-cohort ranking), and $\mathbf{X}_{\mathbf{p},t}$ is a comprehensive set of controls: parent total wealth, non-financial wealth, household income, wealth quartile, labor force participation, household size, head's birth year, education, U.S. state, a cubic in head's age, housing tenure, and race. The specification includes year fixed effects ε_t .

Table 3 presents the results. Column 1 shows that a higher share of college-educated children is associated with significantly higher parent consumption. Columns 2 and 3 use alternative definitions—any child with a BA, or all children with a BA—yielding consistent results. Column 4 uses consumption levels rather than logs; the magnitude implies economically meaningful differences. These effects are net of parent income, wealth, and education, suggesting they are not driven by parental resources alone.

Table 3. Parent Consumption and Child College Status (PSID)

	(1)	(2)	(3)	(4)
	Log C	Log C	Log C	Level C
Share of Children with BA+	0.081*** (0.019)			2470.032*** (866.461)
Number of Children	0.014* (0.008)	0.010 (0.008)	0.016* (0.008)	169.725 (287.036)
Any Child with BA+		0.080*** (0.017)		
All Children with BA+			0.056*** (0.017)	
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Parent income Q FE	Yes	Yes	Yes	Yes
Observations	15,830	15,830	15,830	15,831
R^2	0.558	0.558	0.557	0.357

Notes: OLS regressions of parent household consumption on child college indicators. “Share of Children with BA+” is the fraction of children with a bachelor’s degree or higher. Controls include a cubic in parent age, household income, non-housing wealth, total wealth, wealth quartile, and fixed effects for year, education, parent income quartile, gender, housing tenure, state, labor force status, race, household size, and birth year. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 4 estimates the effect separately by parent income quartile, confirming that the positive relationship between child college status and parent consumption holds across the income distribution.

Table 4. Parent Consumption and Child College Status, by Parent Income Quartile (PSID)

	Parent Income Quartile			
	(1)	(2)	(3)	(4)
	Q1 (Poorest)	Q2	Q3	Q4 (Richest)
Share of Children with BA+	0.136*** (0.040)	0.053* (0.030)	0.016 (0.025)	0.041 (0.029)
Number of Children	0.010 (0.013)	0.003 (0.010)	0.018* (0.009)	0.001 (0.013)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	4,342	3,830	3,931	3,727
R^2	0.442	0.326	0.355	0.415

Notes: OLS regressions of log parent household consumption on share of children with BA+, estimated separately by parent income quartile. Controls as in Table 3 excluding parent income quartile dummies. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

4.3 Parent Wealth Accumulation

The consumption premium has a counterpart in wealth, but the model's prediction for the wealth *stock* is ambiguous. When a child does not attend college, three forces act on parent wealth in opposite directions. A precautionary motive raises desired saving, because the child may need future transfers. A Samaritan's-dilemma distortion lowers the effective return on saving: inside the transfer region $\mathcal{T}$, where $V_{a_p}^p = V_{a_c}^p$, a marginal dollar of parent wealth partly leaks to the child. And realized transfer outflows deplete accumulated wealth. Because these forces offset, the sign of the college-wealth relationship in levels is uninformative about the mechanism; isolating the channels requires the structural model (Section 7).

The model does deliver a sharp prediction for the *flow*. By raising the child's permanent income, college contracts the transfer region, so a parent of a college child faces a smaller expected stream of future transfers and should run down wealth more slowly in retirement (the wealth-decumulation prediction). I test this directly on the rate of decumulation.

Measuring decumulation requires care. The cross-sectional age–wealth profile is biased by differential mortality: poorer households die younger, so the surviving cross-section overstates wealth at older ages (Love et al., 2009; De Nardi et al., 2010). I therefore measure decumulation *within* parent and condition on permanent income, estimating, on the retirement-phase sample (parent age ≥ 60),

$$\Delta W_{p,t} = \gamma_0 + \gamma_1 \text{ShareCollege}_p + \gamma_2 N_p + \delta_{Q_p^{\text{PI}}} + \mathbf{X}'_{p,t} \gamma + \varepsilon_{p,t},$$

where $\Delta W_{p,t}$ is the biennial change in parent wealth and Q_p^{PI} is a time-invariant permanent-income quartile (mean family income ranked within the parent’s birth cohort). A positive γ_1 indicates slower decumulation for parents of college children. I report the change in wealth in levels (OLS and median regression) and in logs, together with a household-level summary: the within-parent slope of $\ln W$ on age over ages 60–80.

Table 5 reports the estimates. The constant γ_0 pins down the baseline pace of drawdown, and it shows that decumulation in this sample is slow: over ages 60–80 the average parent is close to flat, with a mean log-wealth slope of about +1.5% per year, and only parents in the bottom permanent-income quartile decumulate appreciably ($\gamma_0 \approx -3.9\%$ per year in the slope specification, column 4). This limited late-life drawdown is the pattern emphasized by De Nardi et al. (2010). Against this near-flat baseline, the coefficient on the college share is positive in every specification—the sign implied by slower decumulation—but none attains conventional significance: parents of college children show, for example, a \$1,064 larger median biennial change in wealth and a 0.011 higher annual log-wealth slope, both statistically insignificant. The evidence is thus directionally consistent with the wealth-decumulation prediction but too imprecise to be decisive, and I treat it as suggestive, leaving the magnitude of the wealth path to the structural model. The complementary prediction—a short-run dip in parent wealth while the child is enrolled—cannot be tested in the PSID, because the parent–child wealth linkage resolves only once children are observed as adults, leaving no observations in the college-entry window.

This baseline behavior is a useful external check. The slow, income-graded drawdown I estimate matches the two benchmark studies of late-life wealth. Love et al. (2009) find that median wealth in retirement is roughly flat or even rising once annuitized resources are

included, and [De Nardi et al. \(2010\)](#) find that the elderly retain large balances late in life, with the income-rich decumulating more slowly than the income-poor—precisely the gradient in column 4, where only bottom-quartile parents draw wealth down appreciably. That my within-parent wealth measure reproduces both facts is reassuring: the baseline behaves as the literature would predict, so the object of interest is the *college differential* layered on top of it.

The pattern maps onto the model. Parents here hold wealth not only against their own retirement and longevity risk—the motives stressed by [De Nardi et al. \(2010\)](#)—but as a buffer against their children’s future transfer needs, a dynastic precautionary motive in the spirit of [Boar \(2021\)](#), reinforced by the luxury-bequest motive of [De Nardi \(2004\)](#). Both forces are stronger for wealthier parents, who maintain larger transfer regions and bequeath a rising share of wealth, and who therefore decumulate most slowly—the income gradient the data display. College operates on exactly this margin: by raising the child’s permanent income, it contracts the transfer region and shrinks the expected stream of transfers the parent must finance, so a parent of a college child should retain wealth longer. The positive college coefficients are consistent with this channel, but because they are imprecise and the baseline drawdown is itself slow, I leave the magnitude of college’s effect on the parent’s wealth trajectory to the structural model, which isolates the transfer-region channel from the precautionary and bequest motives that also shape decumulation.

Table 5. Parent Wealth Decumulation and Child College Status (PSID)

	(1)	(2)	(3)	(4)
	Log Wealth Change	Wealth Change (OLS)	Wealth Change (Median)	Wealth–Age Slope
Share of Children with BA+	0.023 (0.017)	4,151 (21,568)	1,064 (3,337)	0.011 (0.011)
Number of Children	0.008 (0.007)	-4,531 (6,115)	159 (1,265)	0.008 (0.008)
γ_0	0.003 (0.046)	-57,785 (87,699)	-738 (3,625)	-0.039* (0.022)
Observations	8,766	9,768	9,768	1,422
R^2	0.010	0.002		0.032

Notes: Within-parent measures of wealth decumulation regressed on the share of children with a bachelor’s degree or higher, on the retirement-phase sample (parent age ≥ 60). Column 1: biennial change in log real wealth; Column 2: biennial change in real wealth (OLS); Column 3: the same in median (LAD) regression; Column 4: the per-parent OLS slope of log real wealth on age over ages 60–80. A positive coefficient on the college share indicates *slower* decumulation for parents of college children. The constant γ_0 measures the baseline rate of wealth change for the reference parent (high-school parent in the bottom permanent-income quartile); in Columns 1–3 age is centered at 70, so γ_0 is the (biennial) wealth change evaluated at age 70. All specifications condition on a time-invariant permanent-income quartile (mean family income, ranked within parent birth cohort) and parent education, following [Love et al. \(2009\)](#) and [De Nardi et al. \(2010\)](#); Columns 1–2 add year fixed effects. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

4.4 Dynastic Precautionary Saving

The previous two stylized facts document large differences in parent consumption and wealth associated with child college status. The model identifies the mechanism: college raises the child’s expected income, shrinking the transfer region and the share of states in which the parent must stand ready to transfer. This prediction is closely related to the dynastic precautionary savings channel identified by [Boar \(2021\)](#), who shows that parents accumulate precautionary savings to insure their children against permanent income risk—a 10% increase in a child’s income uncertainty reduces parent consumption by 0.76%. Following her framework, I test whether college attenuates this dynastic precautionary motive.

There is strong independent evidence that college graduates face lower earnings-*growth* risk. [Meghir and Pistaferri \(2004\)](#) estimate (Table 1) that the variance of unexplained earnings growth is 0.065 for college graduates compared to 0.103 for high school graduates—a 37% reduction. This concerns the variance of earnings growth, a distinct object from the

level dispersion of the persistent income shock in the model; as the next paragraph explains, the model reproduces the attenuation through the income *level* and does not require college earnings to be less volatile.

I test this prediction using the following specification, which extends equation (9) of Boar (2021) with a child education interaction:

$$\ln c_{it}^p = \beta_0 + \beta_t + \beta_1 \sigma_{hs}^p + \beta_2 \sigma_{hs}^c + \beta_3 \sigma_{hs}^c \times \mathbf{1}\{e_c = \text{College}\} + \mathbf{X}_{it}^p \beta_4 + \mathbf{X}_{it}^c \beta_5 + \varepsilon_{it} \quad (12)$$

where σ_{hs}^c is the child’s permanent income uncertainty, constructed following Boar’s methodology by estimating sector-specific (age $\times$ industry-occupation) forecast errors of lifetime earnings, and σ_{hs}^p is the parent’s own uncertainty. The key coefficient is β_3 : the model predicts $\beta_2 < 0$ (higher child uncertainty reduces parent consumption) and $\beta_3 > 0$ (college attenuates this effect).

Table 6 presents the results. Column 1 replicates Boar’s baseline specification in the PSID sample, confirming that higher child income uncertainty reduces parent consumption ($\hat{\beta}_2 < 0$). Column 2 adds the education interaction: the positive $\hat{\beta}_3 = 0.012$ indicates that college attenuates the dynastic precautionary motive, though the estimate is imprecise ($p = 0.11$). The net effect of child uncertainty for college-educated children is $\hat{\beta}_2 + \hat{\beta}_3 = -0.020 + 0.012 = -0.008$, substantially smaller than the effect for non-college children ($\hat{\beta}_2 = -0.020$), representing a 62% reduction. College substantially weakens parents’ exposure to child income risk—consistent with the model’s prediction that college shrinks the transfer region (the precautionary-attenuation prediction of Section 3.9).

Why this is consistent with the model. The model rationalizes $\hat{\beta}_3 > 0$ without assuming that college lowers income risk. What governs the parent’s dynastic precautionary motive is not the variance of the child’s income per se, but the probability and size of the child entering the transfer region $\mathcal{T}$ —a region of *low child resources*. Because income enters multiplicatively and the transfer trigger is a low absolute resource level, a permanently higher income shifts the child’s whole resource distribution up and thins the left tail that triggers support, even when the log-variance is unchanged. A college child therefore sits far from $\mathcal{T}$, where the parent’s marginal valuation of child wealth $V_{a_c}^p$ is low and flat, so a given increase

in the child’s income uncertainty moves parent consumption little; a high-school child sits close to $\mathcal{T}$, where V_{ac}^p is steep, so the same uncertainty raises the option value of future transfers and depresses parent consumption. The education gap in the consumption–uncertainty sensitivity is thus a *distance-to-boundary* effect operating through the income level, and it survives even though the externally calibrated income process makes college earnings no less volatile than high-school earnings (Section 5). Panel C of Table 15 confirms the channel in the simulated model: college children spend a smaller share of adult life receiving transfers than high-school children despite facing a weakly more dispersed persistent shock. The lower earnings-growth risk that college graduates enjoy in the data (Meghir and Pistaferri, 2004) reinforces the same pattern, but is a complementary force the model does not rely on.

Table 6. Dynastic Precautionary Savings and Child Education

	(1)	(2)
	Baseline	Education Interaction
σ_{hs}^p (Parent uncertainty)	-0.0149*** (0.00564)	-0.0149*** (0.00565)
σ_{hs}^c (Child uncertainty)	-0.0116*** (0.00370)	-0.0198*** (0.00696)
$\sigma_{hs}^c \times \mathbf{1}\{\text{College}\}$		0.0123 (0.00770)
Parent controls	Yes	Yes
Child controls	Yes	Yes
Year FE	Yes	Yes
Parent income Q FE	No	Yes
Observations	12897	12897
R^2	0.782	0.782

Notes: Dependent variable is log OECD-equivalized non-durable parent consumption. σ_{hs}^p and σ_{hs}^c are permanent income uncertainty measures constructed as $\ln \sqrt{\text{Var}(\varepsilon) + 1}$, where ε is the level-scale forecast error from sector-specific age-income profiles estimated on the full PSID income sample, following Boar (2021). Column 2 adds parent income quartile fixed effects and the college interaction. Additional controls: year fixed effects, parent and child age dummies, parent education, marital status, race, and family size. Standard errors clustered by parent household $\times$ year in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Across Sections 4.2–4.4, college is associated with 8% higher parent consumption (roughly \$2,500 per year), suggestive evidence of slower wealth decumulation, and a substantially weaker precautionary saving response to child income risk. The dynastic precautionary saving result identifies the mechanism directly: college reduces the sensitivity of parent consumption to child income uncertainty by 62%, from -0.020 to -0.008 . This operates through the *option value* of the transfer region: even when transfers are not currently flowing, the *possibility* of future transfers depresses parent consumption. College shrinks this region, releasing

resources.

4.5 Inter-Vivos Transfers

A natural question arises: if college affects parent consumption and wealth so strongly, do we observe correspondingly large differences in actual transfers? As the transfer-margins prediction of Section 3.9 anticipates, the answer requires decomposing transfers into their extensive and intensive margins—and this decomposition maps directly onto the model’s transfer-region structure.

The key feature of inter-vivos transfers in the HRS is that most parent–child pairs report zero transfers in any given wave: only 14.4% of observations involve a positive transfer from parent to child (Table 7). Conditional on a positive transfer, however, amounts are substantial—\$8,161 on average, with a median of \$2,897 and a 90th percentile of \$18,394. This pattern—a large mass at zero with a heavy right tail—is precisely what the model predicts. The transfer region $\mathcal{T}$ is entered infrequently, but when entered, transfers are non-trivial insurance events.

Table 7. Distribution of Inter-Vivos Transfers from Parents to Children (HRS)

Pr(Transfer > 0)	0.144
E(Transfer Transfer > 0)	\$8,161
Median(Transfer Transfer > 0)	\$2,897
P75(Transfer Transfer > 0)	\$7,968
P90(Transfer Transfer > 0)	\$18,394
E(Transfer, unconditional)	\$1,172
Observations	548,239
Observations with Transfer > 0	78,541

Notes: Wave-level statistics for parent-to-child transfers from HRS 1992–2014 (Family-Kids file, waves 1–12). Sample restricted to parent age ≥ 50 , child age ≥ 26 . Transfer amounts in real 2016 US dollars. Unconditional mean includes zeros.

I decompose the within-family college effect along these two margins. Table 8 reports

linear probability models for $\Pr(\text{Transfer} > 0)$ with parent fixed effects:

$$\mathbf{1}[IVT_{ijt} > 0] = \beta_0 + \beta_1 \text{College}_j + \gamma_t + \beta_X \mathbf{X}_{jt} + \delta_{Q_i^p} + \varepsilon_i + \epsilon_{ijt}$$

College reduces the probability of any parent-to-child transfer by 3.1 percentage points—a 21% reduction relative to the mean transfer probability of 14.4% (Table 8, Column 1). Column 2 interacts college with parent income quartile, revealing a steep gradient: the effect of college on transfer probability is -1.0 pp for Q1 parents, -2.8 pp for Q2 ($-0.010 - 0.018$), -3.4 pp for Q3, and -4.2 pp for Q4. Wealthier parents have larger transfer regions ex ante—they can afford to support children in more states of the world—so college shrinks their transfer region by more.

Table 8. Extensive Margin: Effect of College on Transfer Probability (HRS)

	Extensive Margin (Linear Probability)	
	(1)	(2)
	Pr(Parent → Child)	Pr(Parent → Child)
College (BA+)	-0.031*** (0.002)	-0.010*** (0.003)
College × Q2		-0.018*** (0.004)
College × Q3		-0.024*** (0.004)
College × Q4		-0.032*** (0.005)
Parent FE	Yes	Yes
Wave FE	Yes	Yes
Dep. Var. Mean	0.144	0.144
Observations	548,164	548,164
Within R^2	0.012	0.012

Notes: Linear probability models. Dependent variable is an indicator for any positive transfer in the HRS wave. All specifications include parent fixed effects, wave dummies, child cohort controls, and parent income quartile dummies. Standard errors clustered at parent level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 9 conditions on positive transfers and confirms the model’s second prediction for this margin. Conditional on being in the transfer region, college children receive \$557 more per transfer event (Column 1, $p < 0.10$). Column 2 reveals heterogeneity: for Q1 parents, college children receive \$906 *less* conditional on entry, but for wealthier parents the effect is positive and large—\$958 more for Q2 ($-906 + 1,864$), and \$826 more for Q4. This pattern is consistent with the boundary-shift mechanism: because college moves the transfer boundary $\partial\mathcal{T}$ to the left in the (a_c, a_p) space, the states that push a college-educated child into $\mathcal{T}$ involve worse realizations of child income. Conditional on entry, the child is further from the boundary, and the optimal transfer is therefore larger—particularly for wealthier parents

whose transfer regions are larger and whose boundary shift from college is more pronounced.

Table 9. Intensive Margin: Transfer Amount Conditional on Positive Transfer (HRS)

	Intensive Margin (Amount Transfer > 0)	
	(1)	(2)
	Parent → Child	Parent → Child
College (BA+)	557* (288)	-906* (486)
College × Q2		1864*** (607)
College × Q3		965* (542)
College × Q4		1732** (684)
Parent FE	Yes	Yes
Wave FE	Yes	Yes
Dep. Var. Mean	8161	8161
Observations	78,541	78,541
Within R^2	0.002	0.002

Notes: OLS regressions on the subsample with positive transfers. Dependent variable is the real transfer amount (2016 US\$). All specifications include parent fixed effects, wave dummies, child cohort controls, and parent income quartile dummies. Standard errors clustered at parent level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

The reduction in transfer probability implied by college—3.1 percentage points—saves roughly \$253 per year in expected transfers ($0.031 \times \$8,161$), an order of magnitude smaller than the \$2,500 consumption gain documented in Section 4.2. The gap between observed transfer savings and the consumption effect is consistent with the option-value channel identified in Section 4.4: college reduces the precautionary burden parents face, releasing resources beyond what flows through actual transfers.

5 Estimation

I estimate the model in three stages. In the first stage, I set parameters that are well-identified from external data or the existing literature, including the income-process parameters, which I take from [Abbott et al. \(2019\)](#). In the second stage, I map the discrete-time income-process estimates to their continuous-time counterparts. In the third stage, I estimate the remaining twelve structural parameters—altruism, the three psychic-cost parameters, the discount rate, the warm-glow bequest weight and a [De Nardi \(2004\)](#) luxury-bequest shifter, the bequest fraction, the college wage scale, paternalism, the collateral channel, and an intergenerational earnings premium—by the simulated method of moments (SMM), matching 36 targeted data moments.

5.1 Externally Set Parameters

Table 10 reports the parameters set in the first stage. The coefficient of relative risk aversion is $\gamma = 1.5$, following [Abbott et al. \(2019\)](#). The annual real interest rate is $r = 0.03$, following [Daruich and Kozlowski \(2020\)](#). The continuous-time discount rate ρ is estimated internally by SMM to match the wealth-to-income ratio and wealth distribution moments. The college costs, borrowing limits, and financial-aid schedule parameters ($\bar{g}$, $\bar{y}_{\text{efc}}$) are set externally to the [Abbott et al. \(2019\)](#) values, as described in Section 3.7. The income-process parameters—the education-specific ability gradients λ_e , age profiles, OU mean-reversion rates κ_e and diffusions σ_e , and entry dispersions—and the intergenerational ability parameters (ρ_θ , σ_θ , μ_θ) are likewise taken from [Abbott et al. \(2019\)](#). Following [Barczyk and Kredler \(2014a\)](#), the multiplicative wealth-diffusion volatility is fixed at $\sigma_a = 0.03$ per year—the minimal randomization that delivers C^2 value functions and a smooth transfer boundary (Section 3)—rather than estimated, so the wealth distribution is disciplined by saving and bequests. Retirement income is estimated directly from PSID data on households where the respondent is retired and over age 67, computing the average sum of Social Security, SSI, disability, and employer pension income by education group.

Table 10. Externally Set Parameters

Parameter	Description	Value	Source
<i>Preferences</i>			
r	Real interest rate	0.03	Daruich and Kozlowski (2020)
γ	CRRA risk aversion	1.5	Abbott et al. (2019)
$\underline{c}$	Consumption floor (\$000s/yr)	3.9	De Nardi (2004)
σ_a	Wealth-diffusion volatility (/yr)	0.03	Barczyk and Kredler (2014a)
<i>Demographics (years)</i>			
T^{ret}	Parent working, overlap (child 18–42)	24	—
$T - T^{\text{ret}}$	Parent retired, overlap (child 42–48)	6	—
T_{alone}	Child alone, working (48–66)	18	—
$T_{\text{retire},c}$	Child retirement (66–72)	6	—
<i>Ability gradients</i>			
λ_C	Returns to ability, college	0.797	Abbott et al. (2019)
λ_{HS}	Returns to ability, HS	0.517	Abbott et al. (2019)
<i>Earnings process (annual)</i>			
ρ_C^{ann}	Persistence, college	0.966	Abbott et al. (2019)
ρ_{HS}^{ann}	Persistence, HS	0.952	Abbott et al. (2019)
$(\sigma_\eta^{\text{ann}})^2$	Innovation variance (both groups)	0.017	Abbott et al. (2019)
σ_C	OU diffusion, college	0.133	Abbott et al. (2019)
σ_{HS}	OU diffusion, HS	0.134	Abbott et al. (2019)
$\bar{w}$	Avg. annual earnings	\$70k	Census
<i>College costs</i>			
ϕ_C	Annual tuition (before aid)	\$6.7k	Abbott et al. (2019)
τ_C	Time worked in college	0.56	Census
<i>Intergenerational ability</i>			
ρ_θ	Persistence of log ability	0.36	Abbott et al. (2019)
σ_θ	Std. dev. of ability innovation	0.40	Abbott et al. (2019)
μ_θ	Mean of log ability	1.30	Abbott et al. (2019)
<i>Financial aid & borrowing</i>			
$\bar{g}$	Max. annual grant (\$000s)	2.82	Abbott et al. (2019)
$\bar{y}_{\text{efc}}$	EFC income phase-out (\$000s)	72.0	Abbott et al. (2019)
$\bar{b}_0$	Student loan limit (base)	\$23.0k	Federal
ι_s	Student loan premium	0.031	Abbott et al. (2019)
<i>Retirement income (annual)</i>			
SS_C	Soc. Security + pension, college	\$14.3k	PSID
SS_{HS}	Soc. Security + pension, HS	\$13.0k	PSID

Notes: Dollar amounts in 2016 dollars. The continuous-time income process is set externally from Abbott et al. (2019) (Table 3.2, males): persistence maps to the mean-reversion rate via $\kappa_e = -\ln \rho_e^{\text{ann}}$, and the OU diffusion σ_e is chosen so the stationary variance $\sigma_e^2/(2\kappa_e)$ equals the AGMV persistent-shock variance $(\sigma_\eta^{\text{ann}})^2/(1 - (\rho_e^{\text{ann}})^2)$, giving $\sigma_{HS} = 0.134$, $\sigma_C = 0.133$ (Appendix D). The diffusions are *not* estimated. The model carries no transitory income shock. The financial aid schedule $g(y_p^0) = \bar{g} \cdot \max(0, 1 - y_p^0/\bar{y}_{\text{efc}})$ is a function of parent *income* at college entry; $\bar{g}$ and $\bar{y}_{\text{efc}}$ are set externally to Abbott et al. (2019), not estimated. Retirement income from PSID (respondent retired, age > 67). The student loan is the only debt instrument; its base limit $\bar{b}_0$ is cumulative over four college years, and the effective limit $\bar{b}(a_p) = \bar{b}_0 + b_{\text{slope}} \cdot a_p$ has slope b_{slope} estimated by SMM (Table 12). The effective student rate is $r + \iota_s = 0.061$.

5.2 Income Process Estimation

The income process in continuous time is an Ornstein-Uhlenbeck process $dz = -\kappa_e z dt + \sigma_e dW$ with education-specific parameters. The entire income process is set externally from the male estimates in [Abbott et al. \(2019\)](#): I map their persistence to the continuous-time mean-reversion rate κ_e and choose the diffusion σ_e so that the stationary variance $\sigma_e^2/(2\kappa_e)$ reproduces their persistent-shock variance. The diffusion coefficients are therefore *not* estimated within the structural routine; the income process is calibrated first-stage and held fixed.⁵

In discrete time, log labor income is decomposed as in [Abbott et al. \(2019\)](#): $\log \epsilon_j = \lambda_e \log \theta + \gamma_{e,j} + z_j$, where λ_e is the education-specific ability gradient. The ability gradient and age-income profile are taken from [Abbott et al. \(2019\)](#), who estimate λ_e from the NLSY79 by regressing log hourly wages on log AFQT scores, separately by education group, and estimate the deterministic age-income profile $\gamma_{e,j}$ from PSID data using a quadratic in age. The income shock persistence is also taken from [Abbott et al. \(2019\)](#), who estimate it by minimum distance on the autocovariance structure of wage residuals by age and education:

$$\begin{aligned} z_{iat}^e &= \log y_{it} - \widehat{f^e(a_{it})} - \hat{\lambda}_e \log(\text{AFQT}_i) \\ z_{iat}^e &= \rho_e^{\text{ann}} z_{i,a-1,t-1}^e + \eta_{iat}^e \\ \eta_{iat}^e &\sim N(0, (\sigma_\eta^e)^2), \quad z_{i0t}^e \sim N(0, (\sigma_{z_0}^e)^2) \end{aligned}$$

I map the discrete-time persistence to its continuous-time mean-reversion counterpart κ_e and the innovation variance to the OU diffusion σ_e by matching the autocovariance structure; Appendix D derives the mapping formulas. Table 11 reports the discrete-time estimates and their continuous-time images. The mean-reversion rates are similar across education groups ($\kappa_{HS} = 0.049$, $\kappa_C = 0.035$), and the diffusions are nearly identical ($\sigma_{HS} = 0.134$, $\sigma_C = 0.133$). Because college earnings are more persistent, the implied stationary standard deviation of the persistent shock is in fact slightly *higher* for college graduates (0.51 versus 0.43 for high school), so the model’s transfer-region contraction is driven by the income *level*, not by a reduction in income risk (Section 3).

⁵[Abbott et al. \(2019\)](#) estimate the income process separately by gender. I use their male estimates throughout, as the model abstracts from gender differences.

Table 11. Income Process and Age Profile

	High-School	College	Source
<i>Ability gradient</i>			
λ_e	0.517	0.797	AGMV, Tbl. 3.1
<i>Deterministic age profile</i>			
$\gamma_{1,e}$	0.0673	0.1197	AGMV, Tbl. E.1
$\gamma_{2,e} \times 1000$	-0.670	-1.191	AGMV, Tbl. E.1
<i>Persistent shock — discrete-time AR(1)</i>			
ρ_z^{ann}	0.952	0.966	AGMV, Tbl. 3.2
$(\sigma_\eta^{\text{ann}})^2$	0.017	0.017	AGMV, Tbl. 3.2
$\sigma_{z_0}^2$ (entry var.)	0.059	0.094	AGMV, Tbl. 3.2
<i>Persistent shock — continuous-time OU (mapped, fixed)</i>			
κ_e	0.049	0.035	$-\ln \rho_z^{\text{ann}}$
σ_e	0.134	0.133	$\sqrt{2\kappa_e(\sigma_\eta^{\text{ann}})^2/(1 - (\rho_z^{\text{ann}})^2)}$
σ_{z_0} (entry s.d.)	0.243	0.307	$\sqrt{\sigma_{z_0}^2}$
stationary s.d. $\sigma_e/\sqrt{2\kappa_e}$	0.43	0.51	implied
<i>Relative price of college labor (SMM)</i>			
ω_C	—	0.642	SMM (Tbl. 12)

Notes: The table collects all parameters of the income process and reports the source of each. “AGMV” denotes [Abbott et al. \(2019\)](#) (male estimates); the cited table number refers to that paper. The ability gradient λ_e , the deterministic age profile $(\gamma_{1,e}, \gamma_{2,e})$, and the discrete-time persistent-shock parameters $(\rho_z^{\text{ann}}, \sigma_\eta^{\text{ann}}, \sigma_{z_0})$ are taken directly from AGMV. The continuous-time mean-reversion rates κ_e and OU diffusions σ_e are mapped from the AGMV persistence ρ_z^{ann} and innovation variance σ_η^{ann} as described in the text and Appendix D; the model carries no transitory shock ($\sigma_{e,e} = 0$, as in AGMV). The only income-process parameter estimated by SMM is the relative price of college labor ω_C (ω is normalized to one for high school), reproduced here from Table 12. The implied stationary standard deviation of the persistent shock, $\sigma_e/\sqrt{2\kappa_e}$, is 0.43 for high school and 0.51 for college.

Figure D.1 in Appendix D illustrates the implied life-cycle income profiles by education, including the deterministic age-income component, expected income for a median-ability worker, the role of OU shock volatility, and the complementarity between ability and education.

5.3 Ability Gradient

The ability gradient λ_e in the human capital function $h(\theta, e) = (\theta/\bar{\theta})^{\lambda_e}$ governs the returns to cognitive ability by education level. I take the estimates of λ_e directly from [Abbott et al. \(2019\)](#), who estimate them from the NLSY79 by regressing log hourly wages (purged of the PSID age profile) on log AFQT scores, separately by education group. The key estimates are $\hat{\lambda}_{HS} = 0.517$ and $\hat{\lambda}_C = 0.797$, implying that a one-log-point increase in ability raises college earnings by 80% but high-school earnings by only 52%. This complementarity between ability and education is central to the mechanism: the college premium is strongly increasing in ability, so removing parental transfers has heterogeneous effects across the ability distribution. These parameters are reported in Table 10.

5.4 SMM Estimation

The remaining twelve parameters—parent altruism η , psychic cost parameters κ_0 , κ_θ , and σ_κ , the discount rate ρ , the warm-glow weight ϕ_b , the luxury-bequest shifter $\underline{a}_{\text{beq}}$, the bequest fraction α_{beq} , the college wage scale ω_C , the paternalism weight ψ , the collateral channel slope b_{slope} , and the intergenerational earnings premium ζ_{pe} —are estimated by the simulated method of moments (SMM; [McFadden, 1989](#); [Pakes and Pollard, 1989](#)). SMM is well suited to this setting because the model does not admit closed-form expressions for the moments of interest: the coupled HJB system and the endogenous transfer region must be solved numerically. Following [Duffie and Singleton \(1993\)](#) and [Lee and Ingram \(1991\)](#), I simulate the model at each candidate parameter vector and match the simulated moments to their data counterparts.

Let $\mathbf{x} \in \mathbb{R}^{12}$ denote the parameter vector. For each $\mathbf{x}$, I solve the model and simulate $M = 4,000$ dynasties, computing the model counterparts of 36 targeted moments $\mathbf{m}^{\text{sim}}(\mathbf{x})$ (two further income-dispersion moments are computed but left untargeted). The estimator minimizes a block-diagonal percentage-deviation loss:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \sum_g \frac{w_g}{n_g} \sum_{m \in g} \left[\frac{m_m^{\text{sim}}(\mathbf{x}) - m_m^{\text{data}}}{m_m^{\text{data}}} \right]^2, \quad (13)$$

where the targeted moments are partitioned into blocks: college attendance rates by ability

$\times$ parent-wealth quartile ($n_g = 16$, $w_g = 3$), the high-school-to-college income ratio ($n_g = 1$), transfers ($n_g = 5$), intergenerational wealth ($n_g = 10$), decumulation ($n_g = 1$), college graduation by parent-income quartile ($n_g = 4$), and the fraction of children with negative net worth ($n_g = 1$); all blocks other than the college-rate block carry weight $w_g = 1$. The group-size normalization $1/n_g$ ensures that larger groups do not mechanically dominate. College moments receive a weight of three because they are the core target of the model; downweighting them leads to well-fitting wealth distributions but implausibly flat college-ability gradients. While an optimal weighting matrix based on the inverse of the variance-covariance of the moments would be asymptotically efficient, it requires a first-stage consistent estimate that is computationally expensive in this setting. The block-diagonal weighting provides a transparent alternative and is standard in the structural life-cycle literature ([Abbott et al., 2019](#); [Boar, 2021](#)).

5.4.1 Target Moments

The 36 targeted moments are organized into the following blocks, each chosen to provide leverage on specific parameters.

College graduation rates (16 moments). College graduation rates by ability quartile $\times$ parent wealth quartile, from the NLSY97. These moments form the core of the estimation: they trace out how college attainment varies across the joint distribution of ability and parental resources, which is precisely the variation the model is designed to explain.

Income ratio (1 moment). The high-school-to-college mean income ratio, computed from individual gross earnings in the NLSY97. The cross-sectional income standard deviations by education are computed as well but left *untargeted* (the income process is fixed externally to [Abbott et al., 2019](#), so they serve as a validation check rather than a target).⁶

⁶Because each agent in the model represents a single individual—not a household—the income targets use individual gross earnings (wages and salary) from the NLSY97, not household income. This is consistent with the approach in [Boar \(2021\)](#) and [Lee and Seshadri \(2019\)](#), who also target individual labor income in single-agent dynastic models. The income ratio (0.69) is computed from the cross-section of individuals aged 25–37 to maximize sample size, while the (untargeted) standard deviations ($\sigma_{HS} = 27.7K$ and $\sigma_C = 41.7K$) are computed from ages 30–37. All figures are in thousands of 2016 dollars, deflated by the CPI-U. The sample conditions on positive earnings above \$5,000 and defines college graduates as those with a bachelor’s degree or higher.

Transfer moments (5 moments). The mean transfer amount and the college attendance probability by parent wealth quartile. The transfer-probability moments use college attendance as a proxy for the parent-to-child transfer gradient, constructed from the PSID intergenerational sample.⁷

Wealth (10 moments). From the PSID linked sample: the aggregate wealth-to-income ratio, which disciplines the overall level of wealth accumulation; the parent–child wealth rank–rank slope ($\hat{\beta}^{rr} = 0.372$); mean child wealth conditional on parent wealth quartile (4 moments); median wealth for parents and children; and the variance of the inverse hyperbolic sine of wealth for each generation.⁸

Decumulation (1 moment). The parent wealth decumulation rate over ages 60–70.

College graduation by parent income (4 moments). College graduation rates by parent-income quartile, from the NLSY97. These discipline the income gradient of attainment, complementing the wealth gradient in the 16 joint college moments.

Negative net worth (1 moment). The fraction of children aged 35–40 with negative net worth, from the PSID. This moment disciplines the left tail of the child wealth distribution generated by student-loan debt.

5.4.2 Identification

The model is overidentified: twelve parameters are estimated from 36 targeted moments. While all moments jointly contribute to the estimation, the identification logic rests on distinct sources of variation for each parameter group.

Altruism (η). The altruism parameter is identified primarily from the wealth gradient in college attendance, holding ability constant. A higher η increases the size of the transfer

⁷I link children to their parents using the PSID Family Identification Mapping System (FIMS) and observe children as household heads or spouses at ages 35–40, with parent wealth measured when the child was 17–18. All wealth is in thousands of 2016 dollars (CPI-adjusted). Appendix A details the sample construction.

⁸The rank–rank slope of 0.372 is consistent with the range in the literature: [Charles and Hurst \(2003\)](#) estimate an intergenerational wealth elasticity of 0.37, and [Pfeffer and Killewald \(2018\)](#) find rank correlations of 0.40 or higher when children are measured at ages 45–64. My estimate falls in the middle because children in my sample are observed at ages 35–40.

region $\mathcal{T}$, raising the lifetime transfer burden parents face when children do not attend college. This makes college more attractive to wealthy parents, steepening the wealth gradient. The transfer probability moments by parent wealth quartile provide additional leverage: higher altruism implies more frequent transfers, with a wealth gradient shaped by the transfer region boundary.

Psychic cost ($\kappa_0, \kappa_\theta, \sigma_\kappa$). The psychic cost parameters are identified from the ability gradient in college attendance, holding wealth constant. The constant κ_0 governs the overall college rate, the log-ability gradient κ_θ determines how steeply college attendance rises with ability, and the shock scale σ_κ controls the dispersion of college choices among low-ability types: a larger σ_κ generates more college attendance at the bottom of the ability distribution by giving some individuals favorable psychic cost draws.

Discount rate (ρ). The discount rate is pinned down by the aggregate wealth-to-income ratio and median wealth levels. A lower ρ (more patient agents) generates more wealth accumulation, raising both statistics.

Warm-glow weight (ϕ_b). The warm-glow motive is identified from the parent wealth decumulation rate: a stronger warm-glow motive (ϕ_b) slows wealth drawdown in retirement, as parents derive utility from holding wealth at T . The decumulation rate of +2.5% per year (parents ages 60–70 are still accumulating) is consistent with evidence that pre-retirees continue saving due to precautionary motives and warm-glow utility (De Nardi, 2004; Hurd, 1990). The intergenerational rank-rank slope and conditional child wealth moments provide additional leverage on ϕ_b by disciplining how parental wealth transmits across generations.

College wage scale (ω_C). The relative price of college labor ω_C is identified primarily from the college-to-high-school income ratio: without this scaling, the structural premium from ability gradients ($\lambda_C > \lambda_{HS}$) and the steeper college life-cycle profile would generate a college premium far exceeding the data.

Paternalism (ψ). The paternalism weight is identified from the interaction between parent wealth and college attendance. A positive ψ amplifies the wealth gradient by giving the

child an additional reason to attend college: internalizing the parent’s preference for reduced transfer costs.

Bequest fraction (α_{beq}). The bequest fraction is identified from the intergenerational wealth moments: the rank–rank slope, conditional child wealth by parent wealth quartile, and the variance of child wealth. A higher α_{beq} strengthens the intergenerational wealth correlation by channeling more parent wealth to the child at death.

Collateral channel (b_{slope}). The collateral channel slope is identified from the wealth gradient in college attendance, particularly at the top of the parent wealth distribution. A positive b_{slope} relaxes borrowing constraints for children of wealthy parents, raising college enrollment among high-wealth families.

Intergenerational earnings premium (ζ_{pe}). The earnings premium is identified from the intergenerational wealth moments—the rank–rank slope and mean child wealth by parent wealth quartile. Because ζ_{pe} routes the parent–child correlation through inherited earning capacity rather than liquid wealth, it lifts the conditional child-wealth profile and the rank–rank slope without relaxing the child’s age-18 college-financing constraint.

Nonhomothetic bequest shifter. The shifter in the bequest function is identified from the upper tail of the wealth distribution: the variance of the inverse hyperbolic sine of wealth and conditional child wealth in the top parent quartile. Following [De Nardi \(2004\)](#), a positive $\underline{a}_{\text{beq}}$ makes bequests a luxury good, so poor parents bequeath little while the rich bequeath a rising share of wealth, generating the fat right tail.

Table 12. SMM-Estimated Parameters

Parameter	Description	Value
<i>Preferences</i>		
β	Discount factor (annual)	0.96
σ_{cd}	Type-I EV scale (college decision)	1.64
<i>Parental altruism & paternalism</i>		
η	Altruism weight	0.05
ϕ_b	Warm-glow terminal wealth weight	3.08
κ_{beq}	Luxury-bequest shifter (De Nardi, \$000s)	17.7
ξ	Paternalism weight in college decision	0.10
<i>College psychic cost (AGMV-style)</i>		
κ_0	Psychic-cost constant	3.2
κ_θ	Psychic-cost gradient in $\log(\theta)$	-1.4
σ_κ	Psychic-cost idiosyncratic shock S.D.	1.75
<i>Income process</i>		
ω_C	Relative price of college labor	0.64
σ_C	College income diffusion (OU)	0.133
σ_{HS}	High-school income diffusion (OU)	0.134
$\sigma_{\varepsilon,C}$	Transitory income shock S.D., college	0.000
$\sigma_{\varepsilon,HS}$	Transitory income shock S.D., high school	0.000
<i>College financial aid (income-based EFC; fixed, AGMV)</i>		
$\bar{g}$	Maximum annual grant (\$000s)	2.8
$\bar{y}_{\text{efc}}$	EFC phase-out parent-income threshold (\$000s)	72.0
b_1	Collateral channel: borrowing-limit slope in a_p	0.050
<i>Bequests & intergenerational transmission</i>		
α_{beq}	Bequest fraction (share of terminal wealth inherited)	0.30
Δ_{z_0}	Parental-college premium in initial productivity z_0	1.00

Notes: All 19 parameters are estimated jointly by the simulated method of moments (SMM). σ_{cd} is the scale of i.i.d. type-I extreme-value shocks in the college attendance choice. ξ is the paternalism weight: the parent's value difference $V_C^p - V_{HS}^p$ enters the child's college decision. $\kappa(\theta, \varepsilon_\kappa) = \kappa_0 + \kappa_\theta \log(\theta) + \sigma_\kappa \varepsilon_\kappa$ is the psychic cost of college (AGMV specification), decreasing in ability. ϕ_b and κ_{beq} govern the De Nardi (2004) luxury-bequest utility $\phi_b (a_p + \kappa_{beq})^{1-\gamma} / (1-\gamma)$: $\kappa_{beq} > 0$ makes bequests a luxury good (rising marginal propensity to bequeath in wealth), generating the upper tail of the wealth distribution. The same altruism weight η enters both the parent's flow utility and the terminal condition. α_{beq} is the fraction of the parent's terminal wealth inherited by the child at death; b_1 is the slope of the child's borrowing limit in parental wealth (collateral/co-signing channel). Δ_{z_0} is the premium in the child's initial log-productivity draw z_0 when the parent is college-educated: the intergenerational link operates through earning capacity, not transferred liquid wealth, so the college-financing constraint at entry is preserved (see Section 3). ω_C is the relative price of college labor; σ_C and σ_{HS} are the diffusions of the education-specific OU income processes, and $\sigma_{\varepsilon,C}$, $\sigma_{\varepsilon,HS}$ the standard deviations of i.i.d. transitory income shocks. Net college tuition is $\phi - \bar{g} \max\{0, 1 - a_p/\bar{a}_{\text{efc}}\}$: $\bar{g}$ is the maximum grant and $\bar{a}_{\text{efc}}$ the parental-wealth level at which grants fully phase out. Parameters set outside the estimation are reported in the companion table of externally calibrated parameters.

Notes: Parameters estimated by simulated method of moments. The objective function minimizes the percentage-deviation distance between the 36 targeted simulated moments and their data counterparts. See text for the identification argument.

6 Results

6.1 Model Fit

Table 13 reports the model’s fit to the 36 targeted moments. Panel A shows college graduation rates by ability and parent wealth quartile. The model captures the primary features of the data: college graduation rates increase with both ability and parent wealth, and the wealth gradient is steepest for low-ability children. Its fit to the highest-ability children is mixed—over-predicting their attendance among low-wealth families and under-predicting it among the wealthy, where the data show near-universal enrollment (discussed below). Panel B reports income moments and Panel C transfer moments. Panel D reports the intergenerational wealth distribution moments constructed from the PSID parent–child linked sample that discipline the warm-glow weight ϕ_b and the intergenerational wealth transmission channel.

The model reproduces the main features of the data: college graduation rises with ability within every wealth quartile and with parental wealth within most ability quartiles, and the overall graduation rate is close to the data average. The wealth gradient for low-ability children—the margin the mechanism targets—is captured. In the data, low-ability children with parents in the top wealth quartile graduate at 33%, compared to 19% for those in the bottom quartile, a gap of 14 percentage points; the model generates 27% and 15% respectively, a 12-point gradient that closely tracks the data, with levels about four points low throughout. Two discrepancies remain. The model overstates the ability gradient among low-wealth families: for bottom-quartile parents it spreads graduation from 15% at the lowest ability quartile to 65% at the highest—a 50-point gap against 34 points in the data (19% to 53%)—over-predicting attendance for high-ability children of poor parents. Conversely, the high-ability wealth gradient is too flat: high-ability children of top-quartile parents graduate at 62% in the model versus 74% in the data, so the model under-predicts attendance precisely where the data show near-universal enrollment.

Table 13. Targeted Moments: Data and Model

Panel A: College Attainment by Parent Wealth and Child Ability (NLSY97)				
Wealth Quartile \ Ability Quartile	1	2	3	4
1	0.15 (0.19)	0.34 (0.24)	0.38 (0.33)	0.65 (0.53)
2	0.19 (0.24)	0.37 (0.30)	0.45 (0.42)	0.59 (0.53)
3	0.23 (0.26)	0.35 (0.39)	0.50 (0.51)	0.65 (0.63)
4	0.27 (0.33)	0.45 (0.46)	0.56 (0.62)	0.62 (0.74)
Panel B: Income Moments				
	Model	Data		
High-School/College mean income ratio	0.36	0.69		
High-School income S.D. (\$k)	25.7	27.7		
College income S.D. (\$k)	104.6	41.7		
Panel C: Transfer Moments				
	Model	Data		
Mean transfer (\$k/yr)	5.8	6.2		
Pr(transfer > 0) — Wealth Q1	0.27	0.22		
Pr(transfer > 0) — Wealth Q2	0.38	0.31		
Pr(transfer > 0) — Wealth Q3	0.47	0.50		
Pr(transfer > 0) — Wealth Q4	0.59	0.73		
Panel D: Intergenerational Wealth Moments				
	Model	Data		
Wealth-income ratio	1.62	1.87		
Rank-rank slope (IG wealth)	0.23	0.37		
$E[a_c \mid \text{parent wQ1}]$ (\$k)	63.9	67.1		
$E[a_c \mid \text{parent wQ2}]$ (\$k)	88.8	112.1		
$E[a_c \mid \text{parent wQ3}]$ (\$k)	143.5	177.6		
$E[a_c \mid \text{parent wQ4}]$ (\$k)	210.5	340.4		
Median wealth, parents (\$k)	113.8	85.4		
Median wealth, children (\$k)	43.8	51.2		
Var(IHS wealth), parents	0.70	7.25		
Var(IHS wealth), children	10.58	13.48		
Parent wealth growth rate	0.024	0.025		

Notes: Comparison of the targeted moments in the data and the estimated model. Panel A reports college graduation rates by parent wealth quartile (rows) and child ability quartile (columns); numbers without parentheses are model moments, numbers in parentheses are data moments (NLSY97). Panel B: income moments (the income ratio is targeted; the education-specific standard deviations are untargeted). Panel C: transfer moments; Pr(transfer > 0) uses college attendance by parent wealth quartile as proxy for the extensive margin of financial transfers, from the PSID parent-child linked sample. Panel D: intergenerational wealth moments from the same PSID linked sample; all wealth in thousands of 2016 dollars. See Appendix A for sample construction details.

Figure 3 complements Panel D by comparing the full wealth distribution in the model and the data. The model reproduces the right-skewed shape of the empirical wealth distribution and captures the concentration of wealth among higher-income households. The close fit of the wealth distribution is important because parent wealth is a key state variable governing both the college decision and the transfer policy: errors in the wealth distribution would propagate into misspecified college gradients and transfer flows.



Figure 3. Wealth Distribution: Model vs. Data

Notes: Comparison of the parent wealth distribution in the model (simulated) and the data (PSID). Wealth is measured in thousands of 2016 dollars.

6.2 The Role of Parental Altruism in College Attainment

To quantify the importance of parental altruism, I solve the model with $\eta = 0$: parents make no transfers, so children finance college through their own earnings, loans, and grants alone. Table 14 reports college attendance by child ability quartile in the baseline and in this no-altruism economy. The effect is sharply concentrated at the bottom of the ability distribution:

attendance among the lowest-ability children collapses from 20% to 3% when altruism is removed, while attendance among the highest-ability children is essentially unchanged (62% versus 64%) and the middle quartiles fall only modestly; overall attendance declines from 42% to 36%. The pattern reflects where parental support is decisive. High-ability children attend regardless, because the private return to college is high; for low-ability children, by contrast, parental transfers are what make attendance affordable, so withdrawing them all but eliminates their enrollment. Parental altruism thus operates exactly at the margin where the college decision is most sensitive to resources.

Table 14. College Attendance With and Without Parental Altruism

Ability quartile	Baseline Pr(C)	No altruism ($\eta = 0$) Pr(C)	Change (%)
Q1	0.20	0.03	-87
Q2	0.37	0.35	-7
Q3	0.48	0.44	-9
Q4	0.62	0.64	3
Overall	0.42	0.36	-13

Notes: College attendance by child ability quartile in the baseline economy and in a counterfactual with parental altruism shut down ($\eta = 0$), so the parent makes no transfers and the child finances college through own earnings, loans, and grants. The final column reports the percentage change relative to the baseline. The model is re-solved at the estimated parameters with the altruism weight set to zero; statistics are computed from $M = 4,000$ simulated dynasties.

The mechanism works through two channels. First, altruistic parents directly subsidize college costs, reducing the net price their children face. The transfer amount increases with parent wealth but decreases with child ability, because parents use transfers to influence marginal children—those whose college decision depends on financial support. Second, parents internalize that a child without college will generate larger lifetime transfer costs. For wealthy parents, the present value of expected transfers to a high-school child exceeds the cost of college tuition, making college the more efficient investment. However, the child’s perspective introduces an additional force: anticipated parental transfers during and after college subsidize enrollment, while anticipated lifetime transfers also cushion children who

forgo college against income risk. The net effect on college attendance is therefore theoretically ambiguous. Before disentangling these forces, I first describe how college reshapes the transfer region in the baseline economy; Section 7 then introduces the full commitment benchmark and constructs counterfactual economies that eliminate ongoing parent-child interaction.

6.3 How College Reshapes the Transfer Region

In the baseline economy, parental support to adult children operates through the transfer region $\mathcal{T}$. College reduces the child’s exposure to the transfer region by permanently raising expected income through a higher ability gradient ($\lambda_C = 0.797 > \lambda_{HS} = 0.517$) and a steeper life-cycle earnings profile. The higher income level keeps college children’s resources further from the transfer boundary, so that a given negative shock is less likely to pull them into $\mathcal{T}$.

Panel (a) of Figure 4 plots the fraction of simulated individuals in the transfer region at each parent age over the post-college overlap window, by education. High-school children spend a substantially larger fraction of this window receiving transfers; college children exit the region earlier and return less frequently after negative shocks. Panel C of Table 15 quantifies this exposure, reporting the fraction of time receiving transfers separately for the in-college and post-college periods.

Table 15. Transfer Region Exposure and Savings Behavior

	HS Child	College Child
<i>Panel A: Child's own savings effort $s_c^{own} = ra_c + y_c - c_c$ (\$k/yr)</i>		
When receiving transfers	-0.78	-4.95
When not receiving transfers	3.88	13.72
Difference (Samaritan's dilemma)	-4.66	-18.67
<i>Panel B: Over-consumption vs. autarky $\Delta c_c = c_c^{MPE} - c_c^{aut}$ (\$k/yr)</i>		
In transfer region	6.19	11.25
Outside transfer region	8.68	31.81
<i>Panel C: Fraction of time receiving parental transfers</i>		
During college (ages 18–22)	—	0.596
After college (ages 22–48)	0.105	0.018

Notes: Statistics from simulated dynasties ($M = 4,000$). Panel A reports the child's own savings effort $s_c^{own} = \tilde{r}_c a_c + y_c - c_c$, which excludes the mechanical effect of transfers on wealth accumulation. A negative difference indicates the child saves less of their own resources when receiving transfers (the Samaritan's dilemma). Panel B reports the consumption differential relative to an overlap-period autarky benchmark c_c^{aut} : the child's optimal consumption from a lifecycle problem solved at the same age and wealth level with no parent interaction, using the simulated parent wealth to compute the EFC for financial aid. A negative value indicates the child in the baseline economy consumes less than the autarky benchmark at the same state. Panel C reports the fraction of time the child receives parental transfers, separately for the in-college years (ages 18–22, the enrollment-subsidy channel) and the post-college period (ages 22–48, the insurance channel); the in-college figure applies only to college children.

The reduced transfer-region exposure translates into lower lifetime transfer flows. Panel (b) of Figure 4 shows average post-college transfers by parent wealth quartile and child education, and Table 16 reports the same quantities numerically together with the implied reduction in each wealth quartile. College children receive far fewer transfers—reductions exceed 80% in every wealth quartile, and the largest absolute gap is among the wealthiest parents, whose high-school children receive the most support. This reflects both the mechanical effect (higher income reduces the need for support) and the behavioral effect (reduced moral hazard induces more self-insurance). The key insight is that a dollar spent on tuition generates a *permanent* increase in the child's income drift, reducing the probability of entering the transfer region for the remainder of the lifecycle. A dollar transferred directly provides temporary consumption smoothing but does not change the drift, so the child returns to the transfer boundary over

time.

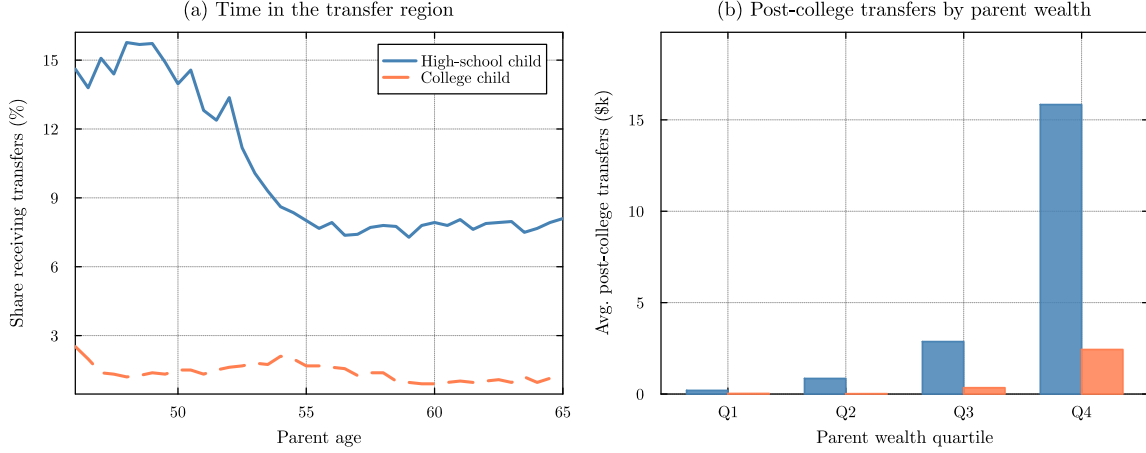


Figure 4. College, Transfer-Region Exposure, and Post-College Transfers

Notes: Simulated dynasties ($M = 4,000$). Panel (a) plots the fraction of children receiving positive transfers ($\tau > 0$) at each parent age over the post-college overlap window—child age 22–48, parent age 46–72—by child education; high-school children spend a substantially larger share of this window in the transfer region. Panel (b) reports average post-college transfers received by the child (excluding the in-college years), by parent wealth quartile and child education.

Table 16. Post-College Transfers by Child Education and Parent Wealth

Parent wealth quartile	High-school child (\$k/yr)	College child (\$k/yr)	Reduction (%)
Q1	0.2	0.0	85
Q2	0.8	0.0	98
Q3	2.9	0.3	88
Q4	15.8	2.4	85
All	4.4	0.8	81

Notes: Average post-college transfers received by the child (excluding the in-college years), by parent wealth quartile and child education, from $M = 4,000$ simulated dynasties; this is the numerical counterpart of Panel (b) of Figure 4. The reduction is $1 - \bar{\tau}_C / \bar{\tau}_{HS}$, the proportional fall in transfers received by college relative to high-school children. The “All” row pools across quartiles. Transfers are in thousands of 2016 dollars per year.

7 Moral Hazard Decomposition

Parental altruism raises college attendance, but it also generates a fundamental tension: the Samaritan’s dilemma. Because children anticipate transfers when their wealth is low, they under-save relative to autarky, imposing larger and more frequent transfer costs on parents. A central question is how this lack of commitment affects the college decision. The answer is theoretically ambiguous. On one hand, anticipated parental support during the college years effectively subsidizes enrollment, making college more attractive to the child. On the other hand, anticipated lifetime transfers cushion non-college children against income risk, reducing their incentive to invest in human capital. Whether college attendance is higher or lower under the Samaritan’s dilemma than under full commitment depends on which force dominates—a quantitative question that the structural model is designed to answer. This section introduces the full commitment benchmark, formalizes both forces, constructs counterfactual economies to isolate them, and reports the quantitative decomposition.

7.1 Full Commitment Benchmark and Two Opposing Forces

To isolate the effects of the Samaritan’s dilemma, I first characterize the efficient allocation under full commitment. Suppose a benevolent planner commits at $t = 0$ to a state-contingent transfer schedule $\{\tau(a_p, a_c, z, t)\}$ for the entire overlap period. Because both agents pool resources under a single plan, the problem reduces to a single-agent optimization over total household wealth $A \equiv a_p + a_c$:

$$\rho V^{\text{FC}}(A, z, t) = \max_{c_{\text{tot}}} \left\{ u_{\text{FC}}(c_{\text{tot}}) + V_A^{\text{FC}} [rA + y_p(t) + y_c(z, t) - c_{\text{tot}}] + \frac{1}{2} \sigma_a^2 A^2 V_{AA}^{\text{FC}} + \mathcal{L}_z V^{\text{FC}} + V_t^{\text{FC}} \right\} \quad (14)$$

Under CRRA preferences, the efficient allocation equates weighted marginal utilities, $u'(c_p) = \eta u'(c_c)$, so consumption shares are time-invariant: $c_c = \eta^{1/\gamma} c_p$. Total consumption $c_{\text{tot}} = c_p(1 + \eta^{1/\gamma})$ yields an aggregated flow utility $u_{\text{FC}}(c_{\text{tot}}) = \frac{c_{\text{tot}}^{1-\gamma}}{1-\gamma} (1 + \eta^{1/\gamma})^\gamma$, and the planner’s HJB reduces to a single-state-variable problem. Full commitment differs from the baseline economy in three ways ([Barczyk and Kredler, 2014a,b](#)): it eliminates overconsumption by both agents, it renders the timing of transfers indeterminate, and—most important for the college decision—it provides insurance without moral hazard. The child’s college choice under

full commitment follows the same probit structure as the baseline (equation 11), with the planner’s continuation values replacing the baseline values.

Relative to full commitment, the Samaritan’s dilemma creates two opposing forces on the child’s college decision. From the parent’s perspective, a college-educated child is cheaper to support: college permanently raises the income level, shifting the child away from the transfer boundary. Under full commitment, the parent could internalize this motive directly by conditioning transfers on e_c . In the baseline economy, however, the parent cannot commit: the transfer policy responds only to the current state (a_p, a_c, z) , and the parent transfers whenever the child is sufficiently poor, regardless of the child’s education choice.

The child chooses education to maximize own continuation utility, taking the parent’s equilibrium transfer policy as given. The continuation values V_C and V_{HS} embed the entire expected path of future transfers. The child does not internalize the cost transfers impose on the parent, generating overconsumption relative to the efficient allocation whenever the child’s state is near or inside $\mathcal{T}$. Two forces emerge from this interaction:

Force 1: Anticipated transfers subsidize enrollment. The child starts college with zero assets and reduced income ($\ell_C \cdot y_c$ net of tuition), placing the initial state deep in the transfer region $\mathcal{T}$. The parent—bound by time consistency—optimally transfers resources that effectively subsidize enrollment costs. This implicit subsidy raises the child’s valuation of the college path relative to full commitment, pushing attendance *above* the efficient level.

Force 2: Anticipated transfers cushion non-college children. Children who forgo college face lower expected income, so they spend substantially more of their adult lives in the transfer region. The parent cannot withhold this support. The anticipation of this lifetime safety net raises the child’s valuation of the high-school path—the child can under-save and over-consume, knowing that parental transfers will materialize when income falls. This cushion reduces the child’s incentive to attend college, pushing attendance *below* the efficient level.

Whether Force 1 or Force 2 dominates depends on ability, parental wealth, and the income process parameters. The decomposition below quantifies these forces and confirms that the sign of the moral hazard effect on college varies across the ability–wealth distribution.

7.2 Counterfactual Economies and Quantitative Decomposition

To isolate the role of commitment from the role of insurance, I construct two counterfactual economies that eliminate ongoing parent-child interaction entirely, replacing the dynamic transfer game with a one-time lump-sum transfer at $t = 0$. After the initial transfer, the child solves the standard lifecycle problem alone—shutting down both the Samaritan’s dilemma and the parental insurance channel. The two counterfactuals differ in whether the parent gives the same transfer in both education states or an education-contingent menu (τ_C, τ_{HS}) , which separates the pure moral hazard channel from the education-targeting channel.⁹

Counterfactual A: Unconditional one-time transfer. The parent chooses τ_0 at $t = 0$ and the child receives it regardless of education choice:

$$\begin{aligned} \max_{\tau_0 \in [0, a_p]} \quad & V_p^{\text{alone}}(a_p - \tau_0) \\ & + \eta \left[\Pr(C \mid \tau_0, \theta) V^{c, \text{alone}}(\tau_0 \mid C) + (1 - \Pr(C \mid \tau_0, \theta)) V^{c, \text{alone}}(\tau_0 \mid HS) \right] \end{aligned} \quad (15)$$

where $V_p^{\text{alone}}(a)$ is the parent’s remaining lifetime utility with no further interaction with the child. The child’s education choice follows the same probit rule as the baseline (equation 11), with the child-alone value gain replacing the coupled values:

$$\Pr(C \mid \tau_0, \theta) = \begin{cases} \Phi \left(\frac{G_0(\tau_0, \theta) / \bar{V} - \kappa_0 - \kappa_\theta \log \theta}{\sigma_\kappa} \right) & \text{if } G_0 > 0, \\ 0 & \text{otherwise,} \end{cases} \quad G_0(\tau_0, \theta) \equiv V^{c, \text{alone}}(\tau_0 \mid C) - V^{c, \text{alone}}(\tau_0 \mid HS) \quad (16)$$

Since both education paths receive the same endowment τ_0 , the paternalism term cancels and the education decision reflects only fundamentals—human capital returns net of psychic cost—and not the structure of ongoing parental support. Comparing baseline attendance with this economy isolates the *pure moral hazard effect*: the distortion to college enrollment caused solely by the child’s anticipation of future transfers.

⁹Appendix H provides a formal derivation of the full commitment benchmark and discusses its relationship to the one-time transfer counterfactuals. The key distinction is that the FC benchmark removes moral hazard while preserving insurance, whereas the one-time transfer counterfactuals remove both.

Counterfactual B: Education-contingent transfer menu. The parent commits at $t = 0$ to a transfer schedule that may differ by the child's education choice, choosing a transfer τ_C paid if the child attends college and a transfer τ_{HS} paid if the child chooses high school:

$$\begin{aligned} \max_{(\tau_C, \tau_{HS}) \in [0, a_p]^2} & \Pr(C \mid \tau_C, \tau_{HS}, \theta) [V_p^{\text{alone}}(a_p - \tau_C) + \eta V^{c, \text{alone}}(\tau_C \mid C)] \\ & + [1 - \Pr(C \mid \tau_C, \tau_{HS}, \theta)] [V_p^{\text{alone}}(a_p - \tau_{HS}) + \eta V^{c, \text{alone}}(\tau_{HS} \mid HS)] \end{aligned} \quad (17)$$

$$\Pr(C \mid \tau_C, \tau_{HS}, \theta) = \begin{cases} \Phi\left(\frac{G_m(\tau_C, \tau_{HS}, \theta)/\bar{V} - \kappa_0 - \kappa_\theta \log \theta}{\sigma_\kappa}\right) & \text{if } G_m > 0, \\ 0 & \text{otherwise,} \end{cases} \quad (18)$$

where the value gain now carries the paternalism term, which no longer cancels because the parent's post-transfer wealth differs across the two branches:

$$G_m(\tau_C, \tau_{HS}, \theta) \equiv [V^{c, \text{alone}}(\tau_C \mid C) + \psi V_p^{\text{alone}}(a_p - \tau_C)] - [V^{c, \text{alone}}(\tau_{HS} \mid HS) + \psi V_p^{\text{alone}}(a_p - \tau_{HS})].$$

where $\kappa(\theta, \varepsilon) \geq 0$ is the psychic cost of college (zero for the HS alternative). This menu is the efficient one-time-transfer benchmark: it nests Counterfactual A (the restriction $\tau_C = \tau_{HS}$) and a college-or-nothing economy (the restriction $\tau_{HS} = 0$), and because it relaxes both restrictions it weakly dominates them in parent welfare. Crucially, the altruistic parent can now support a child in the high-school state directly, so a low-return child is no longer pushed into college merely to capture parental resources. Comparing Counterfactual B with Counterfactual A isolates the *education-targeting effect*: the change in attendance when the parent can direct transfers by education rather than give the same endowment in both states.

Decomposition. Define:

$$\Delta_{\text{MH}}^{\text{pure}}(\theta, a_p) = \Pr(\text{College} \mid \text{baseline}) - \Pr(\text{College} \mid \text{unconditional}) \quad (19)$$

$$\Delta_{\text{targ}}(\theta, a_p) = \Pr(\text{College} \mid \text{unconditional}) - \Pr(\text{College} \mid \text{menu}) \quad (20)$$

$$\Delta_{\text{MH}}^{\text{total}}(\theta, a_p) = \Delta_{\text{MH}}^{\text{pure}} + \Delta_{\text{targ}} \quad (21)$$

A positive $\Delta_{\text{MH}}^{\text{pure}}$ indicates that the baseline economy raises college attendance relative to the no-moral-hazard economy; a negative value indicates that moral hazard depresses college attendance. The education-targeting component Δ_{targ} isolates the change in enrollment when the parent can direct transfers by education: a positive sign indicates that targeting *lowers* attendance, because the parent diverts low-return children out of college and supports them in the high-school state instead.

Results. Table 17 reports the decomposition.

Table 17. Moral Hazard Decomposition of College Attendance

	Overall	Parent Wealth Quartile			
		Q1 (Low)	Q2	Q3	Q4 (High)
Panel A: College Attendance Rate (all ability types)					
Baseline	0.440	0.369	0.439	0.521	0.430
No MH (unconditional)	0.229	0.323	0.188	0.188	0.218
No MH (menu)	0.287	0.323	0.278	0.336	0.211
Δ_{MH}^{pure}	+0.210	+0.045	+0.251	+0.333	+0.212
Δ_{MH}^{total}	+0.153	+0.045	+0.161	+0.185	+0.219
Panel B1: Pure Moral Hazard Effect (Δ_{MH}^{pure}) by Ability					
Low ability		−0.039	+0.045	+0.079	+0.000
Medium ability		+0.354	+0.437	+0.528	+0.458
High ability		+0.832	+0.831	+0.848	+0.041
Panel B2: Education-Targeting Effect (Δ_{targ}) by Ability					
Low ability		+0.000	+0.000	+0.000	+0.000
Medium ability		+0.000	+0.000	+0.000	+0.000
High ability		+0.000	+0.000	+0.000	−0.074
Panel B3: Total Effect ($\Delta_{MH}^{total} = \Delta_{MH}^{pure} + \Delta_{targ}$) by Ability					
Low ability		−0.039	+0.045	+0.079	+0.000
Medium ability		+0.354	+0.437	+0.528	+0.458
High ability		+0.832	+0.831	+0.848	−0.033
Panel C: Optimal One-Time Transfer τ_0^* (\$000s)					
Unconditional:					
Low ability		0.0	0.0	0.0	31.6
Medium ability		0.0	0.0	0.0	31.6
High ability		0.0	0.0	0.0	316.5
Menu, if college (τ_C):					
Low ability		0.0	0.0	0.0	31.6
Medium ability		0.0	0.0	0.0	31.6
High ability		0.0	0.0	0.0	316.5
Menu, if high school (τ_{HS}):					
Low ability		0.0	0.0	0.0	31.6
Medium ability		0.0	0.0	0.0	31.6
High ability		0.0	0.0	0.0	0.0

Notes: Panel A reports average college attendance across ability types under three economies: the baseline, the no-MH economy with an unconditional transfer, and the no-MH economy with the education-contingent menu (τ_C, τ_{HS}). The “Overall” column averages across wealth quartiles. In both no-MH economies, parent and child interact only at $t = 0$; the child solves the lifecycle problem alone thereafter.

$\Delta_{MH}^{pure} = \Pr(\text{College} \mid \text{baseline}) - \Pr(\text{College} \mid \text{unconditional})$ isolates the overconsumption distortion; $\Delta_{MH}^{total} = \Pr(\text{College} \mid \text{baseline}) - \Pr(\text{College} \mid \text{menu})$ adds the education-targeting channel. Panels B1–B3 disaggregate these effects by child ability. Panel C reports the optimal unconditional transfer and the menu transfers τ_C (if college) and τ_{HS} (if high school). Source: simulated model ($M = 4,000$ dynasties).

Aggregate result. In the baseline economy, 44.0% of children attend college. Eliminating ongoing parent-child interaction and replacing it with an unconditional lump-sum transfer *lowers* attendance sharply to 22.9%—a pure moral hazard effect of $\Delta_{MH}^{pure} = +21.0$ percentage points. The anticipated stream of future transfers (Force 1) strongly subsidizes enrollment, so the baseline delivers far more college than the no-moral-hazard economy; on the aggregate level Force 1 dominates Force 2 by a wide margin. Allowing the parent to direct transfers by education—the menu (τ_C, τ_{HS}) —raises attendance to 28.7%, an education-targeting effect of $\Delta_{targ} = -5.8$ percentage points: a college-contingent premium $\tau_C > \tau_{HS}$ lets the parent finance high-ability children whose return justifies the degree, slightly more than offsetting the diversion of low-return children into the high-school state. The total distortion relative to the efficient menu is a sizable *over-enrollment* of $\Delta_{MH}^{total} = +15.3$ percentage points. As the ability decomposition below shows, this over-enrollment is concentrated among medium- and high-ability children, for whom the anticipated transfer stream raises attendance most. This contrasts with a college-or-nothing restriction ($\tau_{HS} = 0$), which mechanically forces attendance upward by withholding all support from non-college children.

Decomposition by ability. The aggregate result masks heterogeneity across the ability distribution; Panels B1–B3 of Table 17 disaggregate the effects by child ability. The targeting channel operates with opposite signs across ability. For low- and medium-ability children the menu sets the college premium $\tau_C - \tau_{HS}$ near zero or supports the high-school state directly ($\tau_{HS} > 0$), so the child stays out of college: the parent prefers to transfer resources rather than finance a degree whose return does not cover the human-capital gain net of the psychic cost. For high-ability children the menu sets $\tau_C > \tau_{HS}$, raising attendance, because the return and the reduction in future transfer costs justify a college-contingent premium. The pure moral hazard channel (Panel B1) is small and mixed in sign for low-ability children but large and positive for medium- and high-ability children, reflecting the enrollment-subsidy force (Force 1): these children anticipate the transfer stream and enroll at much higher rates than under the one-time transfer. Averaged over the ability–wealth distribution the medium- and high-ability positive values dominate, so the pure channel is strongly positive overall. Panel C reports the implied menu transfers (τ_C, τ_{HS}) ; in cells where no child attends in equilibrium the college-state transfer τ_C is off the equilibrium path and not separately identified.

This ability pattern is the complement of the no-altruism comparison in Section 6.2, and the two exercises target different margins. Shutting down altruism entirely (Table 14) removes the support that makes college affordable for *low-ability* children, whose enrollment collapses; the *level* of parental support is what sustains attendance at the bottom of the ability distribution. The moral-hazard channel isolated here removes only the ongoing transfer game while preserving a one-time transfer, so it leaves low-ability enrollment largely intact and instead bites for *medium- and high-ability* children—who would attend with a lump sum but are pushed further by anticipating the ongoing stream. Altruism therefore raises low-ability attendance through the level of support, while the dynamic incentive to draw on that support inflates enrollment higher up the ability distribution; the two effects are distinct and operate on different children.

8 Welfare Analysis

The preceding section established that the Samaritan’s dilemma distorts both college enrollment and savings behavior. This section quantifies its welfare consequences—weighing the cost of moral hazard against the insurance value of ongoing transfers—and evaluates a free-college policy in the presence and absence of commitment.

8.1 Consumption-Equivalent Variation

I measure welfare using the consumption-equivalent variation (CEV): the permanent proportional change in consumption that makes an agent indifferent between the baseline (Samaritan’s dilemma) economy and an alternative. Under CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$, CRRA homogeneity implies that scaling all consumption flows by a factor $(1 + \Delta)$ scales lifetime utility by $(1 + \Delta)^{1-\gamma}$. Inverting gives the CEV

$$\Delta = \left(\frac{V^{\text{alt}}}{V^{\text{base}}} \right)^{\frac{1}{1-\gamma}} - 1, \quad (22)$$

where $\Delta > 0$ means the alternative is welfare-improving.

For the child, $V^c = \int_0^T u(c_c) e^{-\rho t} dt$ and equation (22) applies directly. For the parent, $V^p = \int_0^T [u(c_p) + \eta u(c_c)] e^{-\rho t} dt$ captures both own and altruistic consumption. Because u is

homogeneous of degree $1 - \gamma$, scaling both c_p and c_c by $(1 + \Delta)$ scales V^p by $(1 + \Delta)^{1-\gamma}$. The parent CEV therefore measures the permanent household-wide consumption change—own plus the altruistic component—that achieves indifference.

I compute CEV at each grid point (a_p, θ) by evaluating the value functions at the beginning of the overlap period ($a_c = 0, z = 0, t = 0$), integrating over the college-choice psychic-cost shock. I report unweighted averages across the (a_p, θ) grid unless stated otherwise.

8.2 Welfare Across Commitment Structures

Table 18 reports CEV for three alternative economies relative to the baseline. The first two rows of Panel A correspond to the no-moral-hazard counterfactuals from Section 7: the unconditional one-time transfer eliminates ongoing transfers but gives the child the same endowment in both education states, while the education-contingent menu lets the parent choose separate transfers (τ_C, τ_{HS}) . Both allocations remove the Samaritan’s dilemma by construction—the parent commits to a transfer schedule at $t = 0$ and the child solves an independent problem thereafter. In both, the parent’s own continuation value is the single-agent parent HJB of Appendix J.7, solved with the same consumption floor, bequest motive, and wealth diffusion as the baseline coupled value, so the consumption-equivalent comparison differences like with like rather than against a permanent-income approximation.

Measured by the dynastic (η -weighted) consumption-equivalent variation—the welfare object the parent optimizes—the two one-time-transfer economies are *worse* than the baseline: the unconditional transfer delivers -13.2% and the education-contingent menu -13.1% , while full commitment delivers $+30.1\%$. The contrast is the central result of this section. Both one-time transfers eliminate the Samaritan’s dilemma, but they do so by severing the ongoing relationship entirely, which also removes the insurance that repeated transfers provide against the child’s income risk; the lost insurance outweighs the gain from eliminating over-consumption, so the dynasty is worse off. Full commitment removes the moral hazard while *preserving* insurance—the planner still moves resources to the child state by state—and is the only economy that raises dynastic welfare. The menu weakly dominates the unconditional transfer (-13.1% versus -13.2%), and the parent-own and child-own columns show

that this small edge accrues to the child: matching transfers to education (τ_C, τ_{HS}) raises the child's own value from -4.7% to -2.8% while leaving the parent's own value essentially unchanged (-14.0% versus -13.9%). The full-commitment gain, by contrast, is entirely the parent's—parent-own CEV $+41.8\%$ against the child's -32.7% —because commitment strips the child of the ability to extract resources by under-saving; the positive dynastic figure reflects the parent's valuation of the efficient allocation, which it weights chiefly by its own consumption.

The parent would like to implement full commitment—the one allocation that raises dynastic welfare—but cannot, and the reason is precisely the parent's altruism. Committing to hand the child a one-time transfer and then stand back is not time-consistent. Ex ante the parent would like to promise to transfer τ_0 once and never again, inducing the child to self-insure and eliminating the moral hazard. Ex post, however, whenever the child draws a bad shock and approaches the transfer region, the parent—precisely because $\eta > 0$ —strictly prefers to transfer rather than let the child consume little. The child anticipates this and under-saves, so the promise unravels: the only time-consistent (Markov-perfect) policy is the baseline one, in which the parent transfers whenever the child is sufficiently poor, regardless of education. The commitment allocations are therefore benchmarks the parent would adopt if a commitment technology existed, not policies it can choose unilaterally—the “freedom” the child enjoys in the no-moral-hazard economies is something the altruistic parent cannot credibly grant with cash. This is exactly why college matters: by permanently raising the child's income, a tuition payment moves the child away from the transfer region for the remainder of life, achieving through human capital the commitment that direct transfers cannot.

8.3 Free-College Policy

I consider a policy that eliminates tuition ($\phi = 0$) and finances the revenue shortfall through a lump-sum tax T^{LS} on initial parent wealth. No financial aid is needed when tuition is zero, so $\bar{g} = 0$. The tax is set in general equilibrium to balance the government budget: it must cover tuition for the children who enroll, while enrollment itself responds to the tax through

the parent’s post-tax wealth. The per-dynasty tax therefore solves the fixed point

$$T^{\text{LS}} = 4\phi \cdot \mathbb{E}[\Pr(C \mid a_p - T^{\text{LS}}, \theta)], \quad (23)$$

where ϕ is the annual tuition defined above, 4ϕ is the four-year cost of tuition the government covers per enrolled child, and the expectation is over the (a_p, θ) distribution. Enrollment, transfers, and value functions are all evaluated at the post-tax wealth $a_p - T^{\text{LS}}$. Because enrollment differs across the two behavioral regimes, the budget-balancing tax is solved separately for each. Parents with $a_p < T^{\text{LS}}$ have post-tax wealth clamped at zero; they still benefit because their child pays no tuition.

I solve the free-college economy under two regimes. Under *no commitment*, the model retains its Markov-perfect equilibrium structure—parents and children interact strategically with $\phi = 0$ —and the Samaritan’s dilemma persists. Under *commitment*, the parent commits to a one-time education-contingent menu (τ_C, τ_{HS}) at $t = 0$ and the child solves independently with $\phi = 0$.

Panel B of Table 18 reports the results. Under no commitment, free college raises enrollment from 44.1% to 49.6%, but the dynastic welfare change is essentially nil (-0.4%): the tuition subsidy is largely dissipated because the child continues to anticipate ongoing support and under-saves, and the parent funds the policy through the lump-sum tax (present value \$13.3k). The gain that materializes accrues to the child (own CEV $+3.0\%$) and is roughly offset by the parent’s tax burden (-0.7%). Under commitment, enrollment instead *falls* to 37.4% as the parent channels low-return children toward direct support rather than a subsidized degree; the child’s own welfare gain is much larger ($+11.7\%$), but the parent bears a heavier burden (-14.4%), so the dynastic change remains negative (-12.7%). Free college thus operates mainly as a transfer from parents to children: its value to the child is several times larger when the parent can commit, even though commitment lowers enrollment.

Panel C reports the share of the free-college welfare gain that is dissipated by the lack of commitment, computed as $1 - \Delta^{\text{no cmt}} / \Delta^{\text{cmt}}$, where Δ denotes the CEV gain from free college over the baseline under each regime. Because the parent funds the policy in both regimes, the dynastic gain is negative under each, so the dissipation is measured against the gain that accrues to the child. By that measure 74.4% of the child’s commitment-regime gain is

Table 18. Welfare Analysis: CEV (%) Relative to Baseline

	Parent (own) CEV (%)	Child CEV (%)	Dynastic (+ η) CEV (%)
<i>Panel A: Commitment Structure</i>			
Baseline (Samaritan’s dilemma)	0.00	0.00	0.00
Unconditional transfer (no MH)	-13.95	-4.66	-13.17
Education-contingent menu (no MH)	-14.07	-2.79	-13.12
Full commitment (planner)	41.75	-32.72	30.12
<i>Panel B: Free College Policy ($\phi = 0$, lump-sum tax)</i>			
Free college, no commitment	-0.68	3.01	-0.45
Free college, with commitment	-14.38	11.73	-12.68
<i>Panel C: Share of free-college gain dissipated by lack of commitment</i>			
Share (%)		74.4	

Notes: Consumption-equivalent variation (CEV) measures the permanent proportional change in consumption that achieves indifference between each economy and the baseline. The *Parent (own)* column uses the parent’s own-consumption value V_p^{own} ; the *Child* column uses the child’s own value; the *Dynastic* column uses the η -weighted dynastic value $V_p^{\text{own}} + \eta V_c^{\text{own}}$ that the parent optimizes and is the welfare-relevant aggregate. Panel A compares commitment structures holding tuition at its estimated value. Panel B sets $\phi = 0$ and funds the policy with a budget-balancing lump-sum tax T^{LS} (equation 23) solved in general equilibrium for each regime. Panel C reports the share of the free-college welfare gain dissipated by the lack of commitment, $1 - \Delta^{\text{no cmt}} / \Delta^{\text{cmt}}$. Source: model solution at estimated parameters.

forfeited without commitment—the child captures only about a quarter of what commitment would deliver—and the share rises with parental wealth, from roughly 72% in the bottom wealth quartile to about 86% in the top, because wealthier families rely more heavily on transfers and therefore suffer more from the commitment problem.

9 Conclusion

This paper studies why altruistic parents invest in their children’s college education and how this investment interacts with the moral hazard costs of intergenerational transfers. I develop a dynastic model in which college serves as a commitment device that reduces the Samaritan’s

dilemma, and I test its predictions using matched parent-child data. A set of stylized facts—parent consumption is higher when children attend college, parents of college children appear to decumulate wealth more slowly (though imprecisely estimated), college attenuates dynastic precautionary saving, and college reduces the probability of transfers while leaving conditional transfer amounts unchanged or larger—are consistent with the model’s central mechanism: college shrinks the transfer region, releasing precautionary resources.

The net effect of the Samaritan’s dilemma on college attendance is theoretically ambiguous—the lack of commitment creates an enrollment subsidy (Force 1) that raises attendance and a transfer cushion (Force 2) that depresses it. The structural decomposition shows that the enrollment-subsidy force dominates: relative to a no-commitment benchmark in which the parent makes a single transfer, the Samaritan’s dilemma raises aggregate college enrollment, with the over-enrollment concentrated among medium- and high-ability children who anticipate the ongoing transfer stream. Because letting the parent target transfers by education moves the level only modestly, the distortion the friction imposes on the college margin is primarily one of *scale*—too much enrollment, financed by anticipated support—rather than of composition alone. Whether eliminating the friction improves welfare depends on what replaces it: the ongoing transfer relationship that drives the over-enrollment also insures children against income risk, so severing it with a one-time transfer can leave the dynasty worse off, whereas a commitment that preserves insurance raises welfare. The welfare effects of a free-college policy likewise hinge on whether parents can commit. The same altruism parameter η that generates the Samaritan’s dilemma also drives parents to invest in college as a way to mitigate it.

The framework carries implications for the design of college subsidies and financial aid. Policies that reduce college costs benefit both students and their parents, who face lower life-time transfer obligations. However, the model suggests that the welfare effects of financial aid depend on how policies interact with private transfers: grants that crowd out parental contributions may be less effective at increasing enrollment than those that complement family support. Conversely, policies that expand parents’ capacity to transfer without encouraging education may exacerbate the Samaritan’s dilemma by reducing children’s incentives to invest in their own human capital.

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A PSID Sample Construction

This appendix describes the construction of the analysis sample from the Panel Study of Income Dynamics (PSID). The sample combines four PSID data products: the Family Interview File, the Individual File, the Transition into Adulthood Supplement, and the Family Identification Mapping System. I use the thirteen biennial waves covering 1999 through 2023. The exact survey variable codes underlying each measure are documented in the replication package.

A.1 Data Sources

Family Interview File. The core source is the PSID family-level interview file, which provides household-level information on demographics, income, wealth, employment, and housing tenure for each survey wave from 1999 to 2023. A small number of supplementary measures that are not part of the main interview module—principally liquid savings and detailed industry classifications—are drawn from a companion family-level release and merged by household and wave.

Individual File. The PSID individual file provides person-level information for every sample member, including age, relationship to the household head, position within the family unit, and—beginning in 2013—years of completed education.

Transition into Adulthood Supplement. This supplement covers young adults aged roughly 18 to 28 in the 2015 and 2017 waves. I use it to confirm college enrollment directly, complementing the education-based measures available in the main files.

Family Identification Mapping System. This product provides the intergenerational links between children and their parents. Each child record carries identifiers for up to four parent types—biological father, biological mother, adoptive father, and adoptive mother—together with a generation marker that distinguishes original 1968 sample members from their children, grandchildren, and later descendants. The target sample consists of third-generation and later children (born roughly 1975–2005, reaching college age between 1993 and 2023) matched to their parents.

A.2 Variables Used in the Analysis

Table A.1 lists the measures drawn from the PSID, grouped by concept, together with the survey years over which each is available and the data product from which it is obtained. Variable names in the table are descriptive; they correspond to underlying PSID concepts rather than to any particular survey code.

Table A.1. Measures Drawn from the PSID

Measure	Waves	Source
<i>Panel A: Demographics</i>		
Age of household head	1999–2023	Family File
Age of spouse or partner	1999–2023	Family File
Number of children in the family unit	1999–2023	Family File
Age of youngest child	1999–2023	Family File
State of residence	1999–2023	Family File
Race of household head	1999–2023	Family File
Education of household head	1999–2023	Family File
Education of spouse or partner	1999–2023	Family File
<i>Panel B: Income</i>		
Total family income	1999–2023	Family File
Labor income of household head	1999–2023	Family File
Labor income of spouse or partner	1999–2023	Family File
<i>Panel C: Wealth</i>		
Total net worth (including home equity)	1999–2023	Wealth Module
Net worth excluding home equity	1999–2023	Wealth Module
Home equity	1999–2023	Wealth Module
Retirement accounts (IRAs and annuities)	1999–2023	Wealth Module
Liquid savings	2003–2023	Family File
Stocks and mutual funds	1999–2023	Wealth Module
Vehicles	1999–2023	Wealth Module
Other assets	1999–2023	Wealth Module
Other real estate	1999–2023	Family File
Student loan debt	2011–2023	Family File
Other debts	2013–2023	Family File
<i>Panel D: Employment</i>		
Employment status of head	1999–2023	Family File
Currently working (indicator)	1999–2023	Family File
Reason for leaving last job	1999–2023	Family File
Industry of employment	2003–2023	Family File
Occupation of employment	2003–2023	Family File
<i>Panel E: Consumption</i>		
Food expenditure	1999–2023	Family File
Housing expenditure	1999–2023	Family File
Transportation expenditure	1999–2023	Family File
Health-care expenditure	1999–2023	Family File
Clothing expenditure	2005–2023	Family File
Recreation expenditure	2003–2023	Family File

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Table A.1 (continued)

Measure	Waves	Source
<i>Panel F: Individual-Level</i>		
Relationship to household head	1999–2023	Individual File
Position within the family unit	1999–2023	Individual File
Age of individual	1999–2023	Individual File
Years of completed education	2013–2023	Individual File

Notes: All monetary measures are deflated to 2016 dollars using the CPI-U. Wealth components for 1999–2007 use the PSID imputed wealth supplement; from 2009 onward they are recorded directly in the family interview file. Liquid savings is unavailable before 2003. Industry and occupation classifications change granularity in 2017; I harmonize them to a consistent classification (see below).

A.3 Sample Construction

The analysis sample is built in six sequential steps.

Step 1: Assemble the family-year panel. For each of the thirteen waves I extract the wave-specific family-level measures, place them on a common set of definitions, and stack the waves into a single panel with one observation per family per wave. Supplementary measures available only in the companion family release—liquid savings and detailed industry codes for 2003–2015—are merged by household and wave. I harmonize the industry and occupation classifications across the 2017 change in granularity, retaining a coarser classification that is consistent across all waves, and I define an involuntary job separation as a separation due to a plant closing or a layoff.

Step 2: Assemble the individual-year panel. I reshape the individual file into a person-by-year panel containing each member’s position in the family unit, relationship to the household head, age, and (from 2013 onward) years of completed education. A time-invariant person identifier, constructed from each member’s 1968 family lineage number and within-lineage person number, links individuals across waves and underlies the parent–child match.

Step 3: Construct the household panel. The family-year and individual-year panels are merged by household and wave. I retain current family-unit members who are the household head, the legal spouse, or a cohabiting partner. Including spouses ensures that a parent who is not the household head—typically the mother in a two-parent household—can still be

matched to her children. Rare duplicate person-year records are resolved by giving preference to head status.

Step 4: Attach intergenerational links. Using the Family Identification Mapping System, each child is linked to the household of one parent. When more than one parent is available, the link follows a fixed priority: biological father, then adoptive father, then biological mother, then adoptive mother. Children with no matchable parent are dropped.

Step 5: Identify college attendance. I combine three approaches to maximize coverage across all thirteen waves. For 2013 onward, where years of completed education are recorded at the individual level, a child is coded as attending college if completed education lies strictly between high-school completion and a bachelor’s degree at ages 18–24, or if completed education rises between consecutive waves above the high-school level before age 26. For 1999–2011, where individual education is not yet recorded, children aged 18–22 whose highest education observed in later waves exceeds high school are coded as having attended college. For the 2015 and 2017 waves, the direct enrollment indicators from the Transition into Adulthood Supplement take precedence and override the education-based codes.

Step 6: Form the parent-household sample. The child–parent records are merged with the household panel, collapsed to one observation per parent household and wave, and augmented with the number of children currently enrolled in college. I then merge state-level college costs—in-state public tuition and private tuition by state and year, both from the Integrated Postsecondary Education Data System (IPEDS)—and deflate all monetary measures to 2016 dollars using the CPI-U. Finally, I assign each parent to a permanent-income group based on average real household income observed while the head is aged 26–59 (with overall average income used as a fallback when the head is never observed in that range). The groups are: \$30,000 to \$60,000; \$60,000 to \$100,000; \$100,000 to \$160,000; and above \$160,000. Households with average income below \$30,000 or otherwise unclassifiable are dropped.

A.4 Sample Restrictions

I drop observations meeting any of the following conditions:

1. Missing or invalid demographic information (age of head or spouse not reported).
2. Missing or invalid education for the head or spouse.
3. Top-coded or missing wealth or income (any wealth or income component recorded with a PSID missing or refused code).
4. Negative total net worth.
5. Internally inconsistent portfolios (total net worth below home equity, or below other real estate).
6. Implausible portfolio shares (home-equity share below -40% , other-real-estate share below -40% , or vehicle share above 50% of total net worth).
7. Average real income below \$30,000 or otherwise outside the classification bounds.

A.5 Variable Definitions

Household consumption is the sum of expenditures on food at home, food away from home, housing (rent or imputed rent for homeowners), utilities, transportation, health care, education, and child care; from 2005 onward, clothing, recreation, and vacation are also included. Income is total household income before taxes. Wealth is total net worth: the sum of home equity, other real estate, vehicles, farm and business assets, stocks, checking and savings accounts, bonds, retirement accounts, and other assets, minus all debts. Inter-vivos transfers are recorded separately for transfers from parents to children and from children to parents, and include cash gifts, help with expenses, and regular financial support. The key explanatory variable throughout the empirical analysis is child college attainment, an indicator for whether the child holds a bachelor's degree or higher. To condition on parental resources, I construct within-cohort parent income quartiles, ranking each parent against others born in the same year.

A.6 Key Constructed Variables

Portfolio shares. Each asset class—retirement accounts, home equity, other real estate, and vehicles—is expressed as a share of total net worth and used to screen out internally inconsistent portfolios (see the sample restrictions above).

College-cost instrument. The instrument used in the empirical analysis interacts the (log) in-state tuition in the parent’s state of residence with the number of the parent’s children currently enrolled in college, so that exposure to tuition costs is increasing in the number of children of college age.

Involuntary job separation. An indicator equal to one when the head left the previous job because of a plant closing or a layoff, and zero otherwise.

Real values. All monetary values are deflated to 2016 dollars using the CPI-U for all urban consumers. Log transformations are applied where indicated in the regression specifications.

A.7 Consumption Data

PSID consumption expenditure data are available for every wave from 1999 onward. The consumption measure aggregates spending across six categories: food (at home and away), housing (rent, utilities, property taxes, and insurance), transportation (vehicle costs, fuel, parking, and public transit), health care (insurance premiums and out-of-pocket spending), clothing (available from 2005), and recreation (available from 2003). Clothing and recreation are added to the aggregate only in the years in which they are collected.

A.8 Merge Attrition

Table [A.2](#) documents the sample size at each stage of construction: from the full PSID individual file, through the intergenerational link to parents, to the merge with household-level consumption, income, and wealth. Panel B breaks the parent–child links down by relationship type, and Panel C quantifies the observations lost at each merge stage. The main source of attrition is the requirement that the matched parent appears as a household head or spouse in the family interview file during the sample period.

Table A.2. PSID Sample Construction and Merge Attrition

Stage	Observations	Unique IDs	% Retained
<i>Panel A: Building the Analysis Sample</i>			
Individual file (all persons $\times$ years)	441 158	45 386	—
After college identification	441 152	45 386	100.0
Matched to FIMS (child–parent link)	358 874	34 916	81.3
Matched to parent household (hh_data)	148 541	10 492	41.4
Collapsed to parent-HH $\times$ year	71 896	10 492	—
After sample restrictions	62 320	10 145	86.7
<i>Panel B: Parent Identification Source</i>			
Linked via biological father	297 284		
Linked via adoptive father	4114		
Linked via biological mother	56 856		
No identifiable parent (dropped)	0		
<i>Panel C: Merge Losses</i>			
Individuals not in FIMS	82 278		
Child–years without parent in hh_data	210 333		
Parent–years without state tuition data	16 488		
Final analysis sample	62 320	10 145	
of which: HH–years with child in college	6566		

Notes: Children are linked to parents via the Family Identification Mapping System. The parent–child link follows a hierarchical priority: biological father, adoptive father, biological mother, adoptive mother.

“Unique IDs” refers to unique persons (first two rows) or unique parent households (remaining rows). “% Retained” is relative to the previous stage. All monetary measures are winsorized at the 1st and 99th percentiles and deflated to 2016 dollars.

B NLSY97 Sample Construction

This appendix describes the construction of the NLSY97 analysis sample. The National Longitudinal Survey of Youth 1997 (NLSY97) tracks a nationally representative cohort of 8,984 Americans born between 1980 and 1984, interviewed annually from 1997 through 2011 and biennially thereafter. I use the NLSY97 for two purposes: constructing college graduation rates by ability and parent-wealth quartile (the sixteen cell-level moments used in estimation), and estimating the education-specific income process. The exact survey variable codes underlying each measure are documented in the replication package.

B.1 Data Sources

Baseline and demographics. The baseline survey provides each respondent’s gender, race and ethnicity, birth year, and census region, together with parental education for the biological mother and father and family income during the late 1990s and early 2000s.

Cognitive ability and academic performance. Cognitive ability is measured by the math-verbal percentile from the computer-adaptive aptitude battery administered in 1999. High-school transcripts provide SAT and ACT scores and overall high-school grade point average.

Schooling module. The schooling module records detailed college enrollment and financing information at the college-by-term-by-year level: tuition charged, room and board, grants and scholarships, total loans, out-of-pocket expenses, credit hours attempted, and withdrawal from college.

Income and transfer module. The income module records annual labor income and annual hours worked (used to estimate the income process) and, for the college-age years, total parental transfers and parental allowances.

B.2 Variables Used in the Analysis

Table [B.1](#) lists the measures drawn from the NLSY97, grouped by concept, together with their coverage and source. Variable names are descriptive and correspond to underlying

survey concepts rather than to any particular survey code.

Table B.1. Measures Drawn from the NLSY97

Measure	Years	Source
<i>Panel A: Demographics and Ability</i>		
Gender	1997	Baseline
Race and ethnicity	1997	Baseline
Birth year	1997	Baseline
Math-verbal aptitude percentile	1999	Aptitude battery
SAT verbal and math scores	Cumulative	Transcript
ACT composite score	Cumulative	Transcript
Overall high-school GPA	Cumulative	Transcript
<i>Panel B: Family Background</i>		
Biological mother’s highest grade	1997	Baseline
Biological father’s highest grade	1997	Baseline
Gross family income	1997–2003	Annual
Census region at baseline	1997	Baseline
<i>Panel C: College Enrollment and Costs</i>		
Tuition charged per term	1997–2017	Schooling
Grants and scholarships	1997–2017	Schooling
Total student loans	1997–2011	Schooling
Out-of-pocket college expenses	1998–2001	Schooling
Room and board	1998–2001	Schooling
Credit hours attempted	1997–2017	Schooling
<i>Panel D: Education Attainment</i>		
Highest grade attended	1998–2003	Annual
Highest degree ever received	Cumulative	Cumulative
Withdrawal from college	Annual	Schooling
<i>Panel E: Income Process</i>		
Annual labor income	1997–2011	Income
Annual hours worked (all jobs)	1997–2007	Employment
<i>Panel F: Parental Transfers</i>		
Total parental transfers	1997–2003	Income
Parental allowance	1997–2003	Income

Notes: Cumulative measures are computed by NLS staff from the respondent’s full survey history. The aptitude percentile is the math-verbal score from the computer-adaptive battery administered in 1999. Schooling-module measures are recorded by college enrollment, term, and survey year. Family income refers to the gross household income of the respondent’s family of origin.

B.3 Sample Construction

Step 1: Assemble the respondent panel. I retain core demographics (gender, race, birth year, and census region), parental education for the biological mother and father, cognitive ability (the 1999 math-verbal aptitude percentile), academic performance (SAT and ACT scores and high-school GPA from transcripts), and family income from 1997 to 2003.

Step 2: Build the college-spell panel. The detailed schooling measures are organized into panels indexed by individual, college enrollment, term, and year—tuition, grants and scholarships, total loans, out-of-pocket expenses, room and board, credit hours, and withdrawal—and merged into a unified college-spell panel.

Step 3: Identify education attainment. College attendance is identified from the highest grade attended (available 1998–2003): a respondent is coded as having attended college if the maximum grade attended exceeds high school. College graduation is defined as attaining a bachelor’s degree by age 25, using the cumulative highest-degree measure.

Step 4: Estimate the income process. Using annual labor income and annual hours worked, I construct hourly wages by education group. Following [Abbott et al. \(2019\)](#), I estimate the high-school-to-college income ratio and the education-specific income standard deviations, which provide three of the moments used in estimation.

Step 5: Assemble parental transfers. Parental financial transfers and allowances are organized into a person-year panel for 1997–2003. These measure financial support from parents during the college-age years and discipline the transfer structure in the model.

B.4 Key Constructed Variables

Ability quartiles. I rank respondents by their 1999 math-verbal aptitude percentile and partition them into quartiles, restricting the sample to respondents with a non-missing aptitude score.

Parent-wealth quartiles. Total household net worth at age 17 is used to construct within-cohort parent-wealth quartiles. Together with the ability quartiles, these define the 4×4 grid of college-graduation rates used as the core estimation moments.

College-graduation rates. The sixteen cell-level graduation rates (ability quartile by parent-wealth quartile) are the primary target moments in estimation; they trace how college attainment varies across the joint distribution of ability and parental resources.

Average tuition. The gross annual cost of college (\$12,200) is computed from average tuition reported in the NLSY97 schooling module, following [Abbott et al. \(2019\)](#). The model uses a net annual tuition (after institutional discounts) of $\phi = \$6,700$, with the remaining gap covered by the endogenous financial-aid schedule $g(a_p)$ described in Section [3.7](#).

C HRS Sample Construction

This appendix describes the construction of the HRS analysis sample. The Health and Retirement Study (HRS) is a biennial longitudinal survey of Americans over age 50, conducted by the University of Michigan since 1992. I use the RAND HRS Longitudinal File, which harmonizes variables across all waves (1992–2022), together with four supplements: the RAND Family-Kids file, the RAND Family-Resident file, the Consumption and Activities Mail Survey (CAMS), and the supplemental survey on college support. Bequest information is taken from the HRS exit interview. The exact survey variable codes underlying each measure are documented in the replication package.

C.1 Data Sources

RAND HRS Longitudinal File. The core source is the RAND HRS longitudinal file, in which RAND staff harmonize variable definitions, handle skip patterns, and construct consistent wealth, income, and health measures across the sixteen waves spanning 1992–2022. From this file I draw respondent and spouse demographics, household wealth and its components, household income and its sources, and respondents’ subjective probabilities of leaving a bequest.

RAND Family-Kids file. This supplement records information about each respondent’s children—demographics (age, gender, education, marital status), geographic proximity, financial transfers in both directions, care provision, and bequest intentions—for waves spanning 1992–2014. I organize it into a child-by-wave panel.

RAND Family-Resident file. This supplement provides household composition by wave—the number of living children, sons, and daughters—which I organize into a respondent-by-wave panel.

Consumption and Activities Mail Survey (CAMS). CAMS is a supplemental mail survey that collects detailed consumption and spending data for a subset of HRS households, biennially from 2001 to 2017. It provides total household spending, durable and non-durable spending, housing expenditure, and mortgage payments.

Supplemental survey on college support. This supplement collects retrospective information on parents’ financial contributions to their children’s college education. For each child who attended college it records tuition, room and board, years of college, and the share of each cost paid by the parent, which I use to construct the total parental contribution to college in 2016 dollars.

Exit interview. The HRS exit interview, conducted with a proxy informant after a respondent’s death, records bequest and inheritance information. I use it to construct child-level bequests (dollar transfers and shares of the estate) and housing inheritance (whether each child inherited the main residence and its imputed value).

College tuition data. Regional in-state college tuition by census division and year is merged with HRS respondents using their census division of residence, providing the regional college-cost variation used in the analysis.

C.2 Variables Used in the Analysis

Table C.1 lists the measures drawn from the HRS, grouped by concept, together with their coverage and source. Variable names are descriptive and correspond to underlying survey concepts rather than to any particular survey code.

Table C.1. Measures Drawn from the HRS

Measure	Coverage	Source
<i>Panel A: Respondent Demographics</i>		
Age of respondent	1992–2022	RAND HRS
Age of spouse	1992–2022	RAND HRS
Gender	Time-invariant	RAND HRS
Race	Time-invariant	RAND HRS
Years of education	Time-invariant	RAND HRS
Highest degree	Time-invariant	RAND HRS
Birth year	Time-invariant	RAND HRS
HRS sample cohort	Time-invariant	RAND HRS
Census division of residence	1992–2022	RAND HRS
<i>Panel B: Household Wealth</i>		
Total net worth	1992–2022	RAND HRS
Net home value	1992–2022	RAND HRS
Retirement accounts (IRA/Keogh)	1992–2022	RAND HRS
Checking and savings	1992–2022	RAND HRS

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Table C.1 (continued)

Measure	Coverage	Source
Vehicles	1992–2022	RAND HRS
Business and farm assets	1992–2022	RAND HRS
Other real estate	1992–2022	RAND HRS
<i>Panel C: Income</i>		
Total household income	1992–2022	RAND HRS
Social Security retirement income	1992–2022	RAND HRS
Social Security disability / SSI	1992–2022	RAND HRS
Pension income	1992–2022	RAND HRS
<i>Panel D: Bequest Expectations</i>		
Probability of bequest $\geq$ \$10K	1992–2022	RAND HRS
Probability of bequest $\geq$ \$100K	1992–2022	RAND HRS
Probability of bequest $\geq$ \$500K	1992–2022	RAND HRS
Probability of any bequest	1992–2022	RAND HRS
<i>Panel E: Child Information</i>		
Child’s years of education	1992–2014	Family-Kids
Child’s age	1992–2014	Family-Kids
Child’s gender	Time-invariant	Family-Kids
Child’s marital status	1992–2014	Family-Kids
Child’s income bracket	1992–2014	Family-Kids
Child lives within ten miles	1992–2014	Family-Kids
<i>Panel F: Inter-Vivos Transfers</i>		
Parent gave a transfer (indicator)	1992–2014	Family-Kids
Amount the parent gave	1992–2014	Family-Kids
Child gave a transfer (indicator)	1992–2014	Family-Kids
Amount the child gave	1992–2014	Family-Kids
<i>Panel G: College Support</i>		
Tuition and board paid by the parent	Retrospective	College support
Number of children who attended college	Retrospective	College support
Parent contributed a positive amount	Retrospective	College support
<i>Panel H: Consumption</i>		
Total household consumption	2001–2017	CAMS
Non-durable spending	2001–2017	CAMS
Housing expenditure	2001–2017	CAMS
<i>Panel I: Bequests</i>		
Dollar bequest to child	Exit interview	Exit interview
Share of estate to child	Exit interview	Exit interview
Housing inheritance value	Exit interview	Exit interview

Notes: The Family-Kids and Family-Resident supplements cover waves spanning 1992–2014. Monetary measures from the RAND HRS file are nominal at the source and are deflated to 2016 dollars in the analysis. The college-support measures are retrospective, covering each child’s entire college period. Child income brackets are reported by the parent respondent; I use the bracket midpoint.

C.3 Sample Construction

Step 1: Assemble the respondent panel. I reshape the RAND HRS longitudinal file into a person-by-wave panel containing time-invariant identifiers and demographics (birth

year, gender, race, education, religion) together with wave-varying measures of wealth, income, health, bequest expectations, family structure, employment, and spouse characteristics.

Step 2: Organize the child file. The Family-Kids supplement is reshaped from one record per child to a child-by-wave panel, with a child identifier that links each child to the parent respondent. Measures are grouped into child demographics, child-to-parent transfers, parent-to-child transfers, and bequest intentions.

Step 3: Add household composition. The Family-Resident supplement, which records the number of living children, sons, and daughters by wave, is merged into the respondent panel to provide the child-count breakdown.

Step 4: Add consumption. CAMS is organized into a respondent-by-wave panel of total household spending, durable and non-durable spending, housing expenditure, mortgage payments, and total household consumption. CAMS covers a subsample of HRS households biennially from 2001 to 2017.

Step 5: Construct college support. For each child who attended college, the total parental contribution is the sum of the real tuition paid by the parent (annual tuition, deflated to 2016 dollars, multiplied by years of college and the share of tuition the parent paid) and the real room-and-board paid by the parent (annual room and board, deflated, multiplied by years and the share the parent paid). Records with missing cost or share information are dropped; an explicit “did not contribute” response is recoded to zero. The total family contribution sums across all children in the household.

Step 6: Identify widowhood. I extract each respondent’s year of death and attach it to the spouse’s record, which yields, for each respondent, the year of the partner’s death and supports the analysis of widowhood.

Step 7: Construct bequests from the exit interview. The exit interview provides two kinds of child-level bequest information. Financial bequests are the dollar amounts and shares of the estate transferred to each child; survey missing codes are removed, and when

the estate share is missing but dollar amounts are available, the share is imputed as each child’s portion of total transfers. Housing inheritance is constructed for respondents whose main residence passed to children rather than to a surviving spouse: I identify which children received the home and, when the home was left to all children jointly, divide its imputed value equally among them.

Step 8: Add regional college costs. Average in-state college tuition by census division and year is organized into a region-by-year panel and merged with HRS respondents using their census division of residence.

C.4 Sample Restrictions

I impose the following restrictions, analogous to those applied to the PSID sample:

1. **Age restrictions:** parents over age 50 and children over age 26.
2. **Interview status:** only respondents with completed interviews in a given wave.
3. **Couple households:** for analyses requiring spouse information, I restrict to coupled households.
4. **Missing data:** survey “don’t know,” “refused,” and “missing” responses are carried through the construction stage and handled within each specific analysis, so that wealth and income are not discarded prematurely.

The final HRS analysis sample contains 19,179 parent–child pairs and 98,861 observations.

D Income Process: Discrete-to-Continuous-Time Mapping and Life-Cycle Profiles

This appendix derives the mapping from the discrete-time AR(1) income process estimated by [Abbott et al. \(2019\)](#) to the continuous-time Ornstein-Uhlenbeck (OU) specification used in the model, and presents the implied life-cycle income profiles.

D.1 Mapping Derivation

The discrete-time income shock process is an AR(1) at annual frequency:

$$z_{t+1}^e = \rho_e^{\text{ann}} z_t^e + \eta_{t+1}^e, \quad \eta_{t+1}^e \sim N(0, (\sigma_\eta^e)^2). \quad (24)$$

The continuous-time counterpart is the OU process:

$$dz^e = -\kappa_e z^e dt + \sigma_e dW, \quad (25)$$

where $\kappa_e > 0$ is the mean-reversion rate and σ_e is the diffusion coefficient.

Mean-reversion rate. The exact discretization of (25) at unit time intervals ($\Delta = 1$ year) gives the conditional mean $\mathbb{E}[z_{t+1}^e | z_t^e] = e^{-\kappa_e} z_t^e$. Matching this with the AR(1) coefficient ρ_e^{ann} yields:

$$e^{-\kappa_e} = \rho_e^{\text{ann}} \implies \kappa_e = -\ln(\rho_e^{\text{ann}}). \quad (26)$$

Diffusion coefficient. The conditional variance of the discretized OU process is:

$$\text{Var}(z_{t+1}^e | z_t^e) = \frac{\sigma_e^2}{2\kappa_e} (1 - e^{-2\kappa_e}). \quad (27)$$

The conditional variance of the AR(1) is simply $(\sigma_\eta^e)^2$. Equating the two and using $e^{-\kappa_e} = \rho_e^{\text{ann}}$:

$$\begin{aligned} (\sigma_\eta^e)^2 &= \frac{\sigma_e^2}{2\kappa_e} (1 - (\rho_e^{\text{ann}})^2) \\ \implies \sigma_e &= \sigma_\eta^e \cdot \sqrt{\frac{2\kappa_e}{1 - (\rho_e^{\text{ann}})^2}}. \end{aligned} \tag{28}$$

This mapping preserves both the autocorrelation and the stationary variance $\text{Var}(z^e) = \sigma_e^2/(2\kappa_e)$ of the process. Note that the stationary variance of the OU process equals the unconditional variance of the AR(1), $(\sigma_\eta^e)^2/(1 - (\rho_e^{\text{ann}})^2)$, confirming internal consistency.

Numerical values. Applying (26)–(28) to the male estimates from [Abbott et al. \(2019\)](#) (Table 3.2): for high-school graduates the annual persistence is $\rho_{HS}^{\text{ann}} = 0.952$ with innovation variance $(\sigma_\eta^{HS})^2 = 0.017$, giving $\kappa_{HS} = -\ln(0.952) = 0.049$ and $\sigma_{HS} = \sqrt{2\kappa_{HS}(\sigma_\eta^{HS})^2/(1 - (\rho_{HS}^{\text{ann}})^2)} = 0.134$; for college graduates $\rho_C^{\text{ann}} = 0.966$ with $(\sigma_\eta^C)^2 = 0.017$, giving $\kappa_C = 0.035$ and $\sigma_C = 0.133$. Both the mean-reversion rates and the diffusions are set externally to these AGMV-implied values rather than estimated. Because college earnings are more persistent ($\rho_C > \rho_{HS}$) while the innovation variance is common, the implied stationary variance of the persistent shock, $\sigma_e^2/(2\kappa_e) = (\sigma_\eta^e)^2/(1 - (\rho_e^{\text{ann}})^2)$, is *higher* for college graduates: 0.254 (standard deviation 0.51) versus 0.182 for high school (standard deviation 0.43). The model therefore does not generate the transfer-region contraction through a reduction in income risk; the operative channel is the income level (Section 3).

D.2 Life-Cycle Profiles

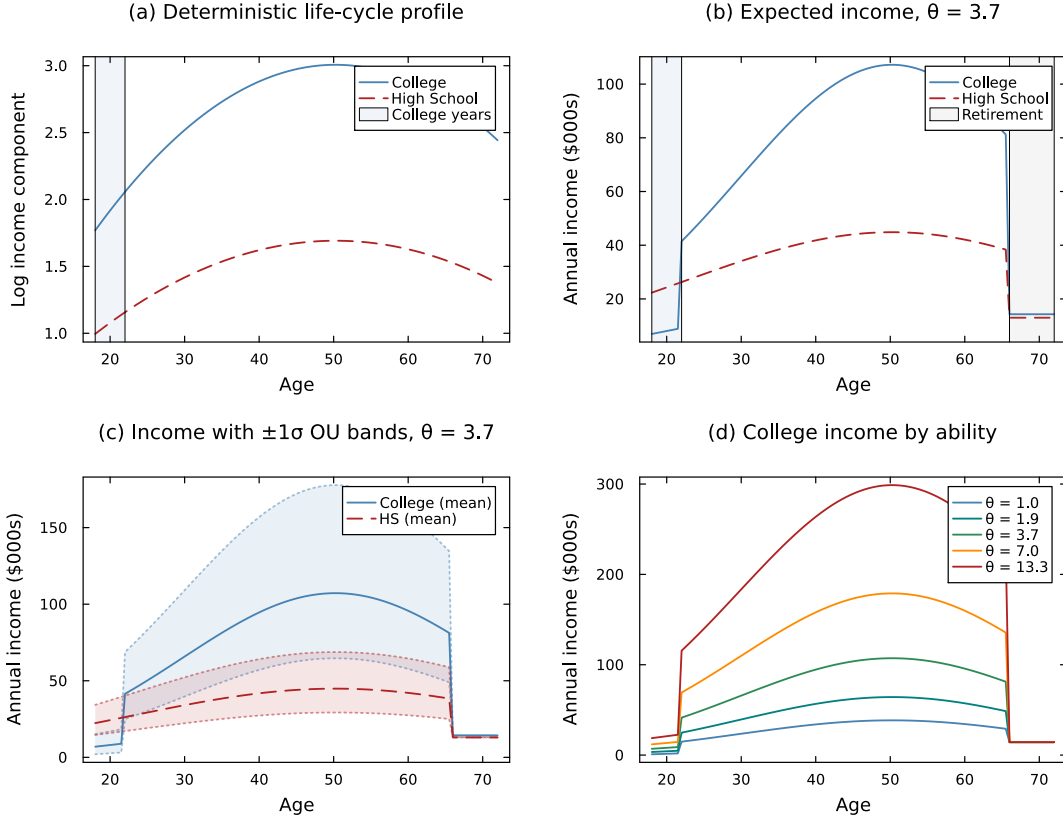


Figure D.1. Income Process: Life-Cycle Profiles by Education

Notes: Panel (a): deterministic age-income component $\gamma(t, e_c) = \gamma_{1,e} t - \gamma_{2,e} t^2$ from [Abbott et al. \(2019\)](#). Panel (b): expected annual income $y_c(t) = w \cdot \theta^{\lambda_e} \cdot \exp(\gamma(t, e_c))$ for median ability ($\theta = 3.9$); college workers earn $\ell_C = 56\%$ of full-time income during ages 18–22 and pay net tuition. Panel (c): ± 1 stationary standard deviation of the OU income shock ($\sigma_e / \sqrt{2\kappa_e}$) around expected income. Panel (d): college income profiles for low ($\theta = 1.5$), medium ($\theta = 3.9$), and high ($\theta = 10.2$) ability workers, illustrating the complementarity between ability and education ($\lambda_C > \lambda_{HS}$).

E Parent Wealth Portfolio

Table E.1 provides a detailed breakdown of parent income and wealth by income group. The table complements the summary variables reported in Table 1 in the main text.

Table E.1. Parent Income and Wealth by Income Group (PSID, 2016 Dollars)

	By Average HH Income Group				All
	\$30–60	\$60–100	\$100–160	\$160+	
Household income	44,223 (21,535)	76,829 (37,964)	120,169 (51,447)	242,277 (253,591)	102,569 (121,904)
Head labor income	23,523 (19,369)	40,976 (28,162)	64,728 (42,902)	123,504 (123,970)	53,930 (64,202)
Total wealth	102,555 (230,260)	200,522 (377,059)	367,076 (536,534)	965,941 (1,052,497)	326,288 (612,283)
Home equity	48,321 (86,276)	76,931 (105,763)	127,798 (136,410)	223,253 (195,989)	102,056 (138,176)
IRA/annuity	6,437 (38,692)	20,026 (68,967)	49,226 (108,878)	107,888 (163,964)	35,668 (98,914)
Non-housing fin. wealth	53,469 (182,888)	121,496 (318,143)	234,059 (453,318)	698,124 (895,431)	214,727 (505,467)
Savings	7,950 (31,182)	16,013 (39,546)	36,321 (61,142)	68,743 (87,368)	25,676 (56,070)
Vehicles	12,549 (16,601)	19,741 (19,952)	24,701 (22,565)	32,096 (28,691)	20,737 (22,256)
Other real estate	7,938 (42,810)	13,934 (57,200)	22,799 (75,043)	55,426 (122,965)	20,513 (73,634)
Student loans	1,736 (7,821)	4,134 (12,628)	7,932 (18,488)	7,755 (20,246)	4,881 (14,759)
Observations	15 096	17 356	13 011	7863	53 326
Unique parent households	2718	2770	1833	1041	8362

Notes: Means with standard deviations in parentheses. All monetary variables deflated to 2016 dollars using the CPI-U and winsorized at the 1st and 99th percentiles. Income groups defined by parent average real household income observed between ages 25 and 60.

F Parent Consumption Detail

Table F.1 decomposes total household consumption into major expenditure categories. Housing constitutes the largest category across all income groups, followed by transportation and food. Discretionary categories—clothing, recreation, and vacation—are only available from the 2005 wave onward and increase sharply with income.

Table F.1. Parent Household Consumption by Income Group (PSID, 2016 Dollars)

	By Average HH Income Group				All
	\$30–60	\$60–100	\$100–160	\$160+	
Total consumption	36,426 (18,374)	46,131 (20,389)	57,243 (24,489)	73,130 (33,277)	50,076 (26,252)
Food	7,915 (4,854)	9,211 (4,944)	10,582 (5,296)	12,872 (6,269)	9,718 (5,470)
Housing	15,030 (9,071)	18,677 (10,757)	24,487 (13,588)	33,215 (18,356)	21,206 (13,873)
Transportation	9,310 (8,213)	11,880 (8,709)	13,811 (9,844)	15,562 (11,999)	12,166 (9,661)
Health care	2,108 (3,027)	3,402 (3,649)	4,195 (4,157)	5,027 (4,832)	3,469 (3,950)
Clothing (2005+)	1,356 (1,704)	1,577 (1,762)	1,912 (1,907)	2,808 (2,626)	1,776 (1,988)
Recreation (2005+)	503 (920)	807 (1,211)	1,178 (1,464)	1,805 (1,897)	957 (1,396)
Vacation (2005+)	835 (1,505)	1,482 (2,020)	2,376 (2,583)	3,886 (3,604)	1,870 (2,554)
Observations	15 096	17 356	13 011	7863	53 326
Unique parent households	2718	2770	1833	1041	8362

Notes: Means with standard deviations in parentheses. Consumption data available from the PSID beginning in 1999. All expenditure categories deflated to 2016 dollars and winsorized at the 1st and 99th percentiles. Clothing, recreation, and vacation are available from 2005 onward only. Total consumption is the sum of all listed categories.

G Additional Robustness: Parent Fixed Effects

As additional robustness for the consumption and wealth stylized facts (Sections 4.2 and 4.3), I exploit parent fixed effects to identify the college effect from within-family (between-sibling) variation. For consumption, the specification uses child-pair-year observations: within a given family and year, siblings who differ in college attainment generate variation in the college indicator while holding all parent-level characteristics constant.

Table G.1 reports parent fixed effects regressions of log parent consumption on the child's college indicator, at the child-pair-year level. The positive and significant coefficient confirms that the consumption difference documented in Section 4.2 is not driven by unobserved parental characteristics: even within the same family, parent consumption is higher in years when the college-educated child's characteristics are more salient.

Table G.1. Parent Consumption and Child College Status: Parent Fixed Effects (PSID)

	(1)	(2)
	OLS + Income Q FE	Parent FE
College (BA+)	0.068*** (0.018)	0.027** (0.013)
Controls	Yes	Yes
Parent FE	No	Yes
Parent income Q FE	Yes	–
Observations	13,841	13,841
R^2 / Within R^2	0.469	0.840

Notes: Regressions of log parent household consumption on child college indicator (BA+), at the child-pair-year level. Column 1: OLS with parent income quartile fixed effects and comprehensive controls. Column 2: parent fixed effects with child age polynomial and education controls. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

H Full Commitment Benchmark

This appendix provides additional details on the full commitment benchmark introduced in Section 7.1 of the main text.

Under full commitment, a benevolent planner commits at $t = 0$ to a state-contingent transfer schedule $\{\tau(a_p, a_c, z, t)\}$ for the entire overlap (parent ages 42–72), solving a single-agent problem over total household wealth $A \equiv a_p + a_c$, whose law of motion carries the same multiplicative wealth diffusion $\sigma_a A dB$ as the decentralized economy (equation 14 in the main text). With CRRA preferences, the efficient allocation equates weighted marginal utilities:

$$u'(c_p) = \eta u'(c_c), \quad \text{so that} \quad c_c = \eta^{1/\gamma} c_p \quad (29)$$

Consumption shares are time-invariant. Substituting yields:

$$u_{\text{FC}}(c_{\text{tot}}) = \frac{c_{\text{tot}}^{1-\gamma}}{1-\gamma} \left(1 + \eta^{1/\gamma}\right)^\gamma \quad (30)$$

Relationship to the one-time transfer counterfactuals. The full commitment benchmark and the one-time transfer counterfactuals (Section 7.2) remove the Samaritan’s dilemma through different mechanisms. Full commitment eliminates moral hazard while preserving state-contingent insurance: the planner adjusts transfers in response to income shocks throughout the overlap period. The one-time transfer counterfactuals eliminate both moral hazard and ongoing insurance: the child receives a lump sum at $t = 0$ and is subsequently on their own. The difference between FC and the one-time transfer economies therefore isolates the *insurance-removal effect*—the change in college attendance and welfare attributable to shutting down ongoing parental support, holding the absence of moral hazard fixed.

Comparing FC with the baseline economy yields the pure moral hazard effect holding insurance constant. As discussed in Section 7.1, this comparison reveals two opposing forces: anticipated transfers subsidize enrollment (raising college attendance above the efficient level) but also cushion non-college children (lowering attendance below the efficient level). The net sign depends on the child’s position in the ability–wealth distribution.

I Equilibrium Properties

This appendix derives the key properties of the Markov-Perfect Equilibrium in the continuous-time coupled problem.

I.1 Optimality Conditions

Following [Barczyk and Kredler \(2014a\)](#), the equilibrium partitions the state space into three regions: a *self-sufficient* region, where the possibility of transfers is remote; an *overconsumption* region, where no transfer currently flows ($\tau^* = 0$) but the state is drifting toward the transfer region; and the *transfer* region, where $\tau^* > 0$. The child's first-order condition $u'(c_c^*) = V_{a_c}^c$ holds throughout. In the self-sufficient region, the child's continuous-time Euler equation takes the standard form $\dot{c}_c/c_c = (r - \rho)/\gamma$. In the overconsumption region, the child anticipates that accumulating wealth will reduce future parental support, so the effective return on saving is $r + \partial\tau^*/\partial a_c$ and the Euler equation becomes:

$$\frac{\dot{c}_c}{c_c} = \frac{1}{\gamma} \left(r - \rho + \frac{\partial\tau^*}{\partial a_c} \right), \quad \frac{\partial\tau^*}{\partial a_c} < 0. \quad (31)$$

This anticipation wedge is the Samaritan's dilemma in continuous time: it lowers the child's return on saving below r and induces over-consumption *before* any transfer flows. Inside the transfer region, by contrast, the parent equalizes the marginal values of own and child wealth ($V_{a_p}^p = V_{a_c}^p$) and acts as a family dictator, setting the child's consumption directly; the child's own Euler equation is therefore not the operative condition there ([Barczyk and Kredler, 2014a](#)).

The parent's first-order condition gives $u'(c_p^*) = V_{a_p}^p$. In the no-transfer region:

$$\frac{\dot{c}_p}{c_p} = \frac{1}{\gamma} (r - \rho) \quad (32)$$

In the transfer region, $V_{a_p}^p = V_{a_c}^p$: the parent equalizes the marginal values of own and child wealth and makes a positive gift $\tau > 0$. The parent funds the gift out of its own resources but continues to optimize its wealth, so its drift $\dot{a}_p = r a_p + y_p - c_p - \tau$ need not be zero—the parent makes a *bounded* transfer (it lifts the child only toward its own optimum) and keeps saving.

I.2 Transfer Region Characterization

The complementary-slackness condition (3) partitions the state space into:

1. **No-transfer region** $\mathcal{N} = \{(a_p, a_c, z, t) : V_{a_p}^p > V_{a_c}^p\}$: Parent and child accumulate wealth independently.
2. **Transfer region** $\mathcal{T} = \{(a_p, a_c, z, t) : V_{a_p}^p = V_{a_c}^p\}$: The parent makes a positive gift that brings the child toward the altruistic target $c_c = \eta^{1/\gamma} c_p$, bounded by the child's own optimum, $\tau^* = \max\{0, \min(c_c^{\text{opt}} - R_c, \eta^{1/\gamma} c_p - R_c)\}$ with $R_c = \tilde{r}_c a_c + y_c$. The parent funds the gift from its resources but continues to save ($\dot{a}_p = r a_p + y_p - c_p - \tau^*$).

The no-transfer region $\mathcal{N}$ subdivides into a *self-sufficient* subregion, where the wealth distribution is balanced and future transfers are remote, and an *overconsumption* subregion bordering $\mathcal{T}$, where no transfer currently flows but the state is drifting toward $\mathcal{T}$ and the anticipation wedge of (31) is active. The boundary of $\mathcal{T}$ is a free boundary determined endogenously. Parents transfer when the child is relatively poor (low a_c), faces negative income shocks (low z), or when the parent is wealthy (high a_p). The region shrinks as the parent approaches death ($t \rightarrow T$).

I.3 The Samaritan's Dilemma

Both agents over-consume relative to the commitment solution. The child's distortion arises in the overconsumption region: anticipating that accumulating wealth would move the state out of $\mathcal{T}$ and eliminate support, the child faces a reduced effective return on saving. This marginal disincentive to save is captured by $\partial\tau^*/\partial a_c$ in (31), and it operates *before* transfers flow, not inside $\mathcal{T}$ itself, where the parent dictates the allocation.

The parent's behavior is also shaped by the lack of commitment: unable to commit to withholding support, the parent transfers whenever the child is sufficiently constrained, and the child anticipates this. Unlike a reflecting-barrier formulation, the parent does not exhaust its wealth in $\mathcal{T}$; it makes a bounded gift—lifting the child only toward its own optimum—and continues to save. The distortion therefore operates through the size and likelihood of transfers, and through the child's reduced self-insurance, rather than through a degenerate parent saving rule.

The continuous-time formulation offers three advantages over discrete-time alternatives. First, transfers are a continuous flow rather than lump-sum payments, eliminating timing discreteness. Second, the free boundary provides a sharp characterization of where moral hazard operates. Third, the multiplicative wealth diffusion $\sigma_a a dB$ (Section 3) renders the value functions C^2 almost everywhere and the free boundary smooth, so the Markov-perfect equilibrium exists in pure strategies and is computed without the non-smoothness—the “blast from the past”—that afflicts deterministic and discrete-time altruism models (Barczyk and Kredler, 2014a, 2021).

I.4 College and Parental Influence

The college decision at $t = 0$ is forward-looking: the child compares lifetime values under each education path, incorporating the entire future trajectory of transfers. Altruistic parents affect the decision through two channels. First, they provide generous transfers during the college years (when child income is low), effectively subsidizing attendance—the transfer region $\mathcal{T}$ is larger when $e_c = C$ during the early years. Second, the continuation value under $e_c = C$ incorporates higher future income, which feeds back into the transfer policy: wealthier children require fewer transfers, freeing parent resources. The net effect generates a positive gradient of college enrollment in parent wealth.

J Solution Algorithm

The model is solved numerically using an upwind finite difference scheme with implicit time-stepping, following [Achdou et al. \(2022\)](#) and the transfer-region methods of [Barczyk and Kredler \(2014a,b\)](#). The algorithm proceeds in six stages: grid construction, solving the child-alone HJB, solving the coupled parent-child HJB system, computing college choice probabilities, forward simulation, and moment computation for SMM estimation.

J.1 State Space and Grid Construction

The model features three continuous state variables during the overlap period: parent wealth $a_p \in [0, \bar{a}_p]$, child wealth $a_c \in [\underline{a}_c, \bar{a}_c]$, and the persistent income shock $z \in [-\bar{z}, \bar{z}]$. In addition, child ability θ and education $e \in \{\text{HS}, \text{C}\}$ are discrete states that index separate HJB problems.

Asset grids. Both wealth grids use exponential spacing to concentrate points near the origin, where policy functions exhibit greater curvature. For N grid points from $a_{\min}$ to $a_{\max}$ with growth factor $g > 0$:

$$a_i = a_{\min} + a_{\max} \cdot \frac{(1+g)^{i-1} - 1}{(1+g)^{N-1} - 1}, \quad i = 1, \dots, N. \quad (33)$$

The parent grid has $N_{a_p} = 30$ points over $[0, 4,000]$ (thousands of dollars). The child grid has $N_{a_c} = 30$ points, with approximately 20% of points allocated to the negative region $[\underline{a}_c, 0)$ using uniform spacing to accommodate student debt during college, and the remaining 80% exponentially spaced over $[0, \bar{a}_c]$. We set the child-grid floor to $\underline{a}_c = -25$, slightly below the cumulative student borrowing limit $\bar{b}_0 = 23$ (Section 3.7) so the constraint is bracketed by interior grid points, and $\bar{a}_c = 2,500$.

Income shock grid. The Ornstein-Uhlenbeck process $dz = -\kappa_e z dt + \sigma_e dW$ is discretized on a uniform grid of $N_z = 9$ points spanning ± 3 stationary standard deviations:

$$z_k \in \left\{ -3 \frac{\sigma_e}{\sqrt{2\kappa_e}}, \dots, +3 \frac{\sigma_e}{\sqrt{2\kappa_e}} \right\}, \quad k = 1, \dots, N_z.$$

The grid width is set to the maximum stationary standard deviation across education types to ensure both HS and college processes are well-covered.

Ability grid. Child ability follows an AR(1) in logs across generations: $\log \theta' = \rho_\theta \log \theta + \mu_\theta(1 - \rho_\theta) + \sigma_\theta \varepsilon$. We discretize this using the Tauchen (1986) method with $N_\theta = 5$ grid points spanning ± 3 stationary standard deviations of $\log \theta$.

Time grid. The backward induction uses $N_t = 60$ equally spaced time steps over the parent's full remaining life ($T = 30$ years, parent ages 42–72, child ages 18–48), yielding $\Delta t = 0.5$ years per step. The implicit scheme is unconditionally stable, so this step size is accurate at half the cost of a finer grid. The coupled parent-child system is solved over the entire horizon $[0, T]$: the parent retires at $T^{\text{ret}} = 24$ (age 66) and switches to social security income, but remains coupled to the child and can continue to transfer until death at T (age 72).

J.2 Stage 1: Child-Alone HJB

After the parent dies (child age 48), the child solves a standard consumption-savings problem independently for the remainder of life ($T_{\text{child alone}} + T_{\text{child retire}} = 24$ years, child ages 48–72, working until 66 and then on social security). The HJB equation, with the same multiplicative wealth diffusion as the coupled problem, is:

$$\rho V^{c, \text{alone}}(a, z, t) = \max_{c \geq 0} \left\{ u(c) + V_a^{c, \text{alone}} [r a + y(t, z) - c] + V_z^{c, \text{alone}} (-\kappa z) + \frac{1}{2} \sigma^2 V_{zz}^{c, \text{alone}} + \frac{1}{2} \sigma_a^2 a^2 V_{aa}^{c, \text{alone}} + V_t^{c, \text{alone}} \right\} \quad (34)$$

We solve this problem separately for each ability type θ and education level $e_c \in \{\text{HS}, \text{C}\}$, giving $2 \times N_\theta = 10$ independent HJB problems that are solved in parallel.

Terminal condition. At the child's death (age 72, i.e. $t = T_{\text{child}}$ on the child-alone clock), the terminal value is $V^c(a, z, T_{\text{child}}) = u(c_T)/\rho$ where $c_T = \max\{r a + \text{SS}_{e_c} + 0.1 a, \epsilon\}$ represents consuming all remaining resources.

Optimal consumption via the first-order condition. At each grid point and time step, the FOC $u'(c^*) = V_a^c$ yields the candidate consumption $c^* = (V_a^c)^{-1/\gamma}$, where γ is the

CRRA coefficient. We compute V_a^c using upwind finite differences: the *forward* difference $\partial_a^+ V$ and *backward* difference $\partial_a^- V$ yield two candidate consumption levels, c^{fwd} and c^{bwd} , with associated savings drifts $s^{\text{fwd}} = ra + y - c^{\text{fwd}}$ and $s^{\text{bwd}} = ra + y - c^{\text{bwd}}$. The upwind scheme selects:

$$c^* = \begin{cases} c^{\text{fwd}} & \text{if } s^{\text{fwd}} > 0 \quad (\text{saving: use forward difference}), \\ c^{\text{bwd}} & \text{if } s^{\text{bwd}} < 0 \quad (\text{dissaving: use backward difference}), \\ ra + y & \text{otherwise (drift } \approx 0: \text{ consume resources)}. \end{cases}$$

This upwind selection ensures numerical stability by using the derivative in the direction from which information flows.

Implicit time stepping. Given optimal policies at time step n , we construct the sparse infinitesimal generator $\mathbb{A}^n$ that encodes the upwind finite-difference operator in a and the discretized OU operator in z (combining upwind drift and central-difference diffusion terms). The implicit update solves:

$$\left[\left(\frac{1}{\Delta t} + \rho \right) \mathbf{I} - \mathbb{A}^n \right] \mathbf{V}^n = \mathbf{u}^n + \frac{1}{\Delta t} \mathbf{V}^{n+1}, \quad (35)$$

where $\mathbf{V}^n$ and $\mathbf{u}^n$ are the vectorized value function and flow utilities at step n , $\mathbf{I}$ is the identity matrix of dimension $N_{ac} \times N_z$, and $\mathbf{V}^{n+1}$ is the value at the next (future) time step. This system is solved via sparse LU factorization.

OU generator. The infinitesimal generator for the OU process on the uniform z -grid combines a central-difference approximation for diffusion with upwind differencing for the drift:

$$\begin{aligned} (\mathbb{A}_z)_{k,k+1} &= \frac{\sigma^2/2}{\Delta z^2} + \frac{\max(-\kappa z_k, 0)}{\Delta z}, \\ (\mathbb{A}_z)_{k,k-1} &= \frac{\sigma^2/2}{\Delta z^2} + \frac{\max(\kappa z_k, 0)}{\Delta z}, \\ (\mathbb{A}_z)_{k,k} &= -(\mathbb{A}_z)_{k,k+1} - (\mathbb{A}_z)_{k,k-1}, \end{aligned} \quad (36)$$

with reflecting boundary conditions at $k = 1$ and $k = N_z$.

Wealth-diffusion generator. The multiplicative wealth shock $\sigma_a a dB$ contributes a second-order term $\frac{1}{2}\sigma_a^2 a^2 V_{aa}$, discretized by a central second difference on the (non-uniform) wealth grid. With diffusion coefficient $D_i = \frac{1}{2}\sigma_a^2 a_i^2$, forward spacing $\Delta_i = a_{i+1} - a_i$ and backward spacing $\Delta_{i-1} = a_i - a_{i-1}$, the generator entries are

$$\begin{aligned} (\mathbb{A}_a)_{i,i+1} &= D_i \frac{2}{\Delta_i(\Delta_i + \Delta_{i-1})}, \\ (\mathbb{A}_a)_{i,i-1} &= D_i \frac{2}{\Delta_{i-1}(\Delta_i + \Delta_{i-1})}, \\ (\mathbb{A}_a)_{i,i} &= -(\mathbb{A}_a)_{i,i+1} - (\mathbb{A}_a)_{i,i-1}. \end{aligned} \tag{37}$$

The coefficient D_i vanishes at $a_i \leq 0$ (no volatility at the borrowing constraint, following [Barczyk and Kredler, 2014a](#)) and at the grid boundaries, so no extra boundary condition is required. Because D_i depends only on the grid, this block is assembled once and reused across all time steps; it preserves the generator's M -matrix structure, so the implicit step remains stable.

J.3 Stage 2: Coupled Parent-Child HJB

During the overlap period, the parent and child interact strategically. The parent's HJB equation is given by (1), with the transfer decision characterized by the complementarity condition (3). That is, the parent transfers whenever the marginal value of an additional dollar in the child's account exceeds the marginal value of retaining it.

Terminal condition. At $t = T$ (parent age 72, the parent's death), the coupled system terminates. The parent's terminal value combines a warm-glow term over remaining wealth and altruistic concern for the child's continuation value (equation 9). The child's terminal value is $V^c(a_p, a_c, z, T) = V^{c,\text{alone}}(a_c + \alpha_{\text{beq}} a_p, z)$, the child-alone value function from Stage 1 evaluated at $t = 0$ with effective wealth including the bequest.

Transfer allocation in closed form. We solve the coupled system backward in time. At each time step n and grid point (i, j, k) the transfer is obtained in *closed form*: because

altruism is one-sided (only the parent gives) and the recipient's marginal value enters through a static first-order condition, the transfer region $\mathcal{T}^n$ and the gift size are determined pointwise, without iterating on the region. The allocation is the continuous-time analog of the rule in [Barczyk and Kredler \(2014a\)](#) and the `GetBKalloc` routine of [Barczyk et al. \(2023\)](#), specialized to one-sided altruism.

1. **Own-consumption first-order conditions.** Each agent's unconstrained optimal consumption is recovered from its own marginal value of wealth using the upwind selection above:

$$c_p = (V_{a_p}^p)^{-1/\gamma}, \quad c_c^{\text{opt}} = (V_{a_c}^c)^{-1/\gamma},$$

with implied parental saving $s_p = ra_p + y_p - c_p$.

2. **Bounded gift.** The static transfer FOC $u'(c_p) = \eta u'(c_c)$ implies a target child consumption $c_c = \eta^{1/\gamma} c_p$, and hence a first-best gift $g_{\text{fb}} = \eta^{1/\gamma} c_p - R_c$, where $R_c = \tilde{r}_c a_c + y_c$ is the child's own resources. The gift is capped at the level that lifts the child to its *own* optimum (the second-best ceiling) and floored at zero:

$$\tau = \max\left\{0, \min\left(c_c^{\text{opt}} - R_c, \eta^{1/\gamma} c_p - R_c\right)\right\}. \quad (38)$$

The point lies in the transfer region, $(i, j, k) \in \mathcal{T}^n$, if and only if $\tau > 0$. The lower bound is the one-sided no-negative-transfer constraint; the upper bound prevents a wealthy parent from transferring more than brings the child to its own optimum, which is what disciplines the gift to a finite, liquidity-insurance role.

3. **Wealth drifts.** The parent funds the gift out of its own saving and continues to optimize its wealth—it is *not* frozen in the transfer region:

$$\dot{a}_p = s_p - \tau, \quad c_c = \min\left(c_c^{\text{opt}}, R_c + \tau\right), \quad \dot{a}_c = R_c + \tau - c_c.$$

Since $\tau \leq c_c^{\text{opt}} - R_c$, the child consumes its resources plus the entire gift, so $\dot{a}_c = 0$ inside the transfer region: the transfer is pure liquidity insurance to a constrained child, consistent with the mechanism in [Section 3.9](#).

4. **Construct the generator and solve.** Given the policies and drifts, assemble the sparse infinitesimal generator $\mathbb{A}^n$ over the full state space (a_p, a_c, z) of dimension $N_{a_p} \times N_{a_c} \times N_z$. The generator combines upwind drift operators in both wealth dimensions, the multiplicative wealth-diffusion blocks in a_p and a_c , and the OU operator in z .

A key computational insight is that the parent and child value functions share the *same* state transitions—both wealth drifts and the OU process are identical for both agents given the equilibrium policies. Therefore, the system matrix

$$\mathbf{B} = \left(\frac{1}{\Delta t} + \rho \right) \mathbf{I} - \mathbb{A}^n$$

is factorized *once* (via sparse LU), and used for two triangular solves with different right-hand sides:

$$\mathbf{B} \mathbf{V}^{p,n} = \mathbf{u}^p + \frac{1}{\Delta t} \mathbf{V}^{p,n+1}, \quad (39)$$

$$\mathbf{B} \mathbf{V}^{c,n} = \mathbf{u}^c + \frac{1}{\Delta t} \mathbf{V}^{c,n+1}, \quad (40)$$

where $\mathbf{u}^p = u(c_p) + \eta u(c_c)$ and $\mathbf{u}^c = u(c_c)$ are the flow utilities.

This procedure is repeated for each combination of child ability θ , child education $e_c \in \{\text{HS}, \text{C}\}$, parent education $e_p \in \{\text{HS}, \text{C}\}$, and the parent's frozen income type z_p (on a grid of $N_{z_p} = 5$ nodes), yielding $N_\theta \times 2 \times 2 \times N_{z_p} = 100$ coupled HJB problems that are solved in parallel across threads.

College-period modifications. When the child is in college (age < 22), two modifications apply: (i) the child earns a fraction ℓ_C of full-time income and pays net tuition $\phi_{\text{net}}(y_p^0) = \phi - \bar{g} \cdot \max(0, 1 - y_p^0/\bar{y}_{\text{efc}})$, which varies continuously with parent *income* at college entry through the EFC system; and (ii) the child may borrow up to $\bar{b}(a_p)$ at the student loan rate $r + \iota_s$.

Borrowing constraints. The child's wealth drift is constrained to be non-negative at the borrowing limit: if $a_c \leq \underline{a}_c + \epsilon$ and $\dot{a}_c < 0$, we set $\dot{a}_c = 0$. During college, $\underline{a}_c = -\bar{b}$; after college, $\underline{a}_c = 0$. The parent faces a natural borrowing constraint $a_p \geq 0$.

J.4 Stage 3: College Choice

At $t = 0$, the college enrollment decision combines the child’s and parent’s values with a paternalism weight $\psi \geq 0$. The probability of attending college follows the probit specification in equation (11): all value functions are evaluated at $a_c = 0$, $z = 0$, and $t = 0$; the paternalism-adjusted value gain $G(a_p, \theta)$ is compared to the realized psychic cost $\kappa(\theta, \varepsilon_\kappa) = (\kappa_0 + \kappa_\theta \log \theta + \sigma_\kappa \varepsilon_\kappa) \cdot \bar{V}$ with $\varepsilon_\kappa \sim N(0, 1)$; and $\bar{V} = \text{mean}_{a_p, \theta} |V_C^c - V_{HS}^c|$ normalizes the cost to the scale of the value gap. Because the cost shock is Gaussian, the attendance probability $\Phi((G/\bar{V} - \kappa_0 - \kappa_\theta \log \theta)/\sigma_\kappa)$ is available in closed form, so no numerical integration is required.

Boundary monotonicity correction. At very low parent wealth ($a_p \approx 0$), the coupled HJB can produce spurious college probabilities due to convergence difficulties at the corner of the state space. We enforce monotonicity by scanning from the bottom of the a_p grid upward: if the college probability at point i exceeds that at point $i + 1$ for the first few grid points, we cap it at the first interior-monotone value. This correction affects at most the bottom third of the grid.

J.5 Stage 4: Dynastic Simulation

We simulate $M = 4,000$ dynasties across $N_{\text{coh}} = 11$ overlapping generations by forward-simulating the model’s discretized law of motion at the same time step used to solve the HJB system.

Initialization. The first cohort begins at median ability $\theta = \theta_{(N_\theta+1)/2}$. Subsequent generations draw ability from the AR(1) process: $\log \theta' = \rho_\theta \log \theta + \mu_\theta(1 - \rho_\theta) + \sigma_\theta \varepsilon$. Each new generation’s parent wealth equals the previous generation’s terminal child wealth (clamped to $[0, \bar{a}_p]$). The initial income shock z_0 is drawn from an education-specific entry distribution $z_0 \sim \mathcal{N}(\zeta_{pe} \mathbf{1}\{e_p = C\}, \sigma_e^2/2\kappa_e + \sigma_{z_0, e_c}^2)$: the variance combines the OU stationary variance with the Abbott et al. (2019) entry-dispersion component σ_{z_0, e_c} , and the mean carries the parent-college earnings premium ζ_{pe} (Section 3).

College decision. For each dynasty m in cohort g , we interpolate the college probability $\Pr(C \mid a_p^{m,g}, \theta^{m,g})$ from the solved grid and draw education status from a Bernoulli distribu-

tion.

Forward simulation. Given education choices and initial conditions, we simulate the state dynamics forward in discrete time steps. At each time step $n = 1, \dots, N_{\text{steps}}$:

1. **Interpolate policies:** trilinear interpolation of (c_p, c_c, τ) from the solved HJB policy grids at the current state (a_p^n, a_c^n, z^n) .
2. **Compute income:** parent income y_p (deterministic or Social Security if retired) and child income y_c (with transitory shock $\varepsilon \sim \mathcal{N}(0, \sigma_{\varepsilon,e}^2)$, net of tuition during college).
3. **Update wealth** (drift plus the multiplicative wealth-diffusion shock):

$$\begin{aligned} a_p^{n+1} &= a_p^n + (r a_p^n + y_p - c_p - \tau) \Delta t + \sigma_a a_p^n \sqrt{\Delta t} \xi_p^n, \\ a_c^{n+1} &= a_c^n + (\tilde{r}_c a_c^n + y_c - c_c + \tau) \Delta t + \sigma_a a_c^n \sqrt{\Delta t} \xi_c^n, \end{aligned} \quad (41)$$

with independent $\xi_p^n, \xi_c^n \sim \mathcal{N}(0, 1)$; the diffusion term is set to zero whenever $a \leq 0$, matching the HJB.

4. **Update income shock:** $z^{n+1} = z^n - \kappa_e z^n \Delta t + \sigma_e \sqrt{\Delta t} \xi^n$, where $\xi^n \sim \mathcal{N}(0, 1)$.
5. **Enforce constraints:** clamp a_p to $[0, \bar{a}_p]$ and a_c to $[\underline{a}_c, \bar{a}_c]$ (with $\underline{a}_c = -\bar{b}$ during college, $\underline{a}_c = 0$ after). Clamp z to the grid bounds.

Reproducibility. All random draws (initial z_0 , college choice, OU innovations, transitory shocks, and the parent and child wealth-diffusion innovations ξ_p, ξ_c) are pre-generated sequentially before the threaded simulation loop, ensuring bit-identical results regardless of the number of parallel threads.

J.6 Stage 5: Moment Computation and SMM Estimation

The model is estimated by Simulated Method of Moments. It computes 40 moments—36 of which are targeted—organized into seven blocks. All moments are computed from a settled cohort (the second-to-last generation) to avoid transient dynamics from initial conditions.

Moment blocks.

1. **College attendance** (16 moments, weight 3): enrollment rates by parent-wealth quartile $\times$ ability quartile.
2. **Income** (1 targeted, 2 untargeted): HS/college income ratio (targeted); cross-sectional income standard deviations for HS and college workers (computed but untargeted, since the income process is fixed externally).
3. **Transfers** (5 moments): mean cumulative transfer during college years, transfer probability $\Pr(\tau > 0)$ by parent-wealth quartile.
4. **Wealth** (10 moments): wealth-to-income ratio, intergenerational rank-rank slope, conditional child wealth $\mathbb{E}[a_c \mid \text{parent wealth } Q]$ (4 moments), median wealth for parents and children, variance of $\text{IHS}(a)$ for parents and children.
5. **Decumulation** (1 moment): parent wealth decumulation rate from age 60 to 70.
6. **College by parent income** (4 moments): college graduation rate by parent-income quartile.
7. **Negative net worth** (1 moment): fraction of children aged 35–40 with negative net worth.

Loss function. The objective uses block-diagonal weighting so that each group contributes proportionally:

$$\mathcal{L}(\boldsymbol{\vartheta}) = \sum_g \frac{w_g}{n_g} \sum_{m \in g} \left(\frac{m_m(\boldsymbol{\vartheta}) - \hat{m}_m}{\hat{m}_m} \right)^2, \quad (42)$$

where n_g is the number of moments in block g and w_g is the group weight (college moments receive $w_g = 3$; all other groups receive $w_g = 1$).

Optimization. We minimize $\mathcal{L}$ over the 12 free parameters using a parallel $(\mu/\mu_w, \lambda)$ -CMA-ES optimizer restricted to the free (non-fixed) subspace. Each candidate vector is evaluated by solving the full model (Stages 1–4) and computing the loss, and multi-start initialization ensures coverage of the parameter space.

J.7 Stage 6: Counterfactual Economies

Full commitment. The parent and child commit to a transfer schedule $\{\tau_t\}_{t \geq 0}$ at $t = 0$. This is solved as a single planner’s problem that jointly maximizes parent and child welfare, eliminating the Samaritan’s dilemma. The resulting college rates isolate the moral hazard effect against the full-commitment benchmark: $\Delta_{MH}^{FC} = \Pr(C \mid \text{baseline}) - \Pr(C \mid FC)$. This full-commitment comparison is distinct from the main-text decomposition, which uses the one-time-transfer economies ($\Delta_{MH}^{\text{pure}}$ and Δ_{targ} in equations 19–21).

No-moral-hazard (unconditional). The parent makes a one-time lump-sum transfer τ_0 at $t = 0$, after which the child solves the child-alone problem for their entire remaining life. The child receives τ_0 regardless of education choice. The parent’s problem is:

$$\max_{\tau_0 \in [0, a_p]} V_p^{\text{alone}}(a_p - \tau_0) + \eta \cdot \mathbb{E}_{\varepsilon_\kappa} \left[\max \left\{ V^{c, \text{alone}}(\tau_0 \mid C) - \kappa(\theta, \varepsilon_\kappa) + \varepsilon_C, V^{c, \text{alone}}(\tau_0 \mid HS) + \varepsilon_{HS} \right\} \right].$$

Here $V_p^{\text{alone}}(\cdot)$ is the value of a single-agent parent problem—a continuous-time consumption-savings HJB over the parent’s remaining life (labor income to age 66, social security thereafter, the consumption floor, the warm-glow bequest, and the same wealth diffusion $\sigma_a a dB$ as the baseline). Solving it as a genuine dynamic program, rather than approximating it by a permanent-income annuity, places the parent’s counterfactual value on the same footing as the baseline coupled value, so the consumption-equivalent comparisons in Section 8 difference like with like. The child-alone value $V^{c, \text{alone}}(\tau_0 \mid e_c)$ is solved over the child’s entire post-transfer life and likewise carries the wealth diffusion.

No-moral-hazard (menu). Same as above, but the parent commits to an education-contingent menu: a transfer τ_C paid if the child attends college and τ_{HS} paid if the child chooses high school, both optimized in $[0, a_p]$. This is the efficient one-time-transfer benchmark; it nests the unconditional transfer ($\tau_C = \tau_{HS}$) and measures the effect of removing moral hazard *plus* optimal education targeting.

Four-way decomposition. The difference in college rates across these allocations decomposes as:

$$\underbrace{\Pr(C \mid \text{BL}) - \Pr(C \mid \text{menu})}_{\Delta_{\text{MH}}^{\text{total}}} = \underbrace{\Pr(C \mid \text{BL}) - \Pr(C \mid \text{uncond.})}_{\Delta_{\text{MH}}^{\text{pure}}} + \underbrace{\Pr(C \mid \text{uncond.}) - \Pr(C \mid \text{menu})}_{\Delta_{\text{targ}}}. \quad (43)$$

The signs are consistent with the main text (Section 7): a positive $\Delta_{\text{MH}}^{\text{pure}}$ indicates that baseline moral hazard raises college attendance relative to the no-moral-hazard economy.